

An Introduction to Mathematical Statistics

Course notes for APPM/MATH 4/5520

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by J.N. Corcoran

University of Colorado, Boulder.¹

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¹Postal Address: Department of Applied Mathematics, University of Colorado,
Boulder, CO 80309-0526, USA; email: corcoran@colorado.edu

Congratulations on deciding to take a course in "MathStat." And, if it wasn't your decision, but instead some annoying or dreaded requirement dictated by your major, consider this your good fortune because MathStat seriously rocks.

The first secret I'm going to let you in on is that MathStat isn't really statistics— it's probability. Yes, there is a difference.

Probability is about the future.

Suppose you have an unfairly weighted coin that, if flipped, will result in "heads" 57% of the time. Suppose that **you are going** to flip this coin 10 times. What is the probability that you **will** see at least 7 heads? How many heads **do you expect** to see?

Did you catch all those future type words and phrases there?

Statistics, on the other hand, is about the past.

That is, suppose that you know next to nothing about the coin but you flipped it 10 times and observed the heads/tails data

$H, H, T, T, T, H, H, T, H, H.$

You can use statistics to attempt figure out (albeit with some uncertainty) whether or not the coin **was** fair in the first place and what that single flip heads probability **might have been**.

Do you see how statistics is about looking **back** to figure out what was going on with the coin?

Much of the subject of statistics is about "reverse engineering" the data generating process in the case that the process involves some randomness or probability. Probability can stand alone as a subject, but in order to do statistics we need to understand probability.

Mathematical Statistics is the probability needed to do statistics.

So you see, you have really enrolled yourself in a probability course and not a statistics course. MathStat is a specialized area of probability and

before you can learn it you need to first know some basics. This is why a probability course, such as APPM 3570 (Applied Probability) or MATH 4510 (Introduction to Probability Theory) is **an important prerequisite**.

Statistics, on the other hand, is not a prerequisite for this course, but if you've studied or used statistical analysis techniques, be prepared to see some really cool connections this semester. If you've ever had a course in statistical analysis where you've estimated things (for example averages or medians) or performed hypothesis tests that involved looking numbers up in, for example, "t-tables", this course will tell you why doing those things was meaningful, when those techniques actually are valid for a problem, and what to do if they aren't. Mathematical Statistics is an intense theoretical course in which sometimes even the link to the "data stuff" will seem completely obscured, but it is, in a word, **beautiful**.

These notes are designed for a course in the department of Applied Mathematics at the University of Colorado, Boulder. The course is cross-listed for graduate students and senior undergraduates. The prerequisite is a basic undergraduate probability course. As the audience background has been varied, and adherence to prerequisites is often tenuous, I have included a "Preliminaries" chapter that may turn more advanced students, who may have stumbled on to these notes from the web, off with its simplicity. Rest assured that things will ramp up quickly after that and that many will want to skip the first chapter altogether. (It may still be useful to skim through for the purpose of picking up notation though!) I believe that these notes are well suited for a graduate course in Mathematical Statistics even in a pure Statistics Department, where I have also had the experience of teaching MathStat. It includes many advanced concepts, examples, and challenging problems.

On a final note, for maximum success in learning MathStat, or any mathematical subject for that matter, **DO NOT TRY TO MEMORIZE THINGS!** I always tell people that I ended up a mathematician because I have an awful memory. On a history exam, if you don't know who did what and in what year, you are out of luck. Mathematics, however, comes from inside of you. If you're stuck on a problem on a math exam, as long as you stay calm and keep breathing, you **can** figure it out. (Always breathe!) If mathematics doesn't "come naturally" to you, I'd **urge** to you to take the extra time and effort to really **understand** the basics of a subject at the beginning of the semester. It will be smooth sailing after that initial period and overall you will spend a lot less time on your MathStat homework. I've dealt with a lot

of “mathphobes” over the years, and time and time again I’ve watched the ones who take this advice eventually no longer need my help and go on to focus on other things from “other classes” to “a life” while the others stay a slave to my extended office hours right up to the final exam! Along these same lines, all of us are guilty at some point or another of searching the web and/or a giant stack of books from the library for examples similar to a homework problem. The next time you do that, try keeping track of all that time you spent that was not spent solving the problem.

“Just do it.” Cheesy? Absolutely, but also very true!

J.N. Corcoran

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Chapter 0

Probability Preliminaries

This chapter is **very** elementary and is not at all representative of the overall level of these notes. This chapter is also a strange sort of introduction to probability in its coverage. Some very basic central concepts and even notation for understanding probability are completely absent. Instead, it contains only the bare minimum of topics needed in order to understand the subsequent chapters. For those students who insist on trying mathematical statistics without a basic probability background, this should help. For others already familiar with things like random variables, “pdfs”, “cdfs”, and expectations, it still may be helpful to only skim Sections 0.1 through 0.7 since they will serve to establish some basic terminology and notation. Sections 0.8-0.10 are recommended reading for everyone!

0.1 Between Zero and One

If you flip a “fair” coin, what is the probability that it will come up “heads”?

Most people have at least a sense of what this question is asking without any formal training in probability. Many will use the words “probability” and “chance” interchangeably and say:

“There is a 50 percent chance of it coming up heads.”

This is not wrong but, **formally**, in mathematics

Probability is a number between 0 and 1.

So, the more precise answer to the original question is $1/2$ or 0.5.

If you roll a “fair” six-sided die, what is the probability that you will get a 5?

In the case of the coin and now the die, the word “fair” is there to say that there is nothing funny going on. The coin isn’t warped, the die isn’t shaved or repainted with extra dots, and the acts of flipping and rolling are not done in a way that favors one outcome over another. For the coin, the two outcomes “heads” and “tails” are **equally likely**, and for the die, the six outcomes are equally likely.

To answer the question, since the outcome we care about (getting a 5) is one outcome out of six equally likely outcomes, the probability of getting a 5 is $1/6$.

If you roll a fair six-sided die, what is the probability that you will get either a 5 or a 6?

Since there are now 2 outcomes out of 6 equally likely outcomes that we care about, the answer is $2/6$ or $1/3$.

 Note

In the case of equally likely outcomes for an experiment, the probability that a certain event occurs is

the number of outcomes in the event of interest
divided by
the total number of possible outcomes.

Thus, in order to compute probabilities in the case of equally likely outcomes, it is important for us to be able to count things.

0.2 Counting

0.2.1 Things in a Row

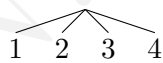
How many different ways can you list the integers 1 through 4 in various orders? Two examples are

3 2 1 4 and 2 4 1 3,

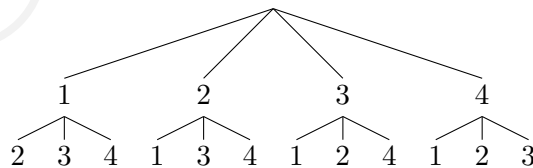
but there are obviously many more possibilities.

Listing things in various orders can help us with the problem of assigning labels to things. For example, suppose you and your three friends are going to assign yourselves administrative titles in your treehouse club. The titles are president, vice president, secretary, and treasurer. You are going to list your names on a piece of paper. The first person on the list will be president, the second will be vice president, and so on. How many different ways can you do this? This is exactly the same as the “1 2 3 4 problem” above—especially so if your names happen to be “1”, “2”, “3”, and “4”.

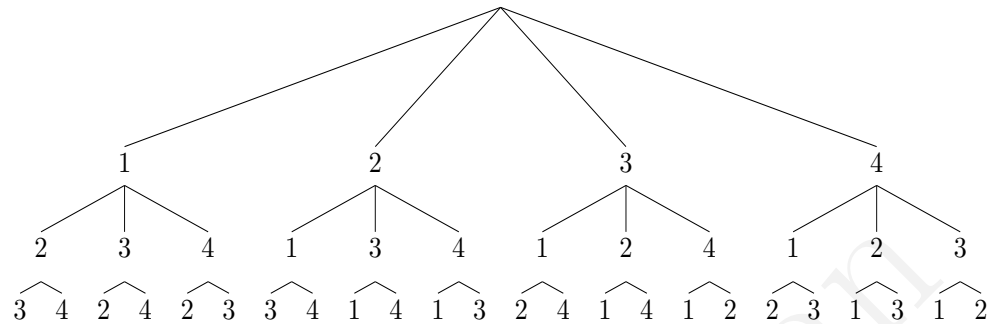
The first number in the list can be 1 or 2 or 3 or 4. Let’s visualize them as branches of a tree.



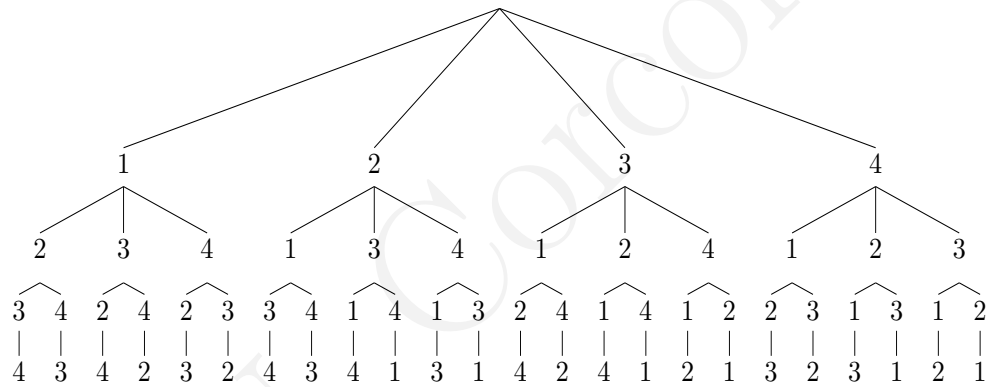
If we start our list with 1, for example, the next number can be 2 or 3 or 4. If we start our list with 2, the next number can be one of 1 or 3 or 4. In general, possibilities for “next numbers”, can be envisioned like this



If our first number is 1 and our second number is 2, our third number can either be 3 or 4. So...more branches.



Finally, if we start our list with 2, 1, 4, for example, we have no choice but to finish up with 3. Thus, we get the final “leaves” on our tree.



Following any path from top to bottom will give a different configuration of the numbers 1, 2, 3, and 4 and all possible configurations are shown. There are 4 branches at the top. Each has 3 offshoots, making $4 \times 3 = 12$ “second level branches”. Each of those has 2 offshoots, making $4 \times 3 \times 2 = 24$ “third level branches” and finally, each of those has one final offshoot making $4 \times 3 \times 2 \times 1 = 24$ total branches representing configurations of 1, 2, 3, and 4. (Of course, we do not need to multiply by 1, but it gives a nice sense of completeness don’t you think?).

**Definition**

For any positive integer n , the symbol “ $n!$ ” is read as “ **n factorial**” and is defined as

$$n! := n(n-1)(n-2) \cdots (2)(1).$$

By convention (so that all will be right with the world and so that certain generalizations will go smoothly) we define $0!$ to be 1.

From the trees above, we have seen that the number of ways to arrange 4 distinct numbers (or people, objects, or “things”) is $4! = (4)(3)(2)(1) = 24$.

**Note**

In general,

$n!$ = the number of ways to arrange n distinct objects.

From our little tree drawing experiment, we have learned at least two other things. First, if the numbers can be “reused”, the number of branches will not decrease. For example, if we roll a fair 6-sided die 4 times we could see outcomes like

3 5 3 4 and 2 1 1 1.

The total number of possible outcomes will be 6 (first branches) times 6 (second branches) times 6 (third branches) times 6 (fourth branches), for a total of $6^4 = 1296$ outcomes.

At this point I could make another fancy box saying something like: “If we perform an experiment with k trials and with n possible outcomes for each trial, we have a total of n^k possible outcomes for the entire experiment.” However, **I won’t and you shouldn’t either**. If you are memorizing things for a math class (outside of definitions), the class will get more and more and more difficult as you pile up information in your head and eventually struggle to access it. If you are memorizing things for a math class, you

are going to be thrown off when problems have only subtle differences from ones you've already seen or solved. On the other hand, if you **understand** and think about what you're doing at each step, the class will get easier and easier even as the material gets more complex!

0.2.2 Choosing Things: Order is Important

Here is the second thing we have learned from our “tree experiment”. Suppose we have to choose 2 numbers out of our 4 numbers and that “**order is important**”. This means that

$$1 \ 4 \quad \text{is different from} \quad 4 \ 1.$$

For the four people in the treehouse club, this scenario could come up if you want to choose only a president and a vice president. You could write two names down and the first in your list will be president, while the second will be vice president. If, as before, the treehouse club's members names are 1, 2, 3, and 4, the choice

$$1 \ 4$$

means that person 1 is president and person 4 is vice president, while the different choice

$$4 \ 1$$

means that person 4 is president and person 1 is vice president.

Since you have 4 choices to put in the first position (envisioned as first branches of a tree) and then 3 leftover people to choose for the second position (second branches), you have a total of $4 \times 3 = 12$ possible ways to assign names to the president and vice president positions in your list.

That was easy enough, but it will be useful to be able to come up with this number using factorials. You might have noticed (or not) that

$$12 = 4 \cdot 3 = \frac{4 \cdot 3 \cdot 2 \cdot 1}{2 \cdot 1} = \frac{4!}{2!},$$

but let's discuss the logic behind this. The point will be a little more clear if we increase the number of treehouse members to five. (The only reason I didn't start with five in the first place is because the tree diagram would have been really wide and too cramped for these pages!)

Suppose there are five members of the treehouse club and that their names are 1, 2, 3, 4, and 5. Further suppose that we want to know how many

different ways we can choose a president and a vice president. We could list all of the members in various orders and choose the first two people in the list to be the president and vice president, respectively. For example, one possible listing of members is

$$4 \ 2 \ 5 \ 3 \ 1.$$

For this listing, 4 will be president and 2 will be vice president.

Another possible listing of members is

$$2 \ 4 \ 5 \ 3 \ 1.$$

In this case, 2 will be president and 4 will be vice president.

In all, there are $5! = 120$ possible listings of these 5 people, but for our purpose of choosing a president and vice president some are redundant. For example

$$4 \ 2 \ 5 \ 3 \ 1 \quad \text{and} \quad 4 \ 2 \ 1 \ 3 \ 5$$

will give the same results for our election. In fact, for this 5 person listing method, we will get the result of 4 as president and 2 as vice president $3! = 6$ different times since there are $3! = 6$ different ways to arrange the three superfluous numbers 1, 3, and 5. So, we need to divide the 120 possible ways to list 5 people by 6. In summary, the number of ways to choose 2 people (numbers/objects/things) from 5 people (numbers/objects/things) when **order is important** is

$$\frac{5!}{3!} = \frac{120}{6} = 20.$$

Note

In general, the number of ways to choose k objects out of n objects when order is important is $n!/(n-k)!$.

What we have just done is to count the number of “permutations” of k objects out of n objects. The quantity we came up with in the end is denoted in many different ways. One of the most popular is ${}_nP_k$. That is,

$${}_nP_k := n!/(n-k)!.$$

You will not need this notation in this course or to understand the rest of these notes. If needed, we will figure out how to choose k things out of n things with logic, just as we did in the treehouse example!

0.2.3 Choosing Things: Order is Not Important

Suppose now that we want to choose two people from our five member treehouse club to fix the ladder. How many different possible “ladder fixing committees” are there? If we were listing out pairs of people, the **order is not important** because having 1 and 4 fix the ladder is the same as having 4 and 1 fix the ladder.

We will take the same approach as in the previous section, where we list out all five names and then take the first two names for our committee.

There are $5! = 120$ ways to list the 5 names. One possibility is

4 2 5 3 1.

This particular listing of names will give us a committee consisting of 4 and 2. As before, the order of the last 3 people is irrelevant and, using all possible listings, we will get 4 and 2 for our committee $3!$ different ways. So, we should divide $5!$ by $3!$ to account for this redundancy. At this point, we have $5!/3!$ lists we care about.

Note that, we still have some redundancy since the outcome

4 2 5 3 1

is now the same as

2 4 5 3 1.

Both of these outcomes mean that our ladder committee consists of 4 and 2. In fact, every pair of people will end up being listed 2 ways. (Note that, for ease of generalization, we will use the factorial notation and say that the number of ways to list 2 people is $2! = 2$.)

So, for our $5!/3!$ surviving lists, we need to divide by $2!$. In summary, the number of ways to choose 2 people (numbers/objects/things) from 5 people (numbers/objects/things) when **order is not important** is

$$\frac{5!/3!}{2!} = \frac{5!}{2!3!}.$$

**Note**

In general, the number of ways to choose k objects out of n objects when order is not important is

$$\frac{n!}{k!(n-k)!}.$$

What we have just done is to count the number of “combinations” of k objects out of n objects. This quantity is almost always denoted by the symbol $\binom{n}{k}$. That is

$$\binom{n}{k} := \frac{n!}{k!(n-k)!}.$$

This symbol is read “**n choose k**” and we will use it a lot!

0.3 Sample Spaces, Events, and Some Simple Probabilities

In probability, we typically use capital Roman letters like A , B and C , for example, to denote “events” which are collections of outcomes from some experiment.

Suppose we perform the fabulously exciting experiment of flipping a fair coin 3 times. Using what I hope is an obvious notation for the sequences of “heads” and “tails” the eight possible outcomes are

$$HHH, HHT, HTH, THH, HTT, THT, TTH, TTT.$$

**Definition**

The set of all possible outcomes of experiment is known as the **sample space** of the experiment.

We will use the symbol Ω to denote the sample space of an experiment.

For this example we would write

$$\Omega = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}.$$

Let us define the event called “ A ” as the one where we see at least one “heads” in our experiment. We can write

A = the event where we see at least one heads,

or, we might use the set notation

$$A = \{HHH, HHT, HTH, THH, HTT, THT, TTH\}$$

to express that A consists of **one** of these outcomes.

What is the probability that the event A occurs? That is, if we are going to flip a fair coin 3 times, what is the probability that we will see at least 1 “heads”?



Notation

We will use $P(A)$ to denote the probability that the event A occurs.

When A is a collection of outcomes at it is here, A “occurring” means that when we performed our experiment it produced one of the outcomes in A .

Since 7 out of the 8 possible outcomes are included in the set A , this probability should be very high. This conclusion, though, is based on the idea that most outcomes are in A , but, it is also based on the idea that the coin is “fair”. If the coin were severely bent or weighted in such a way that “tails” comes up 99.9 percent of the time then we should expect that it is somewhat unlikely that we will see at least 1 heads!

Because heads and tails are “equally likely” for a fair coin, the three toss outcomes, HHH , HHT , etc. are also equally likely. You might be thinking that outcomes like HHH or TTT are less likely than the others for a fair coin. To make this even more dramatic, imagine tossing the coin 20 times. Shouldn’t the outcome of 20 tails be very unlikely? The fact of the matter is that it will not be any more unlikely than any other specific sequence. Just imagine flipping 20 times with the goal of getting $HTTHTHTHTHTHTHTHTHTHT!$

0.3. SAMPLE SPACES, EVENTS, AND SOME SIMPLE PROBABILITIES 11

This would be hard to do! Similarly getting the elusive royal flush poker hand (ace, king, queen, jack and ten all in the same suit) is no more difficult than getting the hand that is exactly the 3 of clubs, 10 of diamonds, 7 of spades, 3 of hearts, and the jack of spades. Actually, it is easier since there are 4 different possible royal flushes— one for each suit!

Let's get back to the question though. If we flip a fair coin 3 times, what is $P(A)$, the probability that we will see at least 1 heads? Since the 8 outcomes in Ω are equally likely and since there are 7 outcomes where we see at least 1 heads, we have a “7 out of 8 chance” of seeing at least one heads. That is,

$$P(A) = 7/8.$$

It might also be convenient (though cumbersome) sometimes for us to write this out as

$$P(\{HHH, HHT, HTH, THH, HTT, THT, TTH\}) = 7/8.$$



Note

In general, if Ω is a set of equally likely outcomes of an experiment and A is a subset of Ω ,

$$P(A) = \frac{\# \text{ outcomes in } A}{\# \text{ outcomes in } \Omega}.$$

This is why establishing some counting techniques in the previous sections was important.

Example 0.3.1

If you randomly select 5 cards from a deck of 52 cards, what is the probability that you will get a royal flush? (ace, king, queen, jack, and 10 in the same suit)

Answer:

First of all, note that order is not important here. If you are playing poker, you might be excited to have 3 aces in your hand but you certainly wouldn't care that one of them was near your thumb and the other two were closer

to your pinky. Therefore, as we learned in Section 0.2.3, the total number of 5 card hands is given by $\binom{52}{5}$.

Since every 5 card hand is equally likely and since there are exactly 4 hands that are a royal flush (one for each of the 4 suits), we have that

$$P(\text{royal flush}) = 4 / \binom{52}{5}$$

Incidentally, after working out the factorials, this is $4/2,598,960 = 1.539077 \times 10^{-6}$! ($\leftarrow$ That is an exclamation point for excitement and not another factorial!) ($\leftarrow$ That is another exclamation point...) ■

0.4 Some Very Brief Words About Independence and “Disjointness”

Independence

In the previous section, we considered flipping a fair coin 3 times. The possible outcomes were

$$\Omega = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}.$$

Although it wasn’t explicitly stated, the implication or assumption here is that the coin is flipped in a random and fair way and that the flips are independent. That is, what happens on the second flip, for example, does not depend on what happened on the first flip. (For example, we are not flipping the coin with a spatula! $T, H, T, H, T, H, T, \dots$)

What is the probability of seeing the result HTH ? Based on the previous section, and because the outcomes are equally likely, we know by counting that the answer is $1/8$. In this section, we will tackle the problem a different way that does not require that we write out and count all the possible outcomes since this may not be a reasonable thing to do for a bigger problem.

Note that the “event” HTH means that we got heads on the first toss of the coin, tails on the second toss, and heads on the third toss. Furthermore, note that the coin doesn’t care about what toss we are on at any point so

$$P(\text{heads on any one toss}) = 1/2$$

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and

$$P(\text{tails on any one toss}) = 1/2.$$

Note that

$$P(HTH) = \frac{1}{8} = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2}.$$



Property

In general, if A and B are independent events

$$P(\underbrace{A \text{ and } B}_{A \cap B}) = P(A) \cdot P(B).$$

Applying this (generalizing to 3 events) to our problem, since the flips are independent, we have that

$$\begin{aligned} P(HTH) &= P(\text{heads on the first flip AND} \\ &\quad \text{tails on the second flip AND} \\ &\quad \text{heads on the third flip}) \\ &\stackrel{\text{indep}}{=} P(\text{heads on the first flip}) \\ &\quad \cdot P(\text{tails on the second flip}) \\ &\quad \cdot P(\text{heads on the third flip}) \\ &= \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}. \end{aligned}$$

(The “*indep*” indicates that independence was used at that point to argue that equality.)

Disjoint Events

Disjoint events are events that can not happen together. For example, in our coin flipping experiment, the event “heads on the first flip” and the event “tails on the first flip” are disjoint. you can have one happen, or the other, but not both. This is different than the relationship between the events “heads on the first flip” and “tails on the second flip” since it is possible that we see both events occur. Some beginners to probability often confuse the ideas of independence and disjointness, but they are very

different things. We can't even begin to talk about events being independent if they can't even exist in the same universe! On the other hand, disjoint events are in a sense very dependent. If you know you got heads on the first flip, then you know for sure that you did not get tails on the first flip.

Suppose that we are going to flip a fair coin 3 times and we want to know the probability that the first two flips are heads or the last two flips are tails. Given all equally likely outcomes

$$\Omega = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\},$$

it is easy to get the answer by counting outcomes of interest here. There are 4 outcomes (HHH , HHT , HTT , and TTT) where we have the desired result, so the desired probability is $4/8 = 1/2$. Note that the events “the first two flips are heads” and “the last two flips are tails” are disjoint. They can't happen at the same time if we are only flipping the coin three times because we would need that middle flip to be simultaneously heads and tails. In other words, the sets of outcomes $\{HHH, HHT\}$ and $\{HTT, TTT\}$ have no overlap. Counting the total number of events in the union

$$\{HHH, HHT, HTT, TTT\} = \{HHH, HHT\} \cup \{HTT, TTT\}$$

is equivalent to counting the events in each of the two subsets and adding the two totals. Thus, from this counting perspective,

$$\begin{aligned} P(\{HHH, HHT, HTT, TTT\}) &= P(\{HHH, HHT\}) + P(\{HTT, TTT\}) \\ &= \frac{2}{8} + \frac{2}{8} = \frac{4}{8} = \frac{1}{2}. \end{aligned}$$

“Disjointness” was key. If the two subset events were not disjoint, we would have double counted some outcomes.



Property

In general, if A and B are **disjoint** events, then

$$P(\underbrace{A \text{ or } B}_{A \cup B}) = P(A) + P(B).$$

This is true for disjoint events even when things are not equally likely and probabilities can not be determined by counting.

0.5 A Brief Review of Random Variables, PDFs, and CDFs

0.5.1 Random Variables, the Bernoulli Distribution, and a Squiggly Line

A **random variable** is a “mapping” from the set of possible outcomes of an experiment, involving some randomness or probability, to real numbers.

Example 0.5.1

Flip a, possibly unfair, coin that has some probability p of coming up “heads” and probability $1 - p$ of coming up “tails”, where p is some fixed number in the interval $[0, 1]$.

Define

$$X = \begin{cases} 1 & , \text{ if “heads”} \\ 0 & , \text{ if “tails”}. \end{cases}$$

Then X is a random variable that takes the value 1 with probability p and 0 with probability $1 - p$.

We can think of it as a mapping from a set Ω into $\mathbb{R}$ (written $X : \Omega \rightarrow \mathbb{R}$) defined as

$$X(\omega) = \begin{cases} 1 & , \text{ if } \omega = H \\ 0 & , \text{ if } \omega = T \end{cases}$$

where $\Omega = \{H, T\}$ is the set of all possible outcomes for this simple experiment. □

This “0-1 type” of random variable comes up so often that it gets a name.

X is called a **Bernoulli random variable with parameter p** . Alternatively, we say that X **has a Bernoulli distribution with parameter p** .

Our shorthand notation for this will be to write

$$X \sim \text{Bernoulli}(p).$$

Sometimes you might instead see $X \sim \text{Bern}(p)$. The symbol “ $\sim$ ” should be read as “has the distribution”.

0.5.2 PDFs for Discrete Random Variables and the Geometric Distribution

There are two very important functions associated with random variables. One is called the

probability density function or “pdf”

and the other is called the

cumulative distribution function or “cdf”.

They are usually denoted by f and F , for the pdf and cdf, respectively.

Most of the random variables we will talk about in this course will be either discrete (taking on possible values from a discrete set) or continuous (taking on possible values in a continuum). This section is about pdfs for discrete random variables. Please note that many authors use the terminology “probability density function” and the acronym “pdf” only when talking about continuous random variables and, in the discrete case, refer to the function about to be defined here as a **probability mass function (pmf)**. In these notes, we will not make this distinction.

First, we consider the discrete case.



Definition

The **probability density function (pdf)** for a discrete random variable X is the function f defined by

$$f(x) := P(X = x).$$

The right-hand side there is used to denote the probability that X takes on the specific value x . X is a random variable and x is a stand in for a specific real number that can be taken on by X .

 Note

In probability and statistics, it is customary to use capital letters to denote random variables and lower case letters to denote specific fixed values.

Example 0.5.2

Suppose that $X \sim \text{Bernoulli}(p)$. What is the pdf of X ?

Answer:

Since $P(X = 1) = p$ and $P(X = 0) = 1 - p$, by our definition of the pdf, we have that $f(1) = p$ and $f(0) = 1 - p$. Furthermore, $f(2.7)$, for example, is $f(2.7) = P(X = 2.7) = 0$, since X can only take on values in $\{0, 1\}$. In summary, the pdf is

$$f(x) = \begin{cases} 1 - p & , \quad x = 0 \\ p & , \quad x = 1 \\ 0 & , \quad \text{otherwise.} \end{cases}$$



In mathematics, we usually call the region on which a function is defined the **domain** of the function. Noticed that the above pdf is defined everywhere (for all real numbers) but is only really “interesting” at the points $x = 0$ and $x = 1$. In general, the region where a function is non-zero is called the **support** of the function.

Let’s now look at another discrete random variable with a distribution used so frequently that it gets a name.

Example 0.5.3

Consider an experiment consisting of a sequence of independent trials of something where each trial can have only two possible outcomes, usually labeled as “success” and “failure”. We will denote success and failure by S and F , respectively.

Let p be the probability of “success” on any one trial. Here, p must live in the interval $[0, 1]$.

Let

$X = \#$ trials until the first success.

Then the possible values for X are $1, 2, 3, \dots$

For example, in an idealized world, you might imagine someone shooting baskets in basketball and X as the number of tries they have to take until they make a basket. (In real life though, the assumptions that the success probability is constant for all trials and the trials are independent would probably be violated as the person could be learning from each failed attempt and making adjustments to get better or maybe their arms are tired and they are getting worse!)

Question: What is the pdf for X ?

Answer:

Note that

$$P(X = 1) = P(S \text{ on first trial}) = p.$$

Continuing,

$$P(X = 2) = P(F \text{ on 1st trial AND } S \text{ on 2nd trial}).$$

By independence of the trials, this is

$$\begin{aligned} P(X = 2) &\stackrel{\text{indep}}{=} P(F \text{ on 1st trial}) \cdot P(S \text{ on second trial}) \\ &= (1 - p) \cdot p. \end{aligned}$$

Similarly,

$$\begin{aligned} P(X = 3) &= P(F \text{ on 1st trial AND } F \text{ on 2nd trial AND } S \text{ on 3rd trial}) \\ &\stackrel{\text{indep}}{=} P(F \text{ on 1st trial}) \cdot P(F \text{ on 2nd trial}) \cdot P(S \text{ on 3rd trial}) \\ &= (1 - p) \cdot (1 - p) \cdot p = (1 - p)^2 \cdot p. \end{aligned}$$

A pattern is forming. Indeed, we have

$$P(X = x) = (1 - p)^{x-1} \cdot p$$

for $x = 1, 2, 3, \dots$

Since $P(X = x) = 0$ for x not in $\{1, 2, 3, \dots\}$, we conclude that the pdf is

$$f(x) = \begin{cases} (1-p)^{x-1} \cdot p & , \quad x = 1, 2, 3, \dots \\ 0 & , \quad \text{otherwise.} \end{cases}$$

Again we have happened across a type of random variable that is so common it gets a name. X is said to be a **geometric random variable with parameter p** . Alternatively, we say that X has a **geometric distribution**. We write

$$X \sim \text{geom}(p). \quad \square$$

Note that, using the same set up with successes and failures on independent trials, some people will define the geometric random variable as

$$X = \# \text{ failures before the first success.}$$

Now, if we see a success on the first trial, X will take the value 0. It is easy to adjust what we did above to show that the pdf is now

$$f(x) = \begin{cases} (1-p)^x \cdot p & , \quad x = 0, 1, 2, \dots \\ 0 & , \quad \text{otherwise.} \end{cases}$$

Although it is **not standard notation**, for this course we will write

$$X \sim \text{geom}_0(p)$$

to denote the random variable having the geometric distribution that starts from 0 and

$$X \sim \text{geom}_1(p)$$

to denote the random variable having the geometric distribution that starts from 1.

0.5.3 PDFs for Continuous Variables and the Exponential Distribution

Let X be a continuous random variable. Then X is assumed to take on one in a whole continuum of values. For example, suppose that we randomly select a student on campus and measure their height in inches. Let X be

this height and assume we can take this measurement with infinite accuracy to as many decimal places as we want. It seems reasonable that we could find someone on campus who is between, say 5'4" and 6'0", and so taking about the probability $P(64 < X < 70)$ makes sense. Because the range of values in the interval $[64, 66]$ is smaller but included in $[64, 70]$, we would expect that

$$P(64 < X < 66) \leq P(64 < X < 70).$$

(Indeed, in “event notation”, if A and B are events such that $A \subseteq B$, we will always have $P(A) \leq P(B)$.)

Back to heights though... As these intervals get smaller, the events become even more rare. It would in fact be very difficult to randomly select a student whose height is between 64 and 64.000001 inches. The likelihood of finding someone to be exactly $64.0\bar{0}$ is vanishingly small!

In fact, we always have the following.



Note

If X is any continuous random variable and x represents a specific number, we will always have that $P(X = x) = 0$.

In this Section, we want to define what is meant by a pdf for a continuous random variable. It can no longer be defined as $f(x) = P(X = x)$, as it was in the discrete case because this is always zero. (I guess we could define it that way but it would be terribly uninteresting!) Instead, we define the pdf for a continuous random variable as follows.



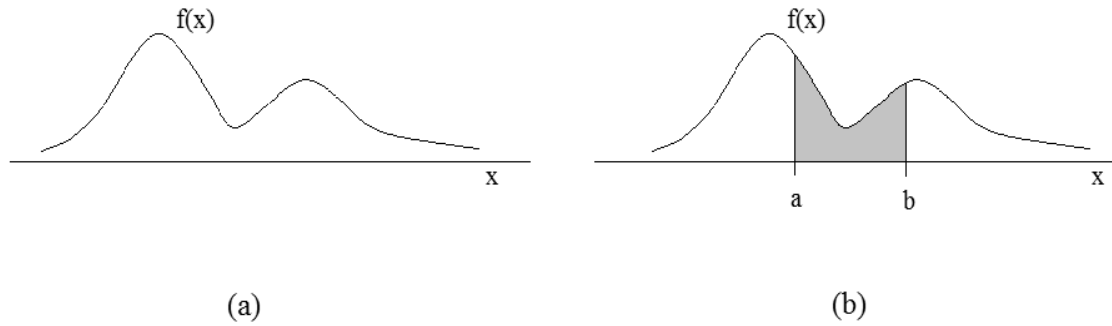
Definition

The **probability density function (pdf)** for a continuous random variable X is a non-negative function f under which area represents probability.

For example, if the continuous random variable X has the pdf depicted in Figure 1(a), then the probability $P(a < X \leq b)$ is the shaded region depicted in Figure 1(b). This is computed as the integral

$$P(a < X \leq b) = \int_a^b f(x) dx.$$

Figure 1: A pdf for a Continuous Random Variable



Since the area of the vertical lines at the boundaries of the region have area zero, we can put them in or take them out without changing the area of the shaded region. Thus,

$$P(a < X \leq b) = P(a \leq X \leq b) = P(a \leq X < b) = P(a < X < b).$$

Furthermore, for any a ,

$$P(X = a) = \int_a^a f(x) dx = 0.$$

The pdf described how the probabilities associated with a random variable are “**distributed**” over the real line. Where is most of the area under the curve “piled up”? That is, where you are most likely to observe the random variable if you record an actual “realization” (numerical observation) of it in the future? Thus, we will often talk about the pdf associated with a “distribution” instead of talking about the random variable itself.

For any pdf f , we must have

- $f(x) \geq 0$ for all x

and, for a pdf for a continuous distribution we must have

- $\int_{-\infty}^{\infty} f(x) dx = 1$

Continuous pdfs are usually defined over all real numbers but may be zero in a lot of places. For example, for the height example given at the beginning of this section, we presumably won't have anyone with a negative height. Conditions for the other end of the real line, for this example, are a bit harder to define so we won't try. As least thought, we can say that

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^0 0 dx + \int_0^{\infty} f(x) dx = \int_0^{\infty} f(x) dx = 1.$$

Throughout these notes, you will see integrals going from $-\infty$ to ∞ when a smaller range for integration may be applicable. This is a “safe bet” for us and allows us to unify many computations. It is a difference between the **domain** (place where it is defined) of a pdf and its **support** (place where it is non-zero).

For a discrete distribution, we must have

$$\sum_x f(x) = 1.$$

Here, the sum is taken over some set of numbers that includes the support for the discrete pdf. We won't write $\sum_{x=-\infty}^{\infty} f(x) = 1$ because this suggests that only integer values are allowed. We may have a discrete random variable that takes on non-integer values such as 0.0, 0.1, 0.2, ..., for example.

Example 0.5.4

Suppose you are standing near the door of a grocery store watching customers arrive.

Suppose further that

- the arrival rate is a constant 15.2 people per minute, and
- the number of arrivals in non-overlapping periods of time are independent.

(Yes, it may not be realistic to have the arrival rate be constant all day. Consider it an overly simplistic model but a model nonetheless!)

Let

X = the time (in minutes) between any two consecutive arrivals.

Then we can show (take APPM 4/5560!) that this continuous random variable X has the pdf

$$f(x) = \begin{cases} 15.2e^{-15.2x} & , \quad x \geq 0 \\ 0 & , \quad x < 0. \end{cases}$$

X is said to be an **exponential random variable with rate 15.2** or to have an **exponential distribution with rate 15.2**.

We will write

$$X \sim \exp(\text{rate} = 15.2).$$

Note that, if people are arriving at a rate of 15.2 per minute, then, on average, the **time** between arrivals is $1/15.2$ minutes. In general, if people are arriving at a rate of λ per minute, then the mean “interarrival time” is $1/\lambda$ minutes.

We have

$$X \sim \exp(\text{rate} = \lambda) \quad \Rightarrow \quad f(x) = \begin{cases} \lambda e^{-\lambda x} & , \quad x \geq 0 \\ 0 & , \quad x < 0. \end{cases}$$

However, many people specify this model in terms of the mean interarrival time and call that mean λ . This would correspond to a rate of $1/\lambda$ and so we write

$$X \sim \exp(\text{mean} = \lambda) \quad \Rightarrow \quad f(x) = \begin{cases} \frac{1}{\lambda} e^{-x/\lambda} & , \quad x \geq 0 \\ 0 & , \quad x < 0. \end{cases}$$

Unfortunately, many texts use the notation $X \sim \exp(\lambda)$ and it is up to you to look through earlier parts of the book to figure out which of these two pdfs they are talking about. In this course, we will almost always be thinking of it in terms of the rate parameter and I will always explicitly write “rate = λ ”. □

0.5.4 CDFs

The acronym **cdf** stands for **cumulative distribution function**.

**Definition**

The **cdf** of a random variable X is usually denoted by $F(x)$ and is defined as

$$F(x) := P(X \leq x).$$

This definition holds for both discrete and continuous random variables.

Example 0.5.5 (Discrete)

Suppose that $X \sim \text{geom}_0(p)$. Note that the “event” that $\{X \leq x\}$ for a discrete random variable is made up of the disjoint events of the form $\{X = u\}$ where u is any number in the support of the random variable that is less than or equal to x .

Since this geometric distribution takes on values in $\{0, 1, 2, \dots\}$, we have, for any integer x ,

$$\begin{aligned} F(x) &\stackrel{\text{def}}{=} P(X \leq x) \stackrel{\text{disj}}{=} \sum_{u \leq x} P(X = u) \stackrel{\text{geom}}{=} \sum_{u=0}^x P(X = u) \\ &= \sum_{u=0}^x (1-p)^u p = p \sum_{u=0}^x (1-p)^u \end{aligned}$$

This is a finite geometric sum. Recall first, from Calculus, the infinite geometric sum

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r} \quad \text{if } |r| < 1.$$

As a memory aid, note that the finite geometric sum, $\sum_{n=0}^N r^n$ is similar in the denominator and has the form

$$\sum_{n=0}^N r^n = \frac{?}{1-r}.$$

Writing the left side out as $1 + r + r^2 + \cdots + r^N$, we see that

$$\begin{aligned}
 ? &= (1 - r)(1 + r + r^2 + \cdots + r^N) \\
 &= (1 + r + r^2 + \cdots + r^N) - r(1 + r + r^2 + \cdots + r^N) \\
 &= (1 + r + r^2 + \cdots + r^N) - (r + r^2 + r^3 + \cdots + r^{N+1}) \\
 &= 1 - r^{N+1}.
 \end{aligned}$$

Thus,

$$\sum_{n=0}^N r^n = \frac{1 - r^{N+1}}{1 - r}.$$

Going back to the cdf for the geometric distribution, we have

$$F(x) = p \sum_{u=0}^x (1 - p)^u$$

which is a finite geometric sum with $r = 1 - p$ and $N = x$. Therefore,

$$F(x) = p \sum_{u=0}^x (1 - p)^u = p \frac{1 - (1 - p)^{x+1}}{1 - (1 - p)} = 1 - (1 - p)^{x+1}.$$

Here, we assumed that x was an integer. Because the distribution is integer valued, we have, for example that

$$F(5.6) := P(X \leq 5.6) = P(X \leq 5) = F(5) = 1 - (1 - p)^6.$$

Because X only takes on non-negative integer values, we have that

$$F(0) = P(X \leq 0) = 1 - (1 - p) = p = P(X = 0)$$

and that, for $x < 0$,

$$F(x) = P(X \leq x) = 0.$$

In summary, if $X \sim \text{geom}_0(p)$, the cdf is

$$F(x) = \begin{cases} 1 - (1 - p)^{\lfloor x \rfloor + 1} & , \quad x \geq 0 \\ 0 & , \quad x < 0 \end{cases}$$

where $\lfloor \cdot \rfloor$ is the greatest integer function. □

Example 0.5.6 (Continuous)

Suppose that $X \sim \text{exp}(\text{rate} = \lambda)$. For any $x \geq 0$,

$$\begin{aligned} F(x) &= P(X \leq x) = \int_{-\infty}^x f(u) du \\ &= \int_0^x \lambda e^{-\lambda u} du = 1 - e^{-\lambda x}. \end{aligned}$$

Notice that the limits of integration changed because the pdf is 0 for $x < 0$. Technically, though it was unwritten, we have

$$\int_{-\infty}^x f(u) du = \int_{-\infty}^0 0 du + \int_0^x \lambda e^{-\lambda u} du$$

Since the exponential distribution only takes on positive values, we have, for $x < 0$, that $F(x) = P(X \leq x) = 0$. In summary, the cdf for the exponential distribution with rate λ is

$$F(x) = \begin{cases} 1 - e^{-\lambda x} & , \quad x \geq 0 \\ 0 & , \quad x < 0. \end{cases}$$

□

PDFs from CDFs

We have seen two examples of how to get from a distribution pdf to its cdf. Suppose we want to reverse the process. That is, given a cdf, can we compute the associated pdf?

Discrete Case:

Rather than write out a formula, it is easiest to consider the cdf for a discrete random variable by example and on a case-by-case basis. Suppose, for example, that we are dealing with a distribution that takes on integer values. Then we can define the pdf at, say 3 by

$$f(3) = P(X = 3) = P(X \leq 3) - P(X \leq 2) = F(3) - F(2).$$

Continuing this example, for general integer values of x , we can define

$$f(x) = P(X \leq x) - P(X \leq x - 1) = F(x) - F(x - 1).$$

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We would also define $f(x) = 0$ whenever x is not an integer or if x is an integer that is outside the support of the distribution.

In theory, it is easy to generalize this method of recovering the pdf from a cdf for discrete random variables that do not live on integers but the notation is kind of messy. We'll just deal with it as it arises.

Continuous Case

It is much easier to move from a cdf to a pdf in the continuous case. Because the cdf is defined as

$$F(x) = \int_{-\infty}^x f(u) du,$$

the Fundamental Theorem of Calculus gives us the following.

★ Property

If X is a continuous random variable with cdf $F(x)$, the pdf is

$$f(x) = \frac{d}{dx}F(x).$$

For example, for the exponential distribution with rate λ , the cdf is $F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$. The pdf is therefore

$$f(x) = \frac{d}{dx}F(x) = \frac{d}{dx}(1 - e^{-\lambda x}) = \lambda e^{-\lambda x}$$

for $x \geq 0$. It is zero elsewhere.

0.6 The Expected Value, Expectation, or Mean of a Random Variable or a Distribution

We will denote the **expected value** (also called the **expectation** or **mean**) of a random variable X as $E[X]$.



Definition (Informal)

The **expected value** of X is a **probability weighted average**.

Example 0.6.1

Suppose that X is discrete, taking values in $\{0, 1, 2\}$ with probabilities

x	0	1	2
$P(X = x)$	0.2	0.7	0.1

Then

$$E[X] = (0)(0.2) + (1)(0.7) + (2)(0.1) = 0.9.$$

**Definition**

In general, for a discrete random variable X with pdf f , the **expected value** is

$$E[X] = \sum_x x \cdot P(X = x) = \sum_x x \cdot f(x).$$

The sum is taken over “everything”. (See the discussion about the support of a distribution in Section 0.5.3).

Example 0.6.2

Let $X \sim \text{geom}_0(p)$. Find $E[X]$.

Answer:

Since

$$f(x) = \begin{cases} (1-p)^x \cdot p & , \quad x = 0, 1, 2, \dots \\ 0 & , \quad \text{otherwise,} \end{cases}$$

we have that

$$E[X] = \sum_x x f(x) = \sum_{x=0}^{\infty} x \cdot (1-p)^x \cdot p.$$

Continuing,

$$E[X] = \sum_{x=0}^{\infty} x \cdot (1-p)^x \cdot p = p \sum_{x=0}^{\infty} x(1-p)^x = p \sum_{x=1}^{\infty} x(1-p)^x \quad (1)$$

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since the sum is zero when $x = 0$. Hmmmm... an infinite sum. The only one of those most of us can remember is the geometric sum

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r} \quad \text{if } |r| < 1$$

which can easily be adjusted to start from different places. For example

$$\sum_{n=3}^{\infty} r^n = r^3 \sum_{n=3}^{\infty} r^{n-3} = r^3 \sum_{n=0}^{\infty} r^n = r^3 \cdot \frac{1}{1-r}.$$

The sum in (1) sort of looks like a geometric sum with $r = 1 - p$. The only problem is that leading x . Check this out:

$$E[X] = p \sum_{x=1}^{\infty} x(1-p)^x = p(1-p) \sum_{x=1}^{\infty} x(1-p)^{x-1}.$$

Now the summand kind of looks like a derivative. Defining $q = 1 - p$, we may rewrite it again as

$$\begin{aligned} E[X] &= p(1-p) \sum_{x=1}^{\infty} x q^{x-1} \\ &= p(1-p) \sum_{x=1}^{\infty} \frac{d}{dq} q^x \\ &= p(1-p) \frac{d}{dq} \sum_{x=1}^{\infty} q^x \end{aligned}$$

which is a nice geometric sum. Thus

$$\begin{aligned} E[X] &= p(1-p) \frac{d}{dq} \sum_{x=1}^{\infty} q^x \\ &= p(1-p) \frac{d}{dq} \frac{q}{1-q} \\ &= p(1-p) \frac{(1-q)(1)-q(-1)}{(1-q)^2} \\ &= p(1-p) \frac{1}{(1-q)^2} = p(1-p) \frac{1}{p^2} = \frac{1-p}{p} \end{aligned}$$

In summary, if $X \sim \text{geom}_0(p)$,

$$E[X] = \frac{1-p}{p}.$$



Now for a continuous example. The sum in the expected value in the discrete case has an obvious analogue as an integral.



Definition

For a continuous random variable X with pdf f , the **expected value** is

$$E[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx.$$

Example 0.6.3

Suppose that $X \sim \text{exp}(\text{rate} = \lambda)$. Find $E[X]$.

Answer:

Since the exponential distribution is a continuous distribution we must use the integral definition for this expected value.

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x \cdot f(x) dx \\ &= \int_{-\infty}^0 x \cdot 0 dx + \int_0^{\infty} x \cdot \lambda e^{-\lambda x} dx \\ &= \int_{-\infty}^{\infty} x \cdot \lambda e^{-\lambda x} dx \end{aligned}$$

Using integration by parts ($\int u dv = uv - \int v du$) with $u = x$ and $dv = \lambda e^{-\lambda x}$, we have

$$E[X] = -xe^{-\lambda x} \Big|_0^{\infty} - \int_0^{\infty} (-e^{-\lambda x}) dx.$$

That first term certainly evaluates to 0 when we plug in $x = 0$. It is also 0 at ∞ (technically a limit as $x \rightarrow \infty$). This is because $e^{-\lambda x}$ goes down to zero (it is a requirement of the exponential distribution that $\lambda > 0$) faster than x blows up, but formally this can be shown using L'Hôpital's Rule.

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We are left with

$$\mathbb{E}[X] = \int_0^\infty e^{-\lambda x} dx = -\frac{1}{\lambda} e^{-\lambda x} \Big|_0^\infty = -\frac{1}{\lambda}(0 - 1) = \frac{1}{\lambda}.$$

Remember, the expected value of a random variable (or, equivalently, of a distribution) is also known as its expectation or mean. The $1/\lambda$ that we found here is consistent with our discussion of rates and means at the end of Section 0.5.3. □

Notation

The **mean** or **expected value** of a random variable X is often denoted with the Greek letter μ . that is,

$$\mu := \mathbb{E}[X].$$

In the case of multiple random variables, say X and Y , a subscript becomes necessary to distinguish the means,

$$\mu_X = \mathbb{E}[X] \quad \text{and} \quad \mu_Y = \mathbb{E}[Y].$$

While people would surely understand what you mean if you instead use lowercase subscripts as in μ_x and μ_y , this represents a really fundamental lack of understanding about what you are doing since lowercase Roman letters in statistics almost always refer to specific numbers and not random variables which can be thought of as “potential but yet unobserved numbers”. For the record, if c is a constant, then $\mathbb{E}[c] = c$. You can think of c as an uninteresting random variable X where $X = c$ with probability 1. Then $\mathbb{E}[c] = \mathbb{E}[X] = c \cdot P(X = c) = c \cdot 1 = c$. For example,

$$\mathbb{E}[3] = 3$$

and, adhering to the convention of random variables being uppercase,

$$\mathbb{E}[x] = x.$$

While we’re at it here, a mean μ is also a constant. When you compute any expectation you have already sort of “summed out” the probability. Thus,

while we may have defined $\mu = E[X]$ for some random variable X , $E[\mu]$ will just be μ again!

0.7 The Variance of a Random Variable or a Distribution

The **variance** of a random variable X is a measure of spread about its mean. We will denote the **variance** of a random variable X as $Var[X]$.



Definition

If we use μ to denote the mean $E[X]$, then the **variance** of X is defined as

$$Var[X] := E[(X - \mu)^2].$$



Notation

A variance is usually also denoted by the symbol σ^2 , with subscripting possible if it becomes necessary to distinguish between the variances for different random variables.

If you were devise your own measure of spread of a random variable X about its mean μ , you might use the expected distance

$$E[|X - \mu|],$$

but we will see in this course that the definition we are using instead is way more convenient from a mathematical perspective.

You may already be familiar with the expression

$$\sigma^2 = Var[X] = E[X^2] - (E[X])^2.$$

We will see in Section 0.10 that the expectation “operator” is in fact a linear operator so that, for random variables X and Y and constants a and b , we have

$$E[aX + bY] = aE[X] + bE[Y].$$

In particular,

$$E[X + Y] = E[X] + E[Y]$$

and, taking $b = -1$,

$$E[X - Y] = E[X] - E[Y].$$

With this in mind, we have that

$$\begin{aligned} \text{Var}[X] &= E[(X - \mu)^2] \\ &= E[X^2 - 2\mu X + \mu^2] \\ &= E[X^2] - 2\mu \underbrace{E[X]}_{\mu} + \mu^2 \quad (\mu \text{ is a constant so } E[\mu^2] = \mu^2) \\ &= E[X^2] - 2\mu^2 + \mu^2 \\ &= E[X^2] - \mu^2 \\ &= E[X^2] - (E[X])^2. \end{aligned}$$

Sometimes it will be convenient to give a measure of spread of random variable (or equivalently, its distribution) in the same units as the original random variable. For example, if X represents the height in inches of a randomly selected student on campus, $\text{Var}[X]$ will be in squared inches. Taking the square root will bring us back to the original units and will sometimes be helpful.



Definition

The **standard deviation** of a random variable X with variance σ^2 is denoted by σ and is defined as

$$\sigma = \sqrt{\sigma^2} = \sqrt{\text{Var}[X]}$$

Note that

$$\sqrt{\text{Var}[X]} = \sqrt{E[(X - \mu)^2]} \neq E[\sqrt{(X - \mu)^2}] = E[|X - \mu|]$$

because an expectation is a sum or integral and sums and integrals just don't work that way with square roots! **We do not have** $E[\sqrt{X}] = \sqrt{E[X]}$ just as **we do not have** $\sqrt{2+3} = \sqrt{2} + \sqrt{3}$!

We will see what the actual relationship between $\sqrt{E[(X - \mu)^2]}$ and $E[\sqrt{(X - \mu)^2}]$ in the exercises for Chapter 2.

0.8 Indicator Notation and a Most Useful Appendix

We have three “named” distributions so far: the Bernoulli, the geometric, and the exponential distributions. The most common named distributions are summarized in Tables A.1 and A.2 of Appendix A. We will eventually talk about all of the columns in these tables. As of right now, the first five columns should make sense. They include names of distributions, corresponding pdfs, a description of the allowable parameter space (For example, p for the Bernoulli and geometric distributions is itself a probability and can range from 0 to 1.), the mean (expected value), and the variance for each distribution. Each pdf in the tables includes an extra part involving an “I”. This is an indicator function which is defined as follows.



Definition

Let A be a set. The function

$$I_A(x) = \begin{cases} 1 & , \text{ if } x \in A \\ 0 & , \text{ if } x \notin A \end{cases}$$

is called an **indicator function**.

Other common notations for this indicator function are $\mathbb{1}_A(x)$, $\chi_A(x)$, and the “Iverson bracket” $[x \in A]$.

With the indicator function, we no longer need to write our pdfs piecewise. For example, if $X \sim \text{geom}_0(p)$, the pdf can now be written as

$$f(x) = (1 - p)^x \cdot p \cdot I_{\{0,1,2,\dots\}}(x).$$

So, for example

$$f(2) = (1 - p)^2 \cdot p \cdot I_{\{0,1,2,\dots\}}(2) = (1 - p)^2 \cdot p \cdot 1 = (1 - p)^2 \cdot p$$

while

$$f(2.5) = (1 - p)^{2.5} \cdot p \cdot I_{\{0,1,2,\dots\}}(2.5) = (1 - p)^{2.5} \cdot p \cdot 0 = 0.$$

In mathematical statistics, indicator functions can be very helpful! I hope to convince you of this soon. After all, writing down pdfs in the piecewise way, where we say “and zero otherwise” is not very hard to do. Furthermore, we could easily get away with just writing down the “important piece” of the piecewise defined pdf and just leave it implied that it is “zero otherwise”. Indeed, there are other reasons for using indicator functions— the first of which we will see shortly. In the meantime, let’s just start getting used to them now. Resistance is futile, or, at the very least, unadvised!

0.9 Joint PDFs, Marginals, and Independence

0.9.1 The Discrete Case



Definition

For discrete random variables X and Y , we define the **joint pdf** as

$$f(x, y) := P(X = x \text{ and } Y = y) \stackrel{\text{notation}}{=} P(X = x, Y = y).$$

Note that the x and y are just place holding dummy variables. For example,

$$P(X = 1, Y = 3) = f(1, 3)$$

and

$$P(X = w, Y = z) = f(w, z).$$

So, if it is necessary to be more specific about which random variables the joint pdf belongs to, we might write it as $f_{X,Y}(x, y)$. Then, it is clear that

$$f_{X,Y}(w, z) = P(X = w, Y = z)$$

and it is even possible to talk about the probability $P(X = y, Y = x)$!

Joint pdfs for discrete random variables are often given in a tabular form.

Example 0.9.1

Suppose that X takes on values in $\{-1, 0, 1\}$ and Y takes on values in $\{-5, 3\}$, the joint probabilities might be written as

		x		
		-1	0	1
y	-5	0.1	0.3	0.2
	3	0.2	0.1	0.1

Reading from the Table, we have, for example,

$$f(0, -5) = P(X = 0, Y = -5) = 0.3.$$

Question: What is $P(X = 0)$?

Answer:

$$P(X = 0) = P(X = 0, Y = -5 \text{ or } X = 0, Y = 3).$$

Since the events $\{X = 0, Y = -5\}$ and $\{X = 0, Y = 3\}$ are disjoint events, we get

$$P(X = 0) = P(X = 0, Y = -5) + P(X = 0, Y = 3) = 0.3 + 0.1 = 0.4$$

Such probabilities for X or Y alone are often tallied up in the margins of the table as

		x			
		-1	0	1	
y	-5	0.1	0.3	0.2	0.6
	3	0.2	0.1	0.1	0.4
		0.3	0.4	0.3	

For this reason, the individual X and Y pdfs are often referred to as the **marginal pdfs** for X and Y ! Denote them as $f_X(x)$ and $f_Y(y)$, respectively. They may be written alone as

		x		
		-1	0	1
$\underbrace{P(X = x)}_{f_X(x)}$		0.3	0.4	0.3

and

	y	
	-5	3
$\underbrace{P(Y = y)}_{f_Y(y)}$	0.6	0.4



0.9.2 The Continuous Case



Definition

For continuous random variables X and Y , the **joint pdf** is a function $f(x, y)$ that defines a surface under which area represents probability.

Analogous to the discrete case where the marginal pdfs for X and Y were arrived at by summing out the values of Y and X , respectively, we have the following in the continuous case.



Definition

If continuous random variables X and Y have joint pdf $f(x, y)$, the **marginal pdfs** for X and Y , denoted by $f_X(x)$ and $f_Y(y)$, respectively are

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

and

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx.$$

As per our discussion about the support of a pdf, the actual region of integration may be much smaller than $(-\infty, \infty)$, and may be dependent on x in the case of $f_X(x)$ and on y in the case of $f_Y(y)$. Specific examples are given in the next Section.

0.9.3 Independence

Mirroring the independence discussed in Section 0.4, random variables X and Y are independent if their joint pdf separates out nicely into an “ x -part” and a “ y -part”. Specifically, it needs to separate out into the product of the marginal pdfs.



Definition

Random variables X and Y are said to be **independent** if

$$f(x, y) = f_X(x) \cdot f_Y(y).$$

Example 0.9.2

Suppose that X and Y are continuous random variables with joint pdf

$$f(x, y) = \begin{cases} xy & , \quad 0 < x < 1, 0 < y < 2 \\ 0 & , \quad \text{otherwise} \end{cases}$$

Alternatively, we could write this with indicators as

$$f(x, y) = xy I_{(0,1)}(x) I_{(0,2)}(y).$$

Since this is a pdf, the total volume under this surface should be 1:

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy &= \int_0^2 \int_0^1 xy dx dy \\ &= \int_0^2 y \underbrace{\int_0^1 x dx}_{1/2} dy \\ &= \frac{1}{2} \int_0^2 y dy = \frac{1}{2} \cdot 2 = 1 \quad \checkmark \end{aligned}$$

Finding probabilities here comes down to finding limits of integration. For example:

- $P(X < 3/4, Y > 1/2) = \int_0^{3/4} \int_{1/2}^2 f(x, y) dy dx = \dots$
- $P(1/2 \leq X \leq 3, 1/4 \leq Y < 1) = \int_{1/2}^3 \int_{1/4}^1 f(x, y) dy dx$
 $= \int_{1/2}^1 \int_{1/4}^1 xy dy dx = \dots$

(Note the limit of integration change to the “relevant part” of the x interval. The pdf is 0 for $x = 1$ to $x = 3$.)

- $P(X \leq Y) = \int_0^1 \int_x^2 xy dy dx = \int_0^2 \int_0^y xy dx dy = \dots$

(If you are having trouble with these limits of integration, draw the rectangle where $0 \leq x \leq 1$ and $0 \leq y \leq 2$ and then shade in the subregion where $x \leq y$. You’ll see that, for any fixed x , y can range from that point on up to its maximum of 2. On the other hand, for any fixed y , x can range from its minimum of 0 up to that y .)

The marginal pdf for X is

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_{-\infty}^0 0 dy + \int_0^2 xy dy + \int_2^{\infty} 0 dy$$

You obviously don’t need to write out those integrals of 0. I was just making the point that you should, in general, integrate over all values of y from $-\infty$ to ∞ . So,

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^2 xy dy = 2x.$$

We are not finished until we write the support (the x values where it is non-zero) of $f_X(x)$:

$$f_X(x) = 2x, \quad 0 < x < 1.$$

The pdf is zero otherwise. We could include an indicator and write

$$f_X(x) = 2x I_{(0,1)}(x),$$

or we could have taken care of this automatically using indicators from the beginning:

$$\begin{aligned}
 f_X(x) &= \int_{-\infty}^{\infty} f(x, y) dy = \int_{-\infty}^{\infty} xy I_{(0,1)}(x) I_{(0,2)}(y) dy \\
 &= x I_{(0,1)}(x) \int_{-\infty}^{\infty} y I_{(0,2)}(y) dy \\
 &= x I_{(0,1)}(x) \underbrace{\int_0^2 y \cdot 1 dy}_2 \\
 &= 2x I_{(0,1)}(x)
 \end{aligned}$$

Wow! So precise and neat! Get on board with indicators now!

The marginal pdf for Y is

$$\begin{aligned}
 f_Y(y) &= \int_{-\infty}^{\infty} f(x, y) dx = \int_{-\infty}^{\infty} xy I_{(0,1)}(x) I_{(0,2)}(y) dx \\
 &= y I_{(0,2)}(y) \int_{-\infty}^{\infty} x I_{(0,1)}(x) dx \\
 &= y I_{(0,2)}(y) \underbrace{\int_0^1 x \cdot 1 dx}_{1/2} \\
 &= \frac{1}{2} y I_{(0,2)}(y).
 \end{aligned}$$

Since $f(x, y) = f_X(x) \cdot f_Y(y)$, we have that X and Y are independent. □

 Note

If we just wanted to show independence, is it really necessary to go through the work of finding the marginal pdfs? Can't we just say that the pdf $f(x, y) = xy$ factors into an “ x -part” and a “ y -part even if we don't know quite how the constants will sort out to make marginal pdfs?

The answer is **yes, if we use indicators!!!** The proof will be an exercise at the end of this Chapter.

Example 0.9.3 (Continuous)

For the last example, we have

$$f(x, y) = xy I_{(0,1)}(x) I_{(0,2)}(y) = \underbrace{(x I_{(0,1)}(x))}_{x\text{-part}} \cdot \underbrace{(y I_{(0,2)}(y))}_{y\text{-part}}$$

$\Rightarrow$ X and Y are independent

even though neither one of the “parts” are valid pdfs. □

Next up is an example of a pdf that seems to factor into an “ x -part” and a “ y -part but where X and Y are not independent. The example shows why we must include the indicators if we want to use this “lazy” (no work to find marginals) way to show independence.

Example 0.9.4

Suppose that X and Y are continuous random variables with joint pdf

$$f(x, y) = \begin{cases} 8xy & , \quad 0 < x < y < 1 \\ 0 & , \quad \text{otherwise} \end{cases}$$

For practice in finding regions of integration, let's check that this pdf integrates to 1.

$$\begin{aligned}
 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dy dx &= \int_0^1 \int_0^x 8xy dy dx \\
 &= \int_0^1 8x \underbrace{\int_0^x y dy}_{x^2/2} dx \\
 &= \int_0^1 4x^3 dx = 1 \quad \checkmark
 \end{aligned}$$

(To get these limits of integration, sketch the region where $0 < x < y < 1$. It is triangular. Take any x between 0 and 1. For each such fixed x , y then ranges from x to 1. If you want to do the integral in the opposite order, take any y between 0 and 1. For each fixed y , you'll see that x ranges from 0 to y . So, the double integral can also be written as $\int_0^1 \int_y^1 8xy dx dy$.)

To find the marginal pdf, $f_X(x)$, for X , think of x as any fixed number between 0 and 1. For any fixed x , y ranges from x to 1. Thus, we get

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_x^1 8xy dy = 8x \int_x^1 y dy = 4x(1 - x^2).$$

This holds for $0 < x < 1$. The pdf is 0 otherwise. We could just tack on an indicator in order to completely describe $f_X(x)$ or we could have used indicators from the beginning in the joint pdf.

Note that the joint pdf could be written with indicators in two ways:

$$f(x, y) = 8xy I_{(0,1)}(y) I_{(0,y)}(x) = 8xy I_{(0,1)}(x) I_{(x,1)}(y).$$

To find the marginal pdf for X , we should use the second representation since it has an indicator that is purely in terms of x which can be easily pulled out of an integral with respect to y :

$$\begin{aligned}
 f_X(x) &= \int_{-\infty}^{\infty} f(x, y) dy = \int_{-\infty}^{\infty} 8xy I_{(0,1)}(x) I_{(x,1)}(y) dy \\
 &= 8x I_{(0,1)}(x) \int_{-\infty}^{\infty} y I_{(x,1)}(y) dy \\
 &= 8x I_{(0,1)}(x) \int_x^1 y \cdot 1 dy = 4x(1 - x^2) I_{(0,1)}(x).
 \end{aligned}$$

The marginal pdf for Y is

$$\begin{aligned} f_Y(y) &= \int_{-\infty}^{\infty} f(x, y) dx = \int_{-\infty}^{\infty} 8xy I_{(0,1)}(y) I_{(0,y)}(x) dx \\ &= 8y I_{(0,1)}(y) \int_{-\infty}^{\infty} x I_{(0,y)}(x) dx \\ &= 8y I_{(0,1)}(y) \int_0^y x \cdot 1 dy = 4y^3 I_{(0,1)}(y). \end{aligned}$$

Clearly, we do not have that $f(x, y) = f_X(x) \cdot f_Y(y)$. Thus, X and Y are **not independent**. We could have seen this without first finding the marginal pdfs since **the indicators in the joint pdf can not be separated into an “ x -part” and a “ y -part”**. ■

While the definition of independence says that X and Y are independent if the joint pdf can be factored into a product of the marginal pdfs, I tried to claim here that maybe we had independence if the joint pdf just factors into an x -part and a y -part and that we can leave it up to someone else to find the marginals. This claim failed for the joint pdf $f(x, y) = 8xy$ because the x and y were still tied together in the condition that $0 < x < y < 1$. However, if we included this constraint in the joint pdf with indicators, we actually can show independence. We have the following claim that is formalized into a Theorem in the Exercises at the end of this Chapter.

★ **Claim**

Random variables X and Y are **independent** if

$$f(x, y) = g(x) \cdot h(y).$$

for some functions g and h , as long as we include indicators in the joint pdf!

Note that, in the previous example, the indicator part of the joint pdf was written in two ways,

$$I_{(0,1)}(y) I_{(0,y)}(x) \quad \text{and} \quad I_{(0,1)}(x) I_{(x,1)}(y),$$

but neither representation completely factors into an x -part and a y -part.

0.9.4 “Twisted” Indicators

Such “twisted indicators”, as in the example at the end of the previous section, where the x and y parts are all mixed together, can sometimes be sorted out and separated. To see whether or not this is the case, it is often helpful by sketching the region where the product of the indicators is equal to 1.

Consider for example,

$$I_{(0,\infty)}(xy) \cdot I_{(0,\infty)}(y - xy).$$

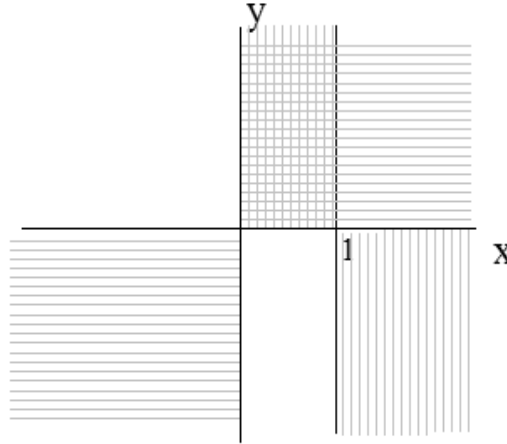
This product of indicators takes the value of 1 whenever both

$$0 < xy < \infty \quad \text{and} \quad 0 < y - xy < \infty,$$

otherwise the product will be 0.

It is helpful to rewrite that second inequality as $0 < y(1 - x) < \infty$. Then we see that it will hold when $y > 0$ and $x < 1$ OR when $y < 0$ and $x > 1$. Figure 2 shows places where $I_{(0,\infty)}(xy)$ equals 1 using horizontal lines and places where $I_{(0,\infty)}(y - xy)$ is 1 using vertical lines.

Figure 2: Sorting Out a Product of Indicators



Both indicators (and hence their product) is 1 on the region where $0 < x < 1$ and $0 < y < \infty$. Thus, we have that

$$I_{(0,\infty)}(xy) \cdot I_{(0,\infty)}(y - xy) = I_{(0,1)}(x) \cdot I_{(0,\infty)}(y).$$

Note that the indicators then factor into an “ x -part” and a “ y -part”. This would help us to establish independence of random variables X and Y , assuming the rest of their joint pdf factors as well.

You will be able to split up indicators like this whenever the region they describe is rectangular. Take a moment to sketch the region that the random variables X and Y live on in Example 0.9.4. It is triangular, and so no amount of rewriting is going to allow us to completely separate the indicators into an x part and a y part!

0.10 Useful Properties of Expectation, Variance, and Covariance

In this section we define what is meant by “covariance” and establish several properties of the expectation and variance operators. All proofs will be given for continuous random variables. They can easily be rewritten for the discrete cases as well.

1. Suppose that X is a random variable with pdf f . Let g be any function. Then

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx. \quad (2)$$

Wait what?!? Does that make sense to you? Shouldn’t we think of $g(X)$ as a new random variable and use the pdf for this new random variable when finding the expectation? For example, maybe we should define the random variable $Y = g(X)$, figure out the pdf, f_Y , for Y and compute

$$\mathbb{E}[g(X)] = \mathbb{E}[Y] = \int_{-\infty}^{\infty} y f_Y(y) dy. \quad (3)$$

Yes, as a matter of fact, we should. However, it turns out that the integrals in (2) and (3) will give us the same thing! So, we’ll generally use (2) since it does not involve the extra work of first finding f_Y . We will prove the equivalence of (2) and (3) in Section 1.6.1 after we have some more machinery built up.

2. Expectation is a linear operator.

Let X, Y be random variables and let a, b be constants. Then

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

To prove this, note that, since the left-hand side is a function of two random variables, it needs to be computed by multiplying against the joint pdf for both random variables. This is a generalization of (2) for the case of a vector-valued random variable $Z = (X, Y)$ and the function $g(Z) = xA + bY$.

We get

$$\begin{aligned}
 \mathbb{E}[aX + bY] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (ax + by) f(x, y) dx dy \\
 &= a \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y) dx dy + b \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x, y) dx dy \\
 &= a \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y) dy dx + b \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x, y) dx dy \\
 &= a \int_{-\infty}^{\infty} x \int_{-\infty}^{\infty} f(x, y) dy dx + b \int_{-\infty}^{\infty} y \int_{-\infty}^{\infty} f(x, y) dx dy \\
 &= a \int_{-\infty}^{\infty} x f_X(x) dx + b \int_{-\infty}^{\infty} y f_Y(y) dy \\
 &= a\mathbb{E}[X] + b\mathbb{E}[Y]
 \end{aligned}$$

★ **Property**

In general, for random variables $X_1, X_2, \dots, X_n$ and constants $a_1, a_2, \dots, a_n$, we have

$$\mathbb{E}\left[\sum_{i=1}^n a_i X_i\right] = \sum_{i=1}^n a_i \mathbb{E}[X_i].$$

3. Expectations of products factor if things are independent.

★ **Property**

If X and Y are independent random variables, then

$$\mathbb{E}[XY] = \mathbb{E}[X] \cdot \mathbb{E}[Y].$$

Proof:

$$\begin{aligned}
 E[XY] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f(x, y) dx dy \\
 &\stackrel{\text{indep}}{=} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_X(x) f_Y(y) dx dy \\
 &= \int_{-\infty}^{\infty} y f_Y(y) \int_{-\infty}^{\infty} x f_X(x) dx dy \\
 &= \int_{-\infty}^{\infty} y f_Y(y) E[X] dy \\
 &= E[X] \int_{-\infty}^{\infty} y f_Y(y) dy \\
 &= E[X] \cdot E[Y]
 \end{aligned}$$

However, **it is important to note** that the reverse is not necessarily true— the expectation factoring like that does not necessarily imply that X and Y are independent. For example, let X take on values $-1, 0$, and 1 , with respective probabilities $1/4, 1/2$, and $1/4$, and let $Y = X^2$. Then, it is easy to see that $E[X] = 0$, and, we also have

$$E[XY] = E[X^3] = (-1)^3(1/4) + (0)^3(1/2) + (1)^3(1/4) = 0.$$

Thus, $E[XY] = E[X] \cdot E[Y]$, but X and Y are most certainly not independent!

It is easy to generalize the above proof to the following.



Property

In general, if X and Y are independent, then for functions g and h ,

$$E[g(X) \cdot h(Y)] = E[g(X)] \cdot E[h(Y)].$$

4. The variance of a random variable X is a measure of spread about its mean. Recall that we use μ to denote the mean $E[X]$ and σ^2 to denote $Var[X]$, and that the variance of X is defined by

$$Var[X] = E[(X - \mu)^2].$$

A useful alternate way to compute a variance is given by the following.



Alternate Definition of Variance

$$\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Proof:

$$\begin{aligned} \text{Var}[X] &= \mathbb{E}[(X - \mu)^2] \\ &= \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - 2\mu \underbrace{\mathbb{E}[X]}_{\mu} + \mu^2 \quad (\mu \text{ is a constant}) \\ &= \mathbb{E}[X^2] - 2\mu^2 + \mu^2 \\ &= \mathbb{E}[X^2] - \mu^2 \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \end{aligned}$$

From either the original definition of variance or from this alternate and usually more computationally convenient formula, it is easy to see that



Property

$$\text{Var}[aX] = a^2 \text{Var}[X]$$

for any constant $a \in \mathbb{R}$. (Convince yourself!)

Let us consider now $\text{Var}[X + Y]$ for two random variables X and Y .

In general, $\text{Var}[X + Y] \neq \text{Var}[X] + \text{Var}[Y]$.

As you might have guessed, the “square” in the definition of variance just makes things “ooky” here. In order to write down an expression for $\text{Var}[X + Y]$, we first need to discuss the concept of covariance.

5. The **covariance** between random variables X and Y is defined as follows.



Definition

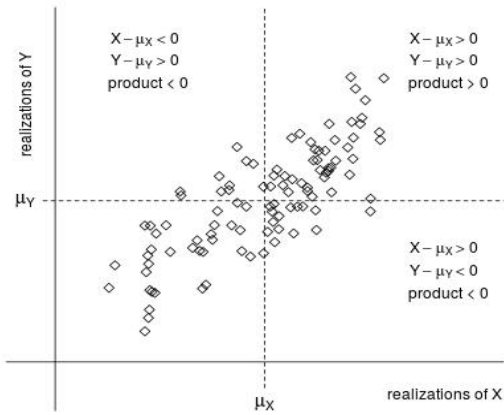
$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$$

where $\mu_X = \mathbb{E}[X]$ and $\mu_Y = \mathbb{E}[Y]$.

Covariance is a measure of the strength of a linear relationship between two random variables. For example, suppose we have random variables X and Y with some realizations depicted as in Figure 3 along with dashed lines representing their means. Note that these means, μ_X and μ_Y , are true expected values based on some knowledge of the distribution of X and Y . These are different from “sample means”, denoted by $\bar{X}$ and $\bar{Y}$, which are computed by averaging the finite number of data values. (Now if you didn’t know μ_X and μ_Y , one’s natural inclination might be to look at some data and estimate them with $\bar{X}$ and $\bar{Y}$, respectively. Assessing whether or not this is a good idea will be a central theme of this text!)

The plot in Figure 3 is separated into four quadrants based on the locations of the distribution means. In this particular case, the depicted realizations of the random (X, Y) pairs are such that the majority of points fall in the quadrants where both X and Y are greater than their means or both are less. In other words, most observations fall where both $X - \mu_X$ and $Y - \mu_Y$ are positive or both are negative. In even more words, most fall in places where the product $(X - \mu_X)(Y - \mu_Y)$ is positive. The covariance between X and Y is defined as the expected value of this product. Recall that this is a probability weighted average of this quantity. If we just averaged the numerical values observed for that product, we would likely produce a positive number (called the

Figure 3: An Apparently Positive Covariance



sample covariance). So, it is likely the true distribution covariance is also positive.

The bottom line is again that covariance is a measure of the strength of a linear relationship between two random variables. If X and Y have a positive covariance, we can roughly say that as one increases the other will tend to increase as well. (Think height and shoe size!) If X and Y have a negative covariance, we can roughly say that as one increases the other will tend to decrease. (Think oil production and gas prices!) A covariance close to zero may appear as a circular cloud of points where it doesn't appear that X and Y are even related as well as a strong distinct relationship where upward and downward trends are able to cancel each other out!

Note that

$$\begin{aligned}
 \text{Cov}(X, Y) &= \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] \\
 &= \mathbb{E}[XY - \mu_Y X - \mu_X Y + \mu_X \mu_Y] \\
 &= \mathbb{E}[XY] - \mu_Y \mathbb{E}[X] - \mu_X \mathbb{E}[Y] + \mu_X \mu_Y \\
 &= \mathbb{E}[XY] - \mu_Y \mu_X - \mu_X \mu_Y + \mu_X \mu_Y \\
 &= \mathbb{E}[XY] - \mu_X \mu_Y \\
 &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]
 \end{aligned}$$

Here are some properties of covariance.

A) Since $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$ we see that, if X and Y are independent, $\text{Cov}(X, Y) = 0$. Again, the reverse is not true. (See the example above in number 2 on this list.)

B) It is routine to show/see that

$$\begin{aligned}
 \text{Cov}(aX, Y) &= a\text{Cov}(X, Y) \\
 \text{Cov}(X + Y, Z) &= \text{Cov}(X, Z) + \text{Cov}(Y, Z) \\
 \text{Cov}(X, Y) &= \text{Cov}(Y, X).
 \end{aligned}$$

That last part allows us to easily conclude that we also have $\text{Cov}(X, bY) = b\text{Cov}(X, Y)$ and $\text{Cov}(X, Y + Z) = \text{Cov}(X, Y) + \text{Cov}(X, Z)$.

C) It is also easy to see, by inspection, that

$$\text{Var}[X] = \text{Cov}(X, X)$$

6. Now that we have defined covariance, we are ready to deal with the variance of a sum.

★ Property

$$\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y] + 2\text{Cov}(X, Y)$$

To see this, write

$$\begin{aligned}
 \text{Var}[X + Y] &= \text{Cov}(X + Y, X + Y) \\
 &= \text{Cov}(X, X) + \text{Cov}(X, Y) + \text{Cov}(Y, X) + \text{Cov}(Y, Y) \\
 &= \text{Cov}(X, X) + 2\text{Cov}(X, Y) + \text{Cov}(Y, Y) \\
 &= \text{Var}[X] + 2\text{Cov}(X, Y) + \text{Var}[Y]
 \end{aligned}$$

Plugging in $-Y$ for Y and noting that $\text{Var}[-Y] = \text{Var}[(-1)Y] = (-1)^2\text{Var}[Y] = \text{Var}[Y]$, we see that

$$\text{Var}[X - Y] = \text{Var}[X] + \text{Var}[Y] - 2\text{Cov}(X, Y).$$

Note that, for independent X and Y , $\text{Cov}(X, Y) = 0$ implies that

$$\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y]$$

and

$$\text{Var}[X - Y] = \text{Var}[X] + \text{Var}[Y].$$

Chapter 0 Exercises

1. Suppose that X and Y are discrete random variables with joint pdf

$$f(x, y) = c \frac{2^{x+y}}{x!y!} I_{\{0,1,2,\dots\}}(x) I_{\{0,1,2,\dots\}}(y).$$

- (a) Find the constant c .
- (b) Find the marginal pdfs of X and Y .
- (c) Are X and Y independent? Explain.

2. Let X and Y have the joint pdf

$$f(x, y) = ce^{-x-y} I_{(0,y)}(x) I_{(0,\infty)}(y)$$

for some constant c .

- (a) Find the value of c that makes this a valid pdf.
 - (b) Find the marginal pdfs for X and Y .
 - (c) Are X and Y independent? Explain.
 - (d) Find the expected value of X .
3. Let $f(x) = ce^x I_{[0,1]}(x) + ce^{-x+2} I_{(1,\infty)}$ where c is some constant.
- (a) Find the constant c that will make f a probability density function.
 - (b) Find the corresponding cumulative distribution function F .
4. A regular tetrahedron (4 sides) with sides numbered one through four is tossed twice. Let X be the larger of the two “down faces”. (The tetrahedron shape does not allow a side to be face up. Just think of this as a four sided die and let X be the larger value obtained in two tosses.)
- (a) Find the pdf of X .
 - (b) Find the expected value of X .
5. Suppose that $X \sim \text{Bernoulli}(p)$. Verify the following quantities that can be found in the table of distributions from Appendix A.
- (a) Find the expected value of X .
 - (b) Find the variance of X .
6. Suppose that $X \sim \exp(\text{rate} = \lambda)$. Verify the following quantities that can be found in the table of distributions from Appendix A.
- (a) Find the expected value of X .
 - (b) Find the variance of X .
7. Suppose that X and Y are independent random variables with $X \sim \text{Bernoulli}(p)$ and $Y \sim \text{Poisson}(\lambda)$. (We have not talked about the Poisson distribution in class or these notes but you can find the necessary pdf in the table of distributions in Appendix A!)
- Find the joint probability $P(X = 1, Y = 3)$.
8. Suppose that X is a continuous random variable with pdf $f(x) = 3x^2 I_{(0,1)}(x)$.

- (a) Find, using the table of distributions from Appendix A, the expected value of X . That is, first identify the distribution by simply matching this pdf with one in the table and then pulling the expected value (mean) from the table without any actual computation.
 - (b) Consider a random rectangle whose sides are of length X and $(1-X)$. Determine the expected value of the area of the rectangle. (You may also quote values from the table of distributions here.)
9. In this chapter, we had a brief discussion about the covariance between random variables X and Y and, in particular, the relationship between X and Y when they have a positive versus negative covariance. The magnitude of a covariance, however, is difficult to interpret on its own and may only be interesting when compared to another covariance. Instead, one may wish to consider a “standardized” version of covariance that is bounded between -1 and 1 .

The (linear) **correlation** between two random variables X and Y is often denoted as $\rho_{X,Y}$ and is defined as

$$\rho_{X,Y} := \frac{\sigma_{X,Y}}{\sigma_X \sigma_Y}$$

where $\sigma_{X,Y} := \text{Cov}(X, Y)$.

- (a) Show that $-1 \leq \rho_{X,Y} \leq 1$.
 - (b) What can one say about the relationship between X and Y if their correlation is a perfect plus or minus 1?
10. Suppose that X and Y are continuous random variables with a joint pdf of the form

$$f(x, y) = [g(x) I_G(x)] \cdot [h(y) I_H(y)]$$

for some functions g and h and some sets of real numbers G and H . As the notation suggests, the sets are constant in that, for example, we do not have $G = G(y)$.

Show that, even though g and h are not necessarily pdfs, X and Y must be independent.

Chapter 1

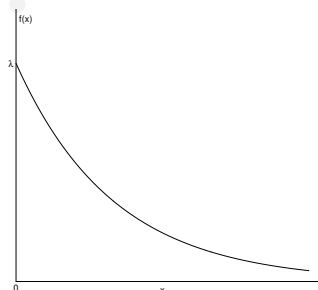
MathStat Preliminaries: Four Important Tools for Mathematical Statistics

1.1 Wait. Where are we going?

Recall the exponential distribution introduced in Section 0.5.3. If $X \sim \text{exp}(\text{rate} = \lambda)$, then X has the pdf

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & , \quad x \geq 0 \\ 0 & , \quad x < 0 \end{cases}$$

which looks like this



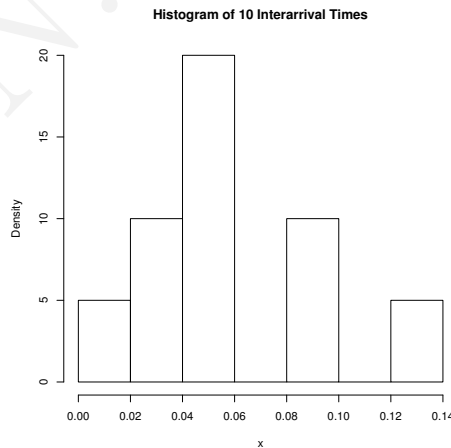
Recall that probabilities associated with X are determined by computing areas under this curve. Since the bulk of the area is on the lower end of the

x -axis, we have higher probabilities of seeing lower values of X . (Though without any scale shown on the axes it is difficult to say what “lower” really means.)

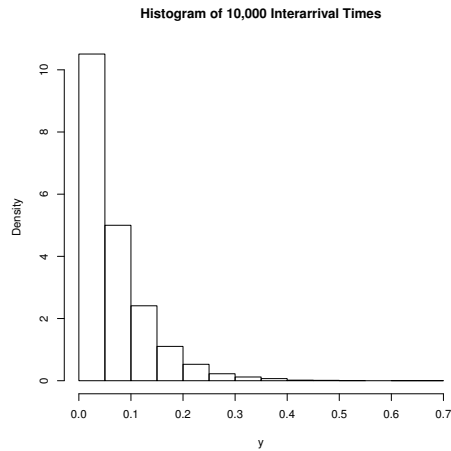
Suppose that we are going (in the future) to observe an actual numerical value for X . Actually, suppose we are going to observe n values for X . Let’s call these, yet unobserved (so still random) values $X_1, X_2, \dots, X_n$. For example, if this is the exponential distribution of Section 0.5.3 that describes a grocery store customer interarrival time, let’s position ourselves at the door and get ready to record the interarrival times for the next n customers.

Once customers start arriving, we will record actual numbers for the interarrival times. These numbers are said to be “realizations” of the random variables $X_1, X_2, \dots$. The probability that a customer interarrival time is less than 0.3 minutes is theoretically given by the area under the curve depicted above over the interval $(0, 0.3)$. If we want to try to estimate this using our “data”, we should compute the proportion of values in our list of realizations that are less than 0.3. In general, if we make a histogram of these realizations, scaled in such a way that the area of a bar over an interval represents the proportion of values in that interval, then the area of that bar should be approximating the desired probability and therefore should be approximating the true area under the pdf.

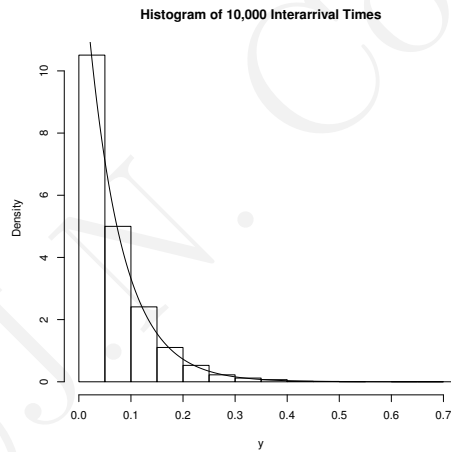
Here is such a histogram of 10 realizations of X where $X \sim \exp(\text{rate} = 15.2)$.



That doesn’t look much like the exponential pdf. However, with more data/realizations, we get the following.



Overlaying the pdf $f(x) = 15.2e^{-15.2x}$ for $x > 0$ gives us a pretty good fit!



Suppose now that we only have those 10,000 data points and the belief that they came from an exponential distribution, but we do not know the value of λ . How would you estimate λ ?

We could use the fact that $1/\lambda$ is the expected value (mean) for the exponential distribution. Recall that this is a probability weighted average. We could just average the 10,000 values we sampled/recorded. They already have their “probability weights” built-in in the way they were generated, with, for example, lower values coming out more often. This average is

known as a **sample mean** and is denoted by $\bar{x}$. If we have yet to record numerical values but are planning on recording 10,000 of them and averaging them, we would denote the sample mean with the $\bar{X}$, which is another random variable.

Using the observation above about the expected value or “distribution mean” and the sample mean, one plan for estimating λ is to think that

$$1/\lambda \approx \bar{X}$$

and to solve for λ . This gives

$$\lambda \approx 1/\bar{X}.$$

So, once we get the data and compute the sample mean, we could flip it over and use that to estimate the unknown λ . (**Important Note:** We are just being “conversational” at this point. It does not make sense to say that $\lambda = 1/\bar{X}$, or that they are even approximately equal, because the left hand side is a fixed (but unknown) constant and the right-hand side is a random variable that has an entire distribution!)

An alternative way to estimate the unknown λ might be to notice that it is the y -intercept on the graph of the pdf. Could we make a more refined histogram with really thin bars and try to estimate λ using the height of the first bar? In the last histogram shown above, the first bar is quite a bit lower than 15.2. If we made each bar half as wide, the first bar would likely shoot up much higher. What if we made them one quarter as wide? Surely we couldn’t keep shrinking them because we only have a finite amount of data and so we would only have a very small number of values in very small intervals. Indeed, the first bar height/area might shrink to zero!

It seems that the “ y -intercept idea” is somehow not as “solid” as the sample mean idea. A large part of mathematical statistics is coming up with ways to estimate parameters after we first quantify what is meant by a “good estimator” and what is meant by a “better estimator”. Before we get started, we are going to need some more tools under our belts.

1.2 “Integrating Without Integrating”, A.K.A. The Best Trick You’ll Learn Today

There are a lot of integrals ahead in this course and yet very few integrations required. For example, if someone asked us to integrate $3e^{-3x}$ from 0 to ∞

1.3. IMPORTANT TOOL I: FINDING DISTRIBUTIONS OF TRANSFORMATIONS OF RANDOM VARIABLES

we could just recognize this as the exponential pdf with rate 3 integrated over its support and say that we know the answer must be 1. Now, if some much meaner person asked us to integrate $5e^{-3x}$ over this region we could force it to look like the known exponential pdf and proceed as follows.

$$\int_0^\infty 5e^{-3x} dx = \frac{5}{3} \underbrace{\int_0^\infty 3e^{-3x} dx}_1 = \frac{5}{3}.$$

As this course goes on, we will develop a large repertoire of recognizable pdfs and be able to pull off this “trick” a lot!

1.3 Important Tool I: Finding Distributions of Transformations of Random Variables

The goal of this Section is to find the pdf of a random variable $Y = g(X)$ when the pdf of X is known and g is a “nice” function. For discrete random variables, the interpretation of a pdf is a simple probability so we will write out what we know, write out what we want we, and “just do it”. For the continuous case, we will derive a general formula.

1.3.1 The Discrete Case and the Binomial Distribution

We will use this opportunity first to introduce (or remind you of) another distribution.

Consider a sequence of n independent trials of an experiment where each trial can result in either “success” (S) or “failure” (F). Suppose that the probability of success remains the same from trial to trial. Call it p where $0 \leq p \leq 1$.

Let

$$X = \# \text{ of successes in } n \text{ trials.}$$

While it looks similar, this is different from the geometric random variable introduced in Chapter 0. There, we continued trials until the first success. Here, we will have a **fixed** number of trials (n) and we are counting up all the successes.



In this case, X is said to have a **binomial distribution** with parameters n and p .

We write

$$X \sim \text{bin}(n, p).$$

As the number of successes in n trials, X can take on values in $\{0, 1, 2, \dots, n\}$. That is, the support of the $\text{bin}(n, p)$ distribution is $\{0, 1, 2, \dots, n\}$.

The pdf is

$$f(x) = P(X = x) = P(SSFSF \dots F \text{ or } SFSFS \dots S \text{ or } \dots)$$

where each listed configuration of outcomes includes exactly x S 's and exactly $n - x$ F 's. Since the outcomes are disjoint, we get

$$f(x) = P(X = x) = P(SSFSF \dots F) + P(SFSFS \dots S) + \dots \quad (1.1)$$

Since the trials are independent,

$$P(SSFSF \dots F) = p \cdot p \cdot (1 - p) \cdot s \cdot (1 - p) \cdots (1 - p) = p^x (1 - p)^{n-x}.$$

In fact, every term in (1.1) gives that same probability! So,

$$f(x) = P(X = x) = c \cdot p^x (1 - p)^{n-x}$$

where c is the number of terms in (1.1).

How many ways are there to write down sequences of n S 's and F 's with exactly x S 's? We need to choose x slots out of n in which to put the S 's. There are

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

different ways to do this. Thus,

$$c = \binom{n}{x}.$$

So, we have seen that $X \sim \text{bin}(n, p)$ means that X has pdf

$$f(x) = P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x} I_{\{0,1,\dots,n\}}(x) \quad (1.2)$$

Check out other properties of the binomial distribution, such as its mean and variance, in the distribution tables in Appendix A.

We are now ready for a transformation.

Example 1.3.1

Suppose that $X \sim \text{bin}(n, p)$. Find the distribution (Name it!) for the random variable $Y := n - X$.

The answer to this one is quite “guessable”, but let’s go through the motions anyway since this will not always be the case.

The pdf for Y is

$$f_Y(y) = P(Y = y) = ?$$

What choice do we have but to use the fact that $Y = n - X$? We don’t know anything else about Y . Just do the problem and don’t worry about a “formula” for transforming the pdf! Let’s go.

$$f_Y(y) = P(Y = y) = P(n - X = y)$$

Now, since $X \sim \text{bin}(n, p)$, we do know how to compute probabilities of the form $P(X = ?)$, so, solve for X here to get

$$f_Y(y) = P(Y = y) = P(n - X = y) = P(X = n - y)$$

and use the pdf for X , plugging in $n - y$ for every x in (1.2):

$$\begin{aligned} f_Y(y) &= P(Y = y) = P(n - X = y) = P(X = n - y) \\ &\stackrel{(1.2)}{=} \binom{n}{n - y} p^{n - y} (1 - p)^{n - (n - y)} I_{\{0, 1, \dots, n\}}(n - y) \\ &= \binom{n}{n - y} p^{n - y} (1 - p)^y I_{\{0, 1, \dots, n\}}(y) \end{aligned}$$

I simplified the exponent $n - (n - y)$ for obvious reasons, but I also simplified the indicator for two reasons.

1. It's good form. After all, you are reporting a function of y . Why would you say "This holds for $n - y$ in the set $\{0, 1, 2, \dots, n\}$." as opposed to saying what y alone can be?
2. It will make it easier to recognize the distribution of Y .

To simplify the indicator $I_{\{0,1,\dots,n\}}(n-y)$, note that it is equal to 1 whenever

$$n - y = 0, 1, 2, \dots, n.$$

But,

$$\begin{aligned} n - y = 0 &\Rightarrow y = n, \\ n - y = 1 &\Rightarrow y = n - 1, \\ &\vdots \\ n - y = n &\Rightarrow y = 0. \end{aligned}$$

So, y takes on values in $\{0, 1, \dots, n\}$.

(Another note on good versus bad "form": Order is unimportant for listing elements in a set, but to say that " y takes on values in $\{n, n - 1, \dots, 0\}$ " is bad form and indicates that you are just "plugging and chugging" through steps of a problem without really thinking about what you are doing.)

In summary,

$$I_{\{0,1,\dots,n\}}(n-y) = I_{\{0,1,\dots,n\}}(y),$$

so the pdf for Y is

$$f_Y(y) = \binom{n}{n-y} p^{n-y} (1-p)^{n-(n-y)} I_{\{0,1,\dots,n\}}(y).$$

When trying to match this up to a distribution in our table of distributions, you will see only two discrete distributions with this type of indicator. One is the discrete uniform distribution whose pdf does not at all resemble this pdf and the other is the binomial distribution. Note that

$$\binom{n}{y} = \frac{n!}{y!(n-y)!} = \frac{n!}{(n-y)!y!} = \frac{n!}{(n-y)!(n-(n-y))!} = \binom{n}{n-y}.$$

Moving the p 's around, we can write the pdf for Y as

$$f_Y(y) = \binom{n}{y} (1-p)^y p^{n-y} I_{\{0,1,\dots,n\}}(y)$$

to see that

$$Y \sim \text{bin}(n, 1 - p).$$

Surprised? Probably not. If X is the number of successes in n trials, then $Y = n - X$ is the number of failures. Relabeling successes as failures, the “success probability” is now $1 - p$. ■

Probability density functions of transformations of random variables do not always turn out to be those of nice “named” distributions. When we ask you to “Find the distribution.”, we will mean a named distribution, otherwise we will just ask you to “Find the pdf.” To be more clear, we will follow “Find the distribution.” with “(Name it!)”

1.3.2 The Continuous Case and the Gamma Distribution

In the discrete case, we didn’t need some sort of general formula for making a transformation of pdfs from one random variable to another. Since pdfs were probabilities, it was easy just to write out what we want and what we know and to go! In the continuous case, the pdf no longer represents probability (i.e. $f(x) \neq P(X = x)$), so the approach from the discrete case will not make sense. The good news though is that we do have probabilities in the cdfs! So, the plan will be to find

$$F_Y(y) = P(Y \leq y) = \dots$$

and then in the end to find the pdf for Y using the fact that

$$f_Y(y) = \frac{d}{dy} F_Y(y).$$

Suppose that X has pdf $f_X(x)$ and cdf $F_X(x)$.

Suppose that Y is defined as $Y = g(X)$.

We will assume that g is invertible. (If it is not, it doesn’t mean we are out of luck, it just means that this approach and the formula we are about to derive won’t work.)

Note that

1. g invertible $\Rightarrow g$ is either strictly increasing or strictly decreasing.
2. If g is increasing (alternatively decreasing) the g^{-1} is also increasing (alternatively decreasing).

To see that second point in the increasing case, note that g increasing means that $x_1 \leq x_2$ implies that $g(x_1) \leq g(x_2)$ and that $x_1 \geq x_2$ implies that $g(x_1) \geq g(x_2)$. (It might help to draw a picture.)

We want to show that g^{-1} is also increasing. Suppose that $x_1 \leq x_2$. We want to show that $g^{-1}(x_1) \leq g^{-1}(x_2)$. Suppose (incorrectly) that

$$g^{-1}(x_1) \geq g^{-1}(x_2).$$

We are going to take g of both sides. Since g is increasing, it preserves the order of the inequality:

$$g(g^{-1}(x_1)) \geq g(g^{-1}(x_2)).$$

Canceling g and g^{-1} gives

$$x_1 \geq x_2.$$

This contradicts the original assumption that $x_1 \leq x_2$. Thus, we must have that $g^{-1}(x_1) \leq g^{-1}(x_2)$ and thus that g^{-1} is increasing.

We are ready to find an expression for the cdf, and then the pdf, for Y .

Case One: g is increasing.

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y)$$

Applying g^{-1} to both sides of that inequality, and using the fact that g increasing $\Rightarrow g^{-1}$ increasing $\Rightarrow g^{-1}$ applied to both sides of an inequality will preserve the order, we get

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(g(X) \leq y) \\ &= P(g^{-1}(g(X)) \leq g^{-1}(y)) \\ &= P(X \leq g^{-1}(y)) \\ &= F_X(g^{-1}(y)) \end{aligned}$$

Thus, the pdf for Y is

$$f_Y(y) = \frac{d}{dy}F_Y(y) = \frac{d}{dy}F_X(g^{-1}(y)) \stackrel{\text{chain rule}}{=} f_X(g^{-1}(y)) \cdot \frac{d}{dy}g^{-1}(y)$$

Note that g^{-1} increasing implies that that derivative is greater than zero.

Case Two: g is decreasing.

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y)$$

Applying g^{-1} to both sides of that inequality, and using the fact that g decreasing $\Rightarrow g^{-1}$ decreasing $\Rightarrow g^{-1}$ applied to both sides of an inequality will flip the inequality, we get

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(g(X) \leq y) \\ &= P(g^{-1}(g(X)) \geq g^{-1}(y)) \\ &= P(X \geq g^{-1}(y)) \\ &= 1 - P(X < g^{-1}(y)) \\ &\stackrel{\text{contin}}{=} 1 - P(X \leq g^{-1}(y)) \\ &= 1 - F_X(g^{-1}(y)) \end{aligned}$$

Thus, the pdf for Y is

$$\begin{aligned} f_Y(y) &= \frac{d}{dy}F_Y(y) = \frac{d}{dy}[1 - F_X(g^{-1}(y))] \\ &\stackrel{\text{chain rule}}{=} [0 - f_X(g^{-1}(y)) \cdot \frac{d}{dy}g^{-1}(y)] \\ &= -f_X(g^{-1}(y)) \cdot \frac{d}{dy}g^{-1}(y) \end{aligned}$$

Note that, since g^{-1} is decreasing (actually strictly so), we have $\frac{d}{dy}g^{-1}(y) < 0$, so, no, we did not end up with a negative pdf! Furthermore, note that $-\frac{d}{dy}g^{-1}(y)$ is positive and equal to $|\frac{d}{dy}g^{-1}(y)|$.

In the previous increasing case, $\frac{d}{dy}g^{-1}(y)$ is positive and equal to $|\frac{d}{dy}g^{-1}(y)|$.

In summary, we can write both the increasing and decreasing cases together as follows.



Continuous Transformation PDF

Let X be a continuous random variable with pdf f_x . Let Y be a random variable defined by $Y = g(X)$ where g is invertible (and differentiable). Then the pdf for Y can be computed as

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{d}{dy}g^{-1}(y) \right|.$$

Example 1.3.2

Let X have a “gamma distribution with parameters α and β ” We write $X \sim \Gamma(\alpha, \beta)$. This means that X is a continuous random variable with pdf



$$f_X(x) = \frac{1}{\Gamma(\alpha)} \beta^\alpha x^{\alpha-1} e^{-\beta x} I_{(0,\infty)}(x)$$

for some parameters $\alpha > 0$ and $\beta > 0$.

Notes:

1. We are using the X subscript on the pdf because we will have multiple pdfs in this problem— one for X and one for a new random variable Y .
2. For some people/books, $X \sim \Gamma(\alpha, \beta)$ means that X has pdf

$$f_X(x) = \frac{1}{\Gamma(\alpha)} (1/\beta)^\alpha x^{\alpha-1} e^{-x/\beta} I_{(0,\infty)}(x).$$

Here, α and β are known as the “shape” and “scale” parameters, respectively.

For our form of the gamma pdf, β is known as the “inverse scale parameter”.

3. The pdf involves the “gamma function”, $\Gamma(\cdot)$. $\Gamma(\alpha)$ is the gamma function evaluated at α and is just a constant. We will define it after finishing this example. The constant $\Gamma(\alpha)$ should not be confused with $\Gamma(\alpha, \beta)$ (two arguments) which is the name of a distribution.

Let $Y = 5X$. Find the distribution of Y . (Name it!)

Answer:

Here we are looking at the transformation $Y = g(X)$ where $g(x) = 5x$.

$$y = g(x) = 5x \quad \Rightarrow \quad x = g^{-1}(y) = y/5$$

So,

$$\begin{aligned} f_Y(y) &= f_X(g^{-1}(y)) \cdot \left| \frac{d}{dy} g^{-1}(y) \right| \\ &= \frac{1}{\Gamma(\alpha)} \beta^\alpha \left(\frac{y}{5}\right)^{\alpha-1} e^{-\beta y/5} \cdot \underbrace{I_{(0,\infty)}(y/5)}_{\parallel I_{(0,\infty)}(y)} \cdot \left|\frac{1}{5}\right| \\ &= \frac{1}{\Gamma(\alpha)} \left(\frac{\beta}{5}\right)^\alpha y^{\alpha-1} e^{-(\beta/5)y} \cdot I_{(0,\infty)}(y) \end{aligned}$$

which is the pdf for the $\Gamma(\alpha, \beta/5)$ distribution. Thus we see that

$$\boxed{Y \sim \Gamma(\alpha, \beta/5).}$$



A pdf is not completely specified without its support. If you do not use indicator functions, you must take a moment to figure out the support of the transformed random variable. Notice that by using the indicator notation, we just made this part of the process automatic!

An Aside: The Gamma Function

The pdf for the gamma distribution was defined using the **gamma function** which is denoted by $\Gamma(\cdot)$.

**Definition**

The gamma function, is defined, for $\alpha > 0$, as

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx.$$

Note that, for any $\beta > 0$,

$$\int_0^{\infty} \beta^{\alpha} x^{\alpha-1} e^{-\beta x} dx = \int_0^{\infty} (\beta x)^{\alpha-1} e^{-\beta x} \beta dx \stackrel{u=\beta x}{=} \int_0^{\infty} u^{\alpha-1} e^{-u} du = \Gamma(\alpha)$$

(Here we have used the fact that $du = \beta dx$ and that if x goes from 0 to ∞ , then $u = \beta x$ also goes from 0 to ∞ since $\beta > 0$.)

Now,

$$\int_0^{\infty} \frac{1}{\Gamma(\alpha)} \beta^{\alpha} x^{\alpha-1} e^{-\beta x} dx = \frac{1}{\Gamma(\alpha)} \int_0^{\infty} \beta^{\alpha} x^{\alpha-1} e^{-\beta x} dx = \frac{1}{\Gamma(\alpha)} \cdot \Gamma(\alpha) = 1,$$

so basically $1/\Gamma(\alpha)$ is the constant that makes $\beta^{\alpha} x^{\alpha-1} e^{-\beta x}$ into a proper pdf over $x \geq 0$!

Properties of the Gamma Function

1. $\Gamma(1) = 1$

$$\text{Proof: } \Gamma(1) = \int_0^{\infty} x^{1-1} e^{-x} dx = \int_0^{\infty} e^{-x} dx = 1.$$

2. For $\alpha > 1$,

$$\Gamma(\alpha) = (\alpha - 1) \cdot \Gamma(\alpha - 1).$$

Proof: $\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx$ Using integration by parts

$$\int u dv = uv - \int v du$$

with $u = x^{\alpha}$ and $dv = e^{-x} dx$ (So $du = (\alpha - 1)x^{\alpha-2} dx$ and $v = \int e^{-x} dx = -e^{-x}$), we have

$$\begin{aligned} \Gamma(\alpha - 1) &= -x^{\alpha-1} e^{-x} \Big|_0^{\infty} + \int_0^{\infty} (\alpha - 1) x^{\alpha-2} e^{-x} dx \\ &= 0 + (\alpha - 1) \int_0^{\infty} x^{\alpha-2} e^{-x} dx = (\alpha - 1) \cdot \Gamma(\alpha - 1) \quad \checkmark \end{aligned}$$

3. If $n \geq 1$ is an integer,

$$\Gamma(n) = (n-1)!.$$

Proof: By repeated application of property 2,

$$\begin{aligned}\Gamma(n) &= (n-1)\Gamma(n-1) = (n-1)(n-2)\Gamma(n-2) \\ &= \cdots = (n-1)(n-2)\cdots(1)\underbrace{\Gamma(1)}_1 = (n-1)!\end{aligned}$$

1.4 Important Tool II: Bivariate Transformations

In this section, we will consider bivariate to bivariate transformations for continuous random variables only. The discrete case will not be as important to use and should be treated in the ad-hoc “just do it” manner as the univariate case in Section 1.3.

Suppose that X_1 and X_2 are continuous random variables with joint pdf $f_{X_1, X_2}(x_1, x_2)$ and suppose that new random variables Y_1 and Y_2 are defined by

$$Y_1 = g_1(X_1, X_2) \quad \text{and} \quad Y_2 = g_2(X_1, X_2).$$

★ Continuous Transformation PDF (Bivariate)

Analogous to the one-dimensional case, the joint pdf for Y_1 and Y_2 is given by

$$f_{Y_1, Y_2}(y_1, y_2) = f_{X_1, X_2}(g_1^{-1}(y_1, y_2), g_2^{-1}(y_1, y_2)) \cdot |J|$$

where $|J|$ is the absolute value of the Jacobian of the transformation which is given by the determinant

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{vmatrix}.$$

We will not go through the derivation of this formula here, as it is too far a tangent into Calculus and will really distract us from Mathematical Statistics! See Appendix B if you wish to read more about Jacobians. It is completely optional (and tedious) reading!

The above formula is for “bivariate-to-bivariate” transformations and has an obvious generalization for the n -dimensional to n -dimensional case. You may also find it useful for a “bivariate-to-univariate” transformation as in the following example.

Example 1.4.1

Let $X_1, X_2 \stackrel{iid}{\sim} \Gamma(\alpha, \beta)$.

This is read as “Let X_1 and X_2 be ‘eye-eye-dee’ $\Gamma(\alpha, \beta)$.” the “iid” stands for **“independent and identically distributed”**. It means that X_1 and X_2 are independent and come from the exact same distribution— in this case, the $\Gamma(\alpha, \beta)$ distribution.

Alternatively, one might say that X_1, X_2 is a **random sample** from the $\Gamma(\alpha, \beta)$ distribution. We’d like to stress again that the words “random sample” include the concept of independence!

Let

$$Y = \frac{X_1}{X_1 + X_2}.$$

Find the distribution (Name it!) of Y .

Why is this example in the “bivariate-to-bivariate” transformation section? It is here because perhaps the easiest way to find this distribution is to think of Y as some Y_1 and then to define a convenient Y_2 for which we can write down the joint pdf for Y_1 and Y_2 after which we can hope to be able to integrate out Y_2 to find the marginal pdf for Y_1 .

Technically, Y_2 can be anything you want, but some choices are more convenient than others— especially for the integration down to the marginal pdf for Y_1 at the end.



If Y_1 is a ratio, it is almost always a good idea to choose Y_2 to be the denominator.

Thus, we will try letting

$$Y_1 = \frac{X_1}{X_1 + X_2} \quad \text{and} \quad Y_2 = X_1 + X_2.$$

Then $y_1 = g_1(x_1, x_2) = x_1/(x_1 + x_2)$ and $y_2 = g_2(x_1, x_2) = x_1 + x_2$. Solving for x_1 and x_2 gives

$$x_1 = y_1 y_2 \quad \text{and} \quad x_2 = y_2 - y_1 y_2,$$

which become the definitions of $g_1^{-1}(y_1, y_2)$ and $g_2^{-1}(y_1, y_2)$, respectively.

The Jacobian is then

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{vmatrix} = \begin{vmatrix} y_2 & y_1 \\ -y_2 & 1 - y_1 \end{vmatrix} = y_2(1 - y_1) - (-y_1 y_2) = y_2.$$

Since $X_1, X_2 \stackrel{iid}{\sim} \Gamma(\alpha, \beta)$, their joint pdf is

$$\begin{aligned} f_{X_1, X_2}(x_1, x_2) &\stackrel{indep}{=} f_{X_1}(x_1) \cdot f_{X_2}(x_2) \\ &= \frac{1}{\Gamma(\alpha)} \beta^\alpha x_1^{\alpha-1} e^{-\beta x_1} \cdot I_{(0, \infty)}(x_1) \cdot \frac{1}{\Gamma(\alpha)} \beta^\alpha x_2^{\alpha-1} e^{-\beta x_2} \cdot I_{(0, \infty)}(x_2) \\ &= \frac{1}{[\Gamma(\alpha)]^2} \beta^{2\alpha} (x_1 x_2)^{\alpha-1} e^{-\beta(x_1 + x_2)} \cdot I_{(0, \infty)}(x_1) \cdot I_{(0, \infty)}(x_2) \end{aligned}$$

Thus, the joint pdf for Y_1 and Y_2 is

$$\begin{aligned} f_{Y_1, Y_2}(y_1, y_2) &= f_{X_1, X_2}(g_1^{-1}(y_1, y_2), g_2^{-1}(y_1, y_2)) \cdot |J| \\ &= f_{X_1, X_2}(y_1 y_2, y_2 - y_1 y_2) \cdot |J| \\ &= \frac{1}{[\Gamma(\alpha)]^2} \beta^{2\alpha} (y_1 y_2^2 - y_1^2 y_2^2)^{\alpha-1} e^{-\beta y_2} \cdot I_{(0, \infty)}(y_1 y_2) \cdot I_{(0, \infty)}(y_2 - y_1 y_2) \cdot |y_2| \end{aligned}$$

We are going to need to simplify this in order to integrate out y_2 and, really, just to be “good citizens of the world”. First, note that the product of

indicators is the one looked at at the very end of Section 0.9.4. From there, we know that

$$I_{(0,\infty)}(y_1 y_2) \cdot I_{(0,\infty)}(y_2 - y_1 y_2) = I_{(0,1)}(y_1) \cdot I_{(0,\infty)}(y_2).$$

Since this product of indicators tells us, in particular, that $y_2 > 0$, we can drop the absolute value on y_2 in the joint pdf for Y_1 and Y_2 . Since we will be integrating out y_2 , let's separate out the $(y_1 y_2^2 - y_1^2 y_2^2)^{\alpha-1}$ part to $(y_2^2)^{\alpha-1} \cdot (y_1 - y_1^2)^{\alpha-1}$. After multiplying in the lone $|y_2| = y_2$ from the Jacobian, we get

$$f_{Y_1, Y_2}(y_1, y_2) = \frac{1}{[\Gamma(\alpha)]^2} \beta^{2\alpha} y_2^{2\alpha-1} [y_1(1-y_1)]^{\alpha-1} e^{-\beta y_2} I_{(0,1)}(y_1) \cdot I_{(0,\infty)}(y_2).$$

Now for the marginal. We'll use our "integrating without integrating" trick. (Actually, even this isn't necessary for this particular problem. Do you see why? The form of the joint pdf is very "nice"!)

$$\begin{aligned} f_{Y_1}(y_1) &= \int_{-\infty}^{\infty} f_{Y_1, Y_2}(y_1, y_2) dy_2 \\ &= \frac{1}{[\Gamma(\alpha)]^2} \beta^{2\alpha} [y_1(1-y_1)]^{\alpha-1} I_{(0,1)}(y_1) \int_0^{\infty} \underbrace{y_2^{2\alpha-1} e^{-\beta y_2}}_{\substack{\text{like a} \\ \Gamma(2\alpha, \beta) \\ \text{pdf}}} dy_2 \\ &= \frac{\Gamma(2\alpha)}{[\Gamma(\alpha)]^2} [y_1(1-y_1)]^{\alpha-1} I_{(0,1)}(y_1) \int_0^{\infty} \underbrace{\frac{1}{\Gamma(2\alpha)} \beta^{2\alpha} y_2^{2\alpha-1} e^{-\beta y_2}}_{\substack{\text{actually a } \Gamma(2\alpha, \beta) \\ \text{pdf, so integrates} \\ \text{to 1}}} dy_2 \\ &= \frac{\Gamma(2\alpha)}{[\Gamma(\alpha)]^2} [y_1(1-y_1)]^{\alpha-1} I_{(0,1)}(y_1) \end{aligned}$$

If you look in your table of distributions (continuous side) you'll see that there are only two possibilities for this distribution based on the indicator. One is the uniform distribution on the interval $(0, 1)$. That is a flat line distribution with height 1 on the interval. It has pdf $f(x) = I_{(0,1)}(x)$. The marginal pdf we have computed is a little more complicated than that.

The second distribution on a finite interval, and, in particular on the interval from $(0, 1)$, is called the **Beta distribution**. This distribution has two parameters, a and b , and is given in the table as having pdf

$$f(x) = \frac{1}{\mathcal{B}(a, b)} x^{a-1} (1-x)^{b-1} dx.$$

We would write $X \sim \text{Beta}(a, b)$.

Our marginal pdf for Y_1 looks a lot like this one (written as a function of y_1). But... what about our “ Γ stuff” versus this thing we have yet to define written as $\mathcal{B}(a, b)$? (This is known as the “Beta function”.)

The “ Γ stuff” and the “Beta stuff” are just constants. They have to match in order for these both to be pdfs! For example, we know that $f(x) = 2e^{-2x} I_{(0,1)}(x)$ is a pdf. It’s the exponential with rate 2 pdf. From this, we know that the function $5e^{-2x} I_{(0,\infty)}(x)$ is **not** a pdf! It can’t integrate to 1 if $2e^{-2x} I_{(0,\infty)}(x)$ integrates to 1.

If you followed our transformational procedures correctly, you are guaranteed to get a pdf in the end. Thus, our pdf

$$f_{Y_1}(y_1) = \frac{\Gamma(2\alpha)}{[\Gamma(\alpha)]^2} [y_1(1-y_1)]^{\alpha-1} I_{(0,1)}(y_1)$$

must be the pdf for the **Beta distribution with parameters $a = \alpha$ and $b = \alpha$** . We write

$$Y \sim \text{Beta}(\alpha, \alpha).$$



1.4.1 The Beta Distribution

The Beta distribution was defined at the end of the previous section. (In fact, the only reason I am making this short extra section is so that you will see “Beta distribution” in the table of contents!)

This distribution is a nice flexible distribution that is suitable to model things that are supposed to be between 0 and 1. (If I didn’t just need to get these notes posted right now, I would make some graphs. Note that when $a = b = 1$, it is the uniform distribution on $(0, 1)$. For $a, b < 1$, it has a

“u-shape”. For $a, b > 1$ it is a unimodal distribution that is tied down to 0 at both endpoints and various vales of a and b in this range can move the “lump” around to wherever you want it!

Just for the record though, we define the Beta function.



Definition

The **Beta function** is defined, for $a, b > 0$, as

$$\mathcal{B}(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx.$$

It is left as an exercise to show the following relationship.



Property

$$\mathcal{B}(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}.$$

1.5 Important Tool II: Mins and Maxes

1.5.1 The Distribution of a Minimum by Example

Let’s just start right off with an example.

Example:

Suppose that

$$X_1, X_2, \dots, X_n \stackrel{iid}{\sim} \exp(\text{rate} = \lambda).$$

Let’s find the distribution of

$$Y = \min(X_1, X_2, \dots, X_n).$$

To find the distribution of a maximum of a minimum, it is most convenient to look at cdfs. The cdf for each individual X is

$$F(x) = \int_{-\infty}^x f(u) du = \int_0^x \lambda e^{-\lambda u} du = 1 - e^{-\lambda x}.$$

The cdf for Y , the minimum, is:

$$F_Y(y) = P(Y \leq y) = P(\min(X_1, X_2, \dots, X_n) \leq y)$$

Note that if the minimum of a group of random variables is less than or equal to y , some in the group will be less than or equal to y while others may be greater than y . Consider the equivalent expression

$$P(\min(X_1, X_2, \dots, X_n) \leq y) = 1 - P(\min(X_1, X_2, \dots, X_n) > y).$$

If the minimum of a group of random variables, $X_1, X_2, \dots, X_n$, is greater than y , we are forced to have every $X_i > y$. On the flip side, if every X_i in $X_1, X_2, \dots, X_n$ is greater than y , then the minimum of this group will definitely be greater than y . In other words, the “event that $\min(X_1, X_2, \dots, X_n) > y$ is equivalent to the event that $X_1 > y$ and $X_2 > y$ and $\dots$ and $X_n > y$. Thus,

The cdf for Y , the minimum, is:

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(\min(X_1, X_2, \dots, X_n) \leq y) \\ &= 1 - P(\min(X_1, X_2, \dots, X_n) > y) \\ &\stackrel{\text{indep}}{=} 1 - P(X_1 > y) \cdot P(X_2 > y) \cdots P(X_n > y). \end{aligned}$$

Since $X_1, X_2, \dots, X_n$ are all identically distributed, all of those probabilities are the same, thus

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(\min(X_1, X_2, \dots, X_n) \leq y) \\ &= 1 - P(\min(X_1, X_2, \dots, X_n) > y) \\ &\stackrel{\text{indep}}{=} 1 - P(X_1 > y) \cdot P(X_2 > y) \cdots P(X_n > y) \\ &\stackrel{\text{ident}}{=} 1 - [P(X_1 > y)]^n \end{aligned}$$

To compute $P(X_1 > y)$, we could integrate the exponential pdf from y to ∞ , or we could use our already computed cdf, $F(x) = P(X_1 \leq x) = 1 - e^{-\lambda x}$:

$$P(X_1 > y) = 1 - P(X_1 \leq y) = 1 - (1 - e^{-\lambda y}) = e^{-\lambda y}.$$

So, the cdf for the minimum Y is

$$F_Y(y) = 1 - [e^{-\lambda y}]^n = 1 - e^{-\lambda n y}.$$

The pdf for the minimum is

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy} [1 - e^{-\lambda n y}] = \lambda n e^{-\lambda n y}.$$

Note that Y , as the minimum of a bunch of values in $(0, \infty)$, Y also lives in $(0, \infty)$. For completeness,

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy} [1 - e^{-\lambda n y}] = \lambda n e^{-\lambda n y} I_{(0, \infty)}(y).$$

So, we have seen that

$$Y \sim \exp(\text{rate} = \lambda n)$$



1.5.2 “Order Statistics” Notation

Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with pdf f and cdf F .



Notation

Define

$$\begin{aligned} X_{(1)} &= \min(X_1, X_2, \dots, X_n) \\ X_{(2)} &= \text{the second smallest of the } X\text{'s} \\ &\vdots \\ X_{(n)} &= \max(X_1, X_2, \dots, X_{(n)}) \end{aligned}$$

The $X_{(1)}, X_{(2)}, \dots, X_{(n)}$ form a new collection of random variables known as the **order statistics** of the original sample. (Another commonly used notation for the order statistics is $X_{1:n}, X_{2:n}, \dots, X_{n:n}$.)

We will discuss order statistics in general in Section ?? (Sorry. Not here yet!) For now, we will just focus on minimums and maximums.

1.5.3 PDFs for Minimums and Maximums

In this section, we will derive the pdf and cdf for minimums and maximums for a random sample from a continuous distribution only. We'll work out the discrete case in the future as needed. (It's a bit messy!)

Minimums:

In the example of Section 1.5.1, we have already basically derived an expression for the pdf of the minimum in terms of f and F , the pdf and cdf respectively for the original random variables.

$$\begin{aligned}
 F_{X_{(1)}}(x) &= P(X_{(1)} \leq x) = P(\min(X_1, X_2, \dots, X_n) \leq x) \\
 &= 1 - P(\min(X_1, X_2, \dots, X_n) > x) \\
 &\stackrel{\text{indep}}{=} 1 - P(X_1 > x) \cdot P(X_2 > x) \cdots P(X_n > x) \\
 &\stackrel{\text{ident}}{=} 1 - [P(X_1 > x)]^n \\
 &= 1 - [1 - F(x)]^n \\
 \Rightarrow f_{X_{(1)}}(x) &= \frac{d}{dx} F_{X_{(1)}}(x) \\
 &= \frac{d}{dx} \{1 - [1 - F(x)]^n\} \\
 &= 0 - n[1 - F(x)]^{n-1} \cdot (-f(x)) \\
 &= n[1 - F(x)]^{n-1} f(x).
 \end{aligned}$$

In summary, if $X_{(1)} = \min(X_1, X_2, \dots, X_n)$ where $X_1, X_2, \dots, X_n$ is a random sample from a continuous distribution with pdf f and cdf F , the pdf for $X_{(1)}$ is

$$f_{X_{(1)}}(x) = n[1 - F(x)]^{n-1} f(x)$$

Please please don't memorize this! Just work it out when you need it. It is quick and things will go a lot more smoothly for you in this course if you rely on understanding what you are doing rather than regurgitating a formula.

Maximums:

Now let's derive an expression for the pdf of the maximum in terms of f and F . Note that we will not have to flip any inequalities because the maximum of a collection of random variables is less than or equal to x if and only if every random variable in the collection is less than or equal to x .

$$\begin{aligned}
 F_{X_{(n)}}(x) &= P(X_{(n)} \leq x) = P(\max(X_1, X_2, \dots, X_n) \leq x) \\
 &\stackrel{indep}{=} P(X_1 \leq x) \cdot P(X_2 \leq x) \cdots P(X_n \leq x) \\
 &\stackrel{ident}{=} [P(X_1 \leq x)]^n \\
 &= [F(x)]^n \\
 \\
 \Rightarrow f_{X_{(n)}}(x) &= \frac{d}{dx} F_{X_{(n)}}(x) \\
 &= \frac{d}{dx} \{[F(x)]^n\} \\
 &= n[F(x)]^{n-1} \cdot f(x)
 \end{aligned}$$

In summary, if $X_{(n)} = \max(X_1, X_2, \dots, X_n)$ where $X_1, X_2, \dots, X_n$ is a random sample from a continuous distribution with pdf f and cdf F , the pdf for $X_{(n)}$ is

$$f_{X_{(n)}}(x) = n[F(x)]^{n-1} \cdot f(x)$$

Again, I'm begging you not to memorize this! It's easy to derive!

For fun, you might want to work out the pdf of the maximum of exponential random variables. The resulting pdf is not from a nice "named" distribution that you will find on your table. Similarly, not all transformations of pdfs for random variables, such as in Section 1.3 will result in recognizable pdfs.

1.6 Important Tool IV: Moment Generating Functions (MGFs)



Definition

For a random variable X , the **moment generating function** (mgf), denoted by $M(t)$ or $M_X(t)$ if necessary, is defined as

$$M(t) = \mathbb{E}[e^{tX}].$$

If it becomes necessary to distinguish between two moment generating functions, for random variables X and Y , we will write $M_X(t)$ and $M_Y(t)$.

Moment generating functions are extremely useful tools for both computing “moments” of distributions which are the expectations $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, $\mathbb{E}[X^3]$, $\dots$, and for identifying distributions of some transformed random variables. Most notably, if you are asked to find the distribution of the “big fat sum” of iid random random variables $X_1, X_2, \dots, X_n$, you are surely going to want to try the moment generating function approach described in this section!

1.6.1 The Expectation of a Function of X

We know how to find the expectation $\mathbb{E}[X]$ but do we know how to find an expectation of a function of X such as $\mathbb{E}[g(X)]$? Before we can find any moment generating functions, we’ll need to know how to do this! We begin with a simple discrete example.

Suppose that X is a discrete random variable, taking on the values -1 , 0 , and 1 , with probabilities

x	-1	0	1
$P(X = x)$	$4/12$	$3/12$	$5/12$

Now suppose that we want to find $\mathbb{E}[X^2]$. Note that this is the expected value of a new random variable $Y := X^2$. Further, note that Y takes on

possible values 0 and 1 and that Y has pdf given by

$$\frac{y}{P(Y=y)} \mid \begin{array}{cc} 0 & 1 \\ 3/12 & 9/12 \end{array}.$$

We now have

$$\mathbb{E}[X^2] = \mathbb{E}[Y] = \sum_{y=0}^1 y \cdot P(Y=y) = 0 \cdot \frac{3}{12} + 1 \cdot \frac{9}{12} = \frac{9}{12}.$$

However, note that

$$\begin{aligned} \mathbb{E}[X^2] &= \mathbb{E}[Y] = 0 \cdot \frac{3}{12} + 1 \cdot \frac{9}{12} \\ &= 0 \cdot \frac{3}{12} + 1 \cdot \left(\frac{4}{12} + \frac{5}{12} \right) \\ &= 0 \cdot \frac{3}{12} + 1 \cdot \frac{4}{12} + 1 \cdot \frac{5}{12} \\ &= (-1)^2 \cdot \frac{4}{12} + 0^2 \cdot \frac{3}{12} + 1^2 \cdot \frac{5}{12} \\ &= \sum_{x=-1}^1 x^2 \cdot P(X=x) \end{aligned}$$

Basically, what we are seeing here is the following.

★ Property

Let X be a random variable with pdf $f_X(x)$. Let $g(x)$ be some function.

- If X is discrete, we have have

$$\mathbb{E}[g(X)] = \sum_x g(x) f_X(x).$$

- If X is continuous, we have have

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx.$$

So, even though $g(X)$ is a new random variable, when finding $\mathbb{E}[g(X)]$ it is unnecessary to find a new pdf! That's kind of surprising if you think about

it... but a lot of people don't think about it— they just do it sort of naturally without realizing how big of a deal this is. This is why this new formula for $E[g(X)]$ (in either the discrete or continuous case) is known as the **Law of the Unconscious Statistician!**

We will now prove the formula for $E[g(X)]$ in the case where X is continuous and g is invertible and differentiable. Proofs of the other cases are omitted.

Suppose that X is continuous and g is a nice invertible and differentiable function. Define $Y = g(X)$. Then, since we know from Section 1.3 that $f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$, we have that

$$\begin{aligned} E[g(X)] &= E[Y] = \int_{-\infty}^{\infty} y \cdot f_Y(y) dy \\ &= \int_{-\infty}^{\infty} y \cdot f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right| dy \end{aligned}$$

Since g is invertible, we know that it is either strictly increasing or strictly decreasing. Let's consider each case separately.

Case One: g is strictly increasing

If g is strictly increasing then g^{-1} is as well. So, the derivative $\frac{d}{dy} g^{-1}(y)$ is always positive and we can drop the absolute value in the above integral. We now have

$$E[g(X)] = \int_{-\infty}^{\infty} y \cdot f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y) dy$$

Consider the change of variable $x = g^{-1}(y)$. Note that $dx = \frac{d}{dy} g^{-1}(y) dy$. Making the substitution, we have

$$E[g(X)] = \int_{?}^{?'} g(x) \cdot f_X(x) dx$$

for some limits of integration $?$ and $?'$.

As for the limits, note that, even the original limits of integration went from $-\infty$ to ∞ , they might actually have ended up being cut off by the support set for the pdf $f_Y(y)$. That is, the pdf $f_Y(y)$ may only be non-zero on some smaller interval. Furthermore, even if $y = g(x)$ does go from $-\infty$ to ∞ , the inverse might not. For example, if $y = g(x) = \ln x$, then y goes from

$-\infty$ to ∞ , but $x = g^{-1}(y)$ only takes values from 0 to ∞ . However, in this example,

$$\int_0^\infty g^{-1}(x) f_X(x) dx = \int_{-\infty}^\infty g^{-1}(x) f_X(x) dx$$

because the pdf for X will be zero outside of $(0, \infty)$.

So, when we make the substitution $x = g^{-1}(y)$, it will be safe to have limits all the way from $-\infty$ to ∞ since g^{-1} is increasing and since the pdf will cut off the limits in the appropriate places. That is, it is safe to use the limits $? = -\infty$ and $? = \infty$ and we have

$$\mathbb{E}[g(X)] = \int_{-\infty}^\infty g(x) f_X(x) dx,$$

as desired.

Case One: g is strictly decreasing

If g is strictly decreasing then g^{-1} is as well. So, the derivative $\frac{d}{dy}g^{-1}(y)$ is always negative and we have that $|\frac{d}{dy}g^{-1}(y)| = -\frac{d}{dy}g^{-1}(y)$. Thus, we have that

$$\begin{aligned} \mathbb{E}[g(X)] &= \int_{-\infty}^\infty y \cdot f_X(g^{-1}(y)) \left| \frac{d}{dy}g^{-1}(y) \right| dy \\ &= - \int_{-\infty}^\infty y \cdot f_X(g^{-1}(y)) \cdot \frac{d}{dy}g^{-1}(y) dy \end{aligned}$$

Making the substitution $x = g^{-1}(y)$ and again noting that $dx = \frac{d}{dy}g^{-1}(y)$, we have that

$$\mathbb{E}[g(X)] = - \int_{?}^{?'} g(x) \cdot f_X(x) dx$$

for some limits of integration $?$ and $?'$.

Since g^{-1} is a decreasing function, as y increases, $x = g^{-1}(y)$ decreases. As per the discussion in the increasing case, even if the random variable X takes on only a limited range of values, we can let x decrease from ∞ all the way to $-\infty$ and the pdf will cut us off appropriately. Thus,

$$\mathbb{E}[g(X)] = - \int_{\infty}^{-\infty} g(x) \cdot f_X(x) dx = \int_{-\infty}^\infty g(x) \cdot f_X(x) dx,$$

as desired.

Now that we know how to compute the expectation of a function of a random variable, we can get back to moment generating functions!

1.6.2 Finding MGFs and the Poisson Distribution

A Discrete Example:

A discrete random variable X is said to have the **Poisson distribution with parameter** λ if X has pdf.



$$f(x) = P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!} I_{\{0,1,2,\dots\}}(x).$$

We write $X \sim \text{Poisson}(\lambda)$.

Recall how we motivated the exponential distribution. We imagined events (arrivals at the door of a grocery store) occurring with two assumptions. They were that

1. the arrival/event rate is a constant λ per unit time, and
2. the number of arrivals/events in non-overlapping periods of time are independent.

Under these assumptions, the exponential distribution describes the amount of time between any two consecutive arrivals/events. One can show (Again, I'd direct you to take APPM 4/5560 where this scenario is a core focus!) that the **number** of arrivals in one unit of time has the Poisson distribution with parameter λ .

So, suppose that $X \sim \text{Poisson}(\lambda)$. Let us find the moment generating function for X . Since this is a discrete random variable, the expectation is computed as a sum and not an integral.

$$\begin{aligned} M_X(t) &= \mathbb{E}[e^{tX}] = \sum_x e^{tx} \cdot P(X = x) \\ &= \sum_{x=0}^{\infty} e^{tx} \frac{e^{-\lambda} \lambda^x}{x!} \\ &= \sum_{x=0}^{\infty} \frac{e^{-\lambda} (\lambda e^t)^x}{x!} \end{aligned}$$

At this point, you could pull out the $e^{-\lambda}$ and recall the Taylor series expansion of e^x is

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}.$$

In our moment generating function expression, the k is actually x and the x is λe^t . So,

$$M_X(t) = e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!} = e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t-1)}.$$

Alternatively, you could do what will become our “standard MathStat trick” that we usually use for integrals. The sum we need to compute looks almost like the sum over all values of the Poisson pdf except, in place of λ , we have λe^t . To make it into the Poisson pdf, we need the “ $e^{-\lambda}$ part” which, in this case is $e^{-\lambda e^t}$. Putting this in (after pulling out the $e^{-\lambda}$) and adjusting for it by putting $e^{\lambda t}$ out front, we have

$$\begin{aligned} M_X(t) &= \sum_{x=0}^{\infty} \frac{e^{-\lambda} (\lambda e^t)^x}{x!} \\ &= e^{\lambda(e^t-1)} \underbrace{\sum_{x=0}^{\infty} \frac{e^{-\lambda e^t} (\lambda e^t)^x}{x!}}_{\substack{\text{sum of Poisson} \\ \text{pdf is 1}}} \\ &= e^{\lambda(e^t-1)}. \end{aligned}$$

(Make sure you can find this in the moment generating column on the table of distributions!)

A Continuous Example:

Let $X \sim \exp(\text{rate} = \lambda)$. Let us find the mgf of X .

$$\begin{aligned} M_X(t) &= E[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} f(x) dx \\ &= \int_0^{\infty} e^{tx} \lambda e^{-\lambda x} dx \\ &= \int_0^{\infty} \lambda e^{-\lambda x + tx} dx \end{aligned}$$

To integrate, it is convenient to factor out the x in that exponent so we are integrating something like e^{cx} . However, by taking out the negative as well, we can see something that looks like the exponential pdf with $\lambda - t$ in place of λ .

$$M_X(t) = \int_0^\infty \lambda e^{-(\lambda-t)x} dx$$

Note that, in order for this integral to converge we must have $\lambda - t > 0$, otherwise the integrand is blowing up. (Alternatively, when comparing this to the exponential pdf, the usual parameter restriction $\lambda > 0$ has become $\lambda - t > 0$.) Either way you look at it, this mgf is only defined for $t < \lambda$.

So, for $t < \lambda$,

$$\begin{aligned} M_X(t) &= \int_0^\infty \lambda e^{-(\lambda-t)x} dx \\ &= \frac{\lambda}{\lambda-t} \int_0^\infty (\lambda-t) e^{-(\lambda-t)x} dx \\ &= \frac{\lambda}{\lambda-t} \cdot 1 = \frac{\lambda}{\lambda-t} \end{aligned}$$

since that integral is that of a pdf over its entire support.

1.6.3 Finding Moments

For a random variable X , or its associated distribution, “**moments**” are expectations of the form $E[X^k]$ for $k = 1, 2, 3, \dots$ (Sometimes such moments are called “non-central moments” as opposed to the “central moments” $E[(X - \mu)^k]$ where X is first “centered” around its mean $\mu = E[X]$.)

Consider taking a derivative with respect to t of the moment generating function $M_X(t)$.

$$M_X(t) = E[e^{tX}] \Rightarrow M'_X(t) = \frac{d}{dt} M_X(t) = \frac{d}{dt} E[e^{tX}] \stackrel{?}{=} E \left[\frac{d}{dt} e^{tX} \right] = E[X e^{tX}]$$

At that question mark, we pulled a derivative inside of an expectation. Since an expectation is a sum or integral, we really just exchanged a derivative and a sum or integral. In general, this may or may not be a valid thing to do. If the sum/integral converges and the sum/integral of the differentiated integrand also converges, they will converge to the same thing. **For this course, things are mostly well behaved and you may assume this exchange is always valid.**

Now,

$$M'_X(0) = \mathbb{E}[Xe^0] = \mathbb{E}[X].$$

So, we see that by differentiating the moment generating function and plugging in a 0 we get the first moment of this distribution!

With a second derivative, we have

$$M''_X(t) = \frac{d}{dt}M'_X(t) = \frac{d}{dt}\mathbb{E}[Xe^{tX}] = \mathbb{E}\left[\frac{d}{dt}Xe^{tX}\right] = \mathbb{E}[X^2e^{tX}].$$

So, plugging in a 0 now gives us

$$M''_X(0) = \mathbb{E}[X^2e^0] = \mathbb{E}[X^2],$$

giving us the second moment of the distribution.

In general, for a random variable X with mgf $M_X(t)$, the k th moment is obtained by $M^{(k)}(0)$, where $M^{(k)}(t)$ is the k th derivative of $M_X(t)$ with respect to t .

Discrete Example:

Let $X \sim \text{Poisson}(\lambda)$. One can show, using the definition of expectation and variance that $\mathbb{E}[X] = \lambda$ and $\text{Var}[X] = \lambda$. While the variance is a bit messy, you should try to show that $\mathbb{E}[X] = \lambda$ using our “summing without summing” trick of rewriting things so that you are ultimately summing a pdf over its entire support.

Let us now find the mean and variance for the Poisson distribution using moment generating functions. Recall that $M_X = \exp[\lambda(e^t - 1)]$. (That “exp” notation is another way to write e to a power.)

So,

$$\begin{aligned} M'_X(t) &= \exp[\lambda(e^t - 1)] \cdot \frac{d}{dt}[\lambda(e^t - 1)] \\ &= \exp[\lambda(e^t - 1)] \cdot \lambda e^t \end{aligned}$$

$$\Rightarrow \mathbb{E}[X] = M'_X(0) = \lambda.$$

Furthermore,

$$\begin{aligned}
 M_X''(t) &= \exp[\lambda(e^t - 1)] \cdot \lambda e^t + \lambda e^t \cdot \exp[\lambda(e^t - 1)] \cdot \lambda e^t \\
 &= \lambda e^t \cdot \exp[\lambda(e^t - 1)] \cdot (1 + \lambda e^t) \\
 &\Rightarrow M_X''(0) = \lambda(1 + \lambda) = \lambda + \lambda^2.
 \end{aligned}$$

So, we have that

$$Var[X] = E[X^2] - (E[X])^2 = \lambda + \lambda^2 - (\lambda)^2 = \lambda,$$

as desired. □

1.6.4 The MGF Uniquely Identifies the Distribution

Here is an exciting claim.

A moment generating function for a random variable X uniquely determines its distribution!

For example, if you compute the mgf for some random variable X and get, for example, $\exp[\lambda(e^t - 1)]$, you know that X must be a $Poisson(\lambda)$ random variable!

This claim is arguably one of the most important reasons for learning about moment generating functions. **It is especially useful for finding the distribution of sums of independent random variables.**

Example:

Let $X_1, X_2, \dots, X_n$ be independent random variables with $X_i \sim Poisson(\lambda_i)$.

Note that these are not “iid” random variables. For this example, the random variables are not identically distributed. Although they all have a Poisson distribution, the changing λ ’s make them different Poisson distributions.

Let us find the distribution of $Y = \sum_{i=1}^n X_i$.

$$\begin{aligned}
M_Y(t) &= \mathbb{E}[e^{tY}] = \mathbb{E}[e^{t\sum X_i}] = \mathbb{E}[\prod_{i=1}^n e^{tX_i}] \\
&\stackrel{\text{indep}}{=} \prod_{i=1}^n \mathbb{E}[e^{tX_i}] = \prod_{i=1}^n M_{X_i}(t) \\
&= \prod_{i=1}^n \exp[\lambda_i(e^t - 1)] \\
&= \exp[(\sum \lambda_i)(e^t - 1)]
\end{aligned}$$

Since this is the mgf of the Poisson distribution with rate $\sum_{i=1}^n \lambda_i$, we can conclude that

$$Y \sim \text{Poisson}(\sum_{i=1}^n \lambda_i).$$

□

That was a lot easier than what would have been a generalized (to n dimensions) Jacobian approach to finding the distribution and even easier than the only slightly less cumbersome “conditioning approach” that you would study in a stochastic process course such as APPM 4/5560. (Just take it already so I can stop the advertisements!)

1.6.5 A Proof in a Simplified Setting

Question:

I have two random variables X and Y with an infinite sequence of matching moments:

$$\mathbb{E}[X] = \mathbb{E}[Y], \quad \mathbb{E}[X^2] = \mathbb{E}[Y^2], \quad \mathbb{E}[X^3] = \mathbb{E}[Y^3], \quad \text{etc ...}$$

Does this mean that X and Y must have the same distribution?

Answer:

Unfortunately, no. For example, consider the following probability density function due to C.C. Heyde:

$$f_a(x) = \frac{1}{x\sqrt{2\pi}} e^{-1/2(\ln x)^2} (1 + a \sin(2\pi \ln(x))) I_{(0,\infty)}(x)$$

where

$$-1 \leq a \leq 1.$$

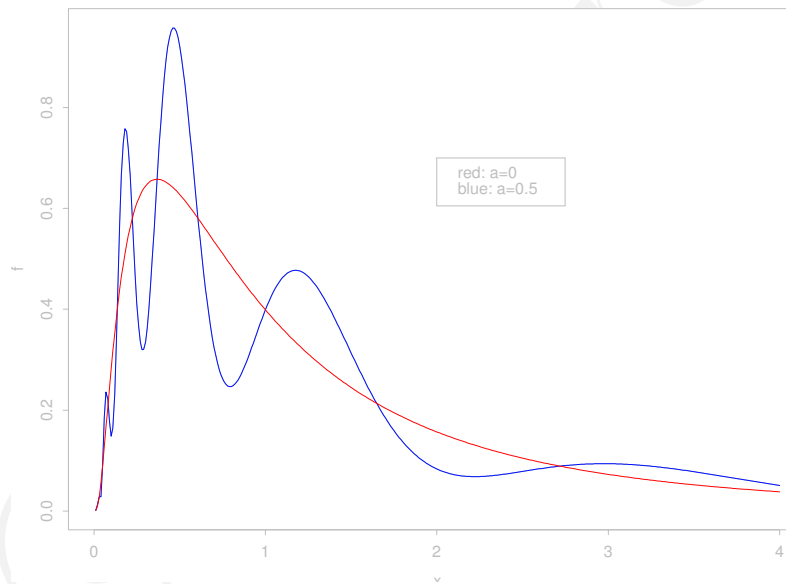
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One can compute (though it is not fun), the moments

$$\begin{aligned} \mathbb{E}[X] &= \sqrt{e}, & \mathbb{E}[X^2] &= e^2, & \mathbb{E}[X^3] &= e^{9/2}, \\ \mathbb{E}[X^4] &= e^8, & \mathbb{E}[X^5] &= e^{25/2}, & \mathbb{E}[X^6] &= e^{18}, & \dots \end{aligned}$$

Note that these moments do not depend on the parameter a ! (This continues for the infinite sequence.) Therefore we have (many!) different distributions, for example $f_{a_1}(x)$ and $f_{a_2}(x)$ where $a_1 \neq a_2$ with the same infinite sequence of moments.

Figure 1.1: Densities for Two Distributions with the Same Infinite Sequence of Moments



Question:

Okay, well is there any moment criterion for identifying distributions that would ensure that two distributions are identical?

Answer:

Yes! If random variables X and Y both have moment generating functions $M_X(t)$ and $M_Y(t)$ that exist in some neighborhood of zero and if $M_X(t) = M_Y(t)$ for all t in this neighborhood, then X and Y have the same distributions!

We will prove the claim here in a simplified setting. The general proof of this can be found in Feller (An Introduction to Probability Theory and Its Applications, Vol. 2, 1971, pg 233). It is an inversion problem involving Laplace transform theory. (Did you notice that the mgf bears a striking resemblance to the Laplace transform?)

Proof of a special case:

Suppose that X and Y are random variables both taking only possible values in $\{0, 1, 2, \dots, n\}$.

Further, suppose that X and Y have the same mgf for all t :

$$\sum_{x=0}^n e^{tx} f_X(x) = \sum_{y=0}^n e^{ty} f_Y(y).$$

For simplicity, we will let $s = e^t$ and we will define $c_i = f_X(i) - f_Y(i)$ for $i = 0, 1, \dots, n$.

Now

$$\begin{aligned} \sum_{x=0}^n e^{tx} f_X(x) - \sum_{y=0}^n e^{ty} f_Y(y) &= 0 \\ \Downarrow \\ \sum_{x=0}^n s^x f_X(x) - \sum_{y=0}^n s^y f_Y(y) &= 0 \\ \Downarrow \\ \sum_{x=0}^n s^x f_X(x) - \sum_{x=0}^n s^x f_Y(x) &= 0 \end{aligned}$$

$$\Downarrow$$

$$\sum_{x=0}^n s^x [f_X(x) - f_Y(x)] = 0$$

$$\Downarrow$$

$$\sum_{x=0}^n s^x c_x = 0 \quad \forall s > 0$$

The above is simply a polynomial in s with coefficients $c_0, c_1, \dots, c_n$. The only way it can be zero for all values of s is if $c_0 = c_1 = \dots = c_n = 0$.

So, we have that

$$0 = c_i = f_X(i) - f_Y(i) \quad \text{for } i = 0, 1, \dots, n.$$

Therefore

$$f_X(i) = f_Y(i) \quad \text{for } i = 0, 1, \dots, n.$$

In other words the density functions for X and Y are exactly the same. In other other words, X and Y have the same distributions!

1.6.6 Sums of iid Random Variables

Let $X_1, X_2, \dots, X_n$ be iid random variables from any distribution. Let $Y = \sum_{i=1}^n X_i$.

Since all of the X_i share the same pdf, let's call it f , the expectations $\mathbb{E}[e^{tX_i}]$ and $\mathbb{E}[e^{tX_j}]$ are the same. They will both be the sum or integral of e^{tx} times the pdf $f(x)$. Thus, by going through the same calculations as in the Poisson example above, it is easy to see that

$$M_Y(t) = [M_{X_1}(t)]^n.$$

This will be so useful that it deserves a box.

If $X_1, X_2, \dots, X_n$ be iid random variables from a distribution with moment generating function $M_X(t)$. The sum $Y = \sum_{i=1}^n X_i$ has moment generating function

$$M_Y(t) = [M_X(t)]^n.$$

Here, X is used to denote any of the X_i .

Example:

Suppose that $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} \text{Poisson}(\lambda)$. What is the distribution of $Y = \sum_{i=1}^n X_i$?

$$M_Y(t) = [M_X(t)]^n = [\exp[\lambda(e^t - 1)]]^n = \exp[\lambda n(e^t - 1)],$$

which is the mgf for the Poisson distribution with rate λn . So, we know that $Y \sim \text{Poisson}(\lambda n)$. □

For our last example, we now consider one particular sum of random variables that we will see and use quite often.

Example:

Let $X_1, X_2, \dots, X_n$ be a random sample from the exponential distribution with rate λ . That is, let $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} \exp(\text{rate} = \lambda)$. What is the distribution of $Y = \sum_{i=1}^n X_i$?

$$M_Y(t) = [M_X(t)]^n = \left(\frac{\lambda}{\lambda - t} \right)^n$$

Comparing this to moment generating functions from the table of distributions in Appendix A, we see that

$$Y \sim \Gamma(\lambda, n).$$

□

Chapter 1 Exercises

1. Consider a sequence of independent trials where each trial can result in one of two possible outcomes: success or failure. Suppose that the probability of success on any one trial is p for some $0 \leq p \leq 1$. Recall that, in Chapter 0, we defined the geometric random variable as counting the number of trials until the first success.

Now, let X be the number of trials until the r th success where r is some positive integer. Then X is said to be a “negative binomial” random variable. Just as we had two different types of geometric distributions, we may define two different types of negative binomial distributions.

- (a) Find the correct pdf on your table of distributions. Then argue how to derive this pdf from the scenario described above.
 - (b) Find the distribution of $Y = X - r$. (Name it!) What is the interpretation of Y in terms of the success/failure experiment?
2. Let $X \sim \Gamma(\alpha, \beta)$. Find $E[X]$ by “integrating without integrating”.
 3. Let g be a continuous, invertible, and increasing function. Show that g^{-1} is also increasing.
 4. Let $X \sim \text{unif}(0, 1)$, find the pdf of $Y = e^X$.
 5. Let $X \sim \exp(\text{rate} = \lambda)$. Find the distribution of $Y = e^{-X}$. (Name it!)
 6. Let X be a random variable with the gamma distribution with parameters α and β , (ie: $X \sim \Gamma(\alpha, \beta)$).
 - (a) Let $c > 0$ be a constant. Find the distribution of $Y = cX$. (Name it!)
 - (b) Now consider a random sample, $X_1, X_2, \dots, X_n$, gamma distribution with parameters α and β , (ie: $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} \Gamma(\alpha, \beta)$). Find the distribution of $Y = \sum_{i=1}^n X_i$. (Name it!)
 7. Suppose that X has a (continuous) Pareto distribution with parameter γ . (i.e.: $X \sim \text{Pareto}(\gamma)$) Find the distribution of $Y = \ln(X + 1)$. (Name it!)
 8. Suppose that X has a beta distribution, $X \sim \text{beta}(a, b)$.

- (a) Find the pdf for $Y = -\ln X$.
 - (b) If a is a positive integer and if $b = 1$, what is the distribution of Y . (Name it!)
9. Let X have the uniform distribution over the interval $(-\pi/2, \pi/2)$. Show that $Y = \tan X$ has a Cauchy distribution.
10. A signal is sent to a rotating antenna, and the angle of the antenna at the time the signal is received can be assumed to be uniformly distributed from 0 to 2π , $\Theta \sim \text{unif}(0, 2\pi)$. The signal can be received if $Y = \tan \Theta > y_0$ for some fixed y_0 . (For example, $y \geq 1$ corresponds to the angles $45^\circ < \Theta < 90^\circ$ and $225^\circ < \Theta < 270^\circ$.)
Find the distribution of Y . (Name it!)
11. Let $X \sim \text{unif}(0, 1)$.
- (a) Find the distribution of $Y = -\ln X$.
 - (b) Find a transformation $Y = g(X)$ so that $Y \sim \exp(\text{rate} = \lambda)$.
12. Let X_1 and X_2 have the joint pdf
- $$f_{X_1, X_2}(x_1, x_2) = 2e^{-x_1 - x_2} I_{(0, x_2)}(x_1) I_{(0, \infty)}(x_2).$$
- Find the joint pdf of $Y_1 = 2X_1$ and $Y_2 = X_2 - X_1$, and argue that Y_1 and Y_2 are independent.
13. Let $X_1, X_2 \sim \text{iid exp}(\text{rate} = 1)$. What is the pdf for X_1/X_2 ? (The name of the resulting distribution is the “F distribution”.)
14. Suppose that $X_1, X_2 \stackrel{\text{iid}}{\sim} N(0, 1)$. Find the distribution of X_1/X_2 . (Name it!)
15. Let $X_1, X_2 \stackrel{\text{iid}}{\sim} \exp(\text{rate} = \lambda)$. Find the distribution of $Y = \frac{X_1}{X_1 + X_2}$. (Name it!)
16. Suppose that $X_1, X_2, \dots, X_n$ is a random sample from the uniform distribution over the interval $(0, 1)$.
- (a) Find the distribution of $\min(X_1, X_2, \dots, X_n)$. (Name it!)
 - (b) Find the distribution of $\max(X_1, X_2, \dots, X_n)$. (Name it!)
17. Let X and Y be independent random variables with cdfs F_X and F_Y , respectively.

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- (a) Let $Z = \max(X, Y)$. Find the cdf for Z in terms of the cdfs for X and Y .
- (b) Let $Z = \min(X, Y)$. Find the cdf for Z in terms of the cdfs for X and Y .

18. Show the following relationship between the beta and gamma functions

$$\mathcal{B}(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

Here, $a, b > 0$ are fixed parameters.

19. Let X be a continuous random variable with pdf $f_X(x)$. Let g be a continuous, increasing, invertible function. We have seen that the pdf for Y is given by

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy}g^{-1}(y).$$

Show that this is a valid pdf.

20. Suppose that X is a continuous random variable with pdf $f(x)$. Let $Y = X^2$. (Note that this is not a one-to-one, invertible transformation.)

Find an expression for the pdf of Y in terms of the pdf of X .

21. Suppose that X_1 and X_2 are independent random variables and that $Y_1 = g_1(X_1)$ and $Y_2 = g_2(X_2)$. Then Y_1 and Y_2 are independent. Sounds reasonable yes?

Prove this in the case that X_1 and X_2 are continuous and g_1 and g_2 are invertible.

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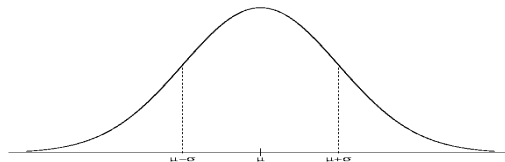
Chapter 2

Qualities of Estimators: Defining Good, Better, and Best

If you knew any continuous distribution before coming into this course, it was probably the **normal distribution**, also known as the **Gaussian distribution**. It's pdf is the “bell curve” that we all know and ~~over~~use love. It is parameterized by the mean μ and variance σ^2 for X , and we write $X \sim N(\mu, \sigma^2)$. The pdf is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2},$$

which looks like



If you do a little Calculus, you'll see that the points of inflection, where the pdf switches concavity, are precisely σ units away from the mean μ .

Note that there is no indicator because this pdf is defined for all $-\infty < x < \infty$. Thus, as there is no “and zero otherwise”, there is no need to describe

the support with an indicator. If we did, it would be $I_{(-\infty),\infty}(x)$, which is always 1. For fun, why don't you multiply every function you write down today by 1!

Estimation:

Let $X_1, X_2, \dots, X_n$ be a random sample from the $N(\mu, 1)$ distribution. If n is large and if you make a histogram of your sampled values, you should see the histogram roughly matching up with the pdf for the $N(\mu, 1)$ distribution. In particular, you will see the majority of sampled values towards the center (which is μ) and not so many out “in the tails’.”

Question: How might you estimate the mean μ ?

Answer: The most “natural” answer would be to use the **sample mean**:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

The true mean μ for the distribution is a probability weighted average. The sample mean is the average of the sampled values and it already has the “probability weighted” part “built in” since the values are falling mostly in the high probability regions and not so much in the low probability regions.

We will use the notation $\hat{\mu}$ to denote an estimator for μ . So, we write

$$\hat{\mu} = \bar{X}.$$

Note that the X in “ X bar” is capitalized. At this point we are considering it as a random variable. We are **going**, in the **future**, to sample values $X_1, X_2, \dots, X_n$ from this normal distribution and then we are **going** to average them. Once we do, we will have observations $x_1, x_2, \dots, x_n$ which have sample mean $\bar{x}$. The word “estimator” is used to refer to the random variable $\bar{X}$ while the word “estimate” is used to refer to the actual observation $\bar{x}$.

Estimating the mean μ with the sample mean $\bar{X}$ seems like the “obvious” thing to do. However, the following questions arise.

1. How “good” is this estimator? What does “good” even mean?

2. Can we do better?
3. How would you go about estimating parameters with a less obvious interpretation, such as the α from the $\Gamma(\alpha, \beta)$ distribution?

Before we can propose good estimators for parameters of various distributions, we need to be able to talk about several measures of the quality of an estimator so that we know where to aim.

2.1 Notation, Statistics, and Unbiasedness

We will use θ to denote a general parameter. It might represent μ in a normal distribution, λ in the exponential distribution, β in a gamma distribution, or a number of other things. It might even represent a vector of parameters such as $\theta = (\mu, \sigma^2)$ or $\theta = (\alpha, \beta)$.

From now on, we really want to emphasize that our pdfs have parameters. So, instead of writing $f(x)$, we will write $f(x; \theta)$. The semicolon will be used to separate variables and parameters. For example, when referring to a joint pdf for $X_1, X_2, \dots, X_n$ with parameters α and β , we might write $f(x_1, x_2, \dots, x_n; \alpha, \beta)$.

Let $X_1, X_2, \dots, X_n$ be a random sample from the distribution with pdf $f(x; \theta)$.

The Goal:

The goal is to estimate the parameter θ or some function of the parameter $\tau(\theta)$ based on what we see in the random sample $X_1, X_2, \dots, X_n$.

Our estimator will be a **statistic** which is defined as some function of the data. A statistic is a random variable that we will denote

$$T = t(X_1, X_2, \dots, X_n) = t(\vec{X}).$$

When estimating $\tau(\theta)$ using the statistic $T = t(\vec{X})$, one thing we might want is for the estimator to be “on average” correct.

We say that T is an **unbiased** estimator for $\tau(\theta)$ if

$$\mathbb{E}[T] = \tau(\theta).$$

If you are not completely comfortable with properties of expectations and with computing variances, now would be a good time to go back and review Section 0.10.

Example:

Let $X_1, X_2, \dots, X_n$ be a random sample from any distribution with mean μ and variance σ^2 .

(Note that μ and σ^2 denote a generic mean and generic variance. We are not necessarily talking about the normal distribution, for example, where the symbols μ and σ^2 are already used. Suppose that the random sample is from the $\Gamma(\alpha, \beta)$ distribution. Then $\mu = \alpha/\beta$ and $\sigma^2 = \alpha/\beta^2$.)

Suppose we want to estimate the mean μ with the sample mean $\bar{X}$. This is a new random variable with its own distribution and own properties! Its mean is

$$\begin{aligned} \mu_{\bar{X}} &= \mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] \\ &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] \stackrel{\text{ident}}{=} \frac{1}{n} \sum_{i=1}^n \mu \\ &= \frac{1}{n} n\mu = \mu \end{aligned}$$

This is true for the sample mean for a random sample from any distribution!

Let $X_1, X_2, \dots, X_n$ be a random sample from any distribution with mean μ . Then

$$\mathbb{E}[\bar{X}] = \mu.$$

What can we say about the variance of $\bar{X}$ in terms of the original variance for the distribution? Variance is a measure of spread. We see that, when using $\bar{X}$ to estimate μ , on average we will be correct. However, if this estimator has a large variance, we will see wild swings between values if we take many samples. Often you only get one sample to work with, so this

“high variability” isn’t good! For example, suppose the true (but unknown to you) mean is $\mu = 15.3$. Suppose you estimate it with some estimator $\hat{\mu}$. Would you rather this estimator produce values like $-117.8, 124.29, 8.55$, and so on, that, when themselves averaged give you 15.3 or would you rather it produced values like $15.1, 15.8, 15.4$? Remember, you only get to see one estimate so, even though the first estimator gives you the correct value on average, the second estimator is much more desirable! In fact, you might actually prefer a little “bias” in your estimator if it is “almost unbiased” and has small variance.

Example:

Let $X_1, X_2, \dots, X_n$ be a random sample from any distribution with mean μ and variance σ^2 .

Suppose, as in the previous example, we chose to estimate μ with $\bar{X}$. What is the variance of this estimator?

$$\begin{aligned}\sigma_{\bar{X}}^2 &= \text{Var}[\bar{X}] = \text{Var}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] \\ &= \frac{1}{n^2} \text{Var}\left[\sum_{i=1}^n X_i\right] \stackrel{\text{indep}}{=} \frac{1}{n^2} \sum_{i=1}^n \text{Var}[X_i] \\ &\stackrel{\text{ident}}{=} \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{1}{n^2} n \sigma^2 = \frac{\sigma^2}{n}\end{aligned}$$

So, no matter what distribution we are considering, $\bar{X}$ is always an unbiased estimator for the mean μ of the distribution. Also, the variability of $\bar{X}$ is smaller than the variability of a single X_i . For example, scores on Calc I exams tend to vary between single digit scores and perfect scores, but if you looked at the averages for each recitation section, you wouldn’t expect to see such variability!

Note that we have derived an expression for the variance of the sample mean for a random sample from any distribution!

Let $X_1, X_2, \dots, X_n$ be a random sample from any distribution with variance σ^2 . Then

$$\text{Var}[\bar{X}] = \frac{\sigma^2}{n}.$$

2.2 Mean Squared Error and Bias

We know that if $\hat{\theta}$ is an unbiased estimator of θ then $E[\hat{\theta}] = \theta$.

The **bias** of an estimator $\hat{\theta}$ of θ is denoted and defined as

$$B(\hat{\theta}) = E[\hat{\theta}] - \theta.$$

The **mean square error** (MSE) of the estimator $\hat{\theta}$ of θ is

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2].$$

By the definitions we can see the following

If $\hat{\theta}$ is “unbiased for θ ”, then

$$MSE(\hat{\theta}) = Var[\hat{\theta}].$$

In general though, MSE and variance are related as follows.

$$\begin{aligned} Var[\hat{\theta}] &= E[(\hat{\theta} - E[\hat{\theta}])^2] = E[(\hat{\theta} - \theta + \theta - E[\hat{\theta}])^2] \\ &= E[(\hat{\theta} - \theta - B(\hat{\theta}))^2] \end{aligned}$$

Squaring that out and running the expectation across, but keeping the $\hat{\theta} - \theta$ together gives

$$\begin{aligned} Var[\hat{\theta}] &= E[(\hat{\theta} - \theta)^2] - 2B(\hat{\theta}) \underbrace{E[\hat{\theta} - \theta]}_{\substack{E[\hat{\theta}] - \theta \\ || \\ B(\hat{\theta})}} + (B(\hat{\theta}))^2 \\ &= E[(\hat{\theta} - \theta)^2] - (B(\hat{\theta}))^2 \\ &= MSE(\hat{\theta}) - (B(\hat{\theta}))^2 \end{aligned}$$

From this relationship, we can again see that if $\hat{\theta}$ is unbiased for θ , $Var[\hat{\theta}] = MSE(\hat{\theta})$.

An Estimation Example:

Let $X_1, X_2, \dots, X_n$ be a random sample from the $exp(rate = \lambda)$ distribution. Assume that $n \geq 3$.

We will consider estimating $\theta := 1/\lambda$ and then λ directly.

Since $\theta = 1/\lambda$ is the mean for this distribution, we know that $\hat{\theta} = \bar{X}$ is an unbiased estimator for θ . It's MSE is

$$MSE(\hat{\theta}) \stackrel{\text{unbiased}}{=} Var[\hat{\theta}] = Var[\bar{X}] = \frac{Var[X_1]}{n} = \frac{(1/\lambda^2)}{n} = \frac{1}{n\lambda^2}$$

Now consider estimating λ itself. While it is perfectly reasonable to use $1/\bar{X}$, we might not want to do this if, for example, we want an unbiased estimator for λ .

Is $\hat{\lambda} := 1/\bar{X}$ an unbiased estimator of λ ?

$$E[\hat{\lambda}] = E\left[\frac{1}{\bar{X}}\right]$$

but THIS IS IN NO WAY SHAPE OR FORM EQUAL TO

$$\frac{1}{E[\bar{X}]}!!!!$$

Instead,

$$E[\hat{\lambda}] = E\left[\frac{1}{\bar{X}}\right] = E\left[\frac{n}{\sum X_i}\right] = nE\left[\frac{1}{\sum X_i}\right] = nE\left[\frac{1}{Y}\right]$$

where $Y = \sum_{i=1}^n X_i \sim \Gamma(n, \lambda)$.

So,

$$\begin{aligned}
 \mathbb{E}\left[\frac{1}{Y}\right] &= \int_{-\infty}^{\infty} \frac{1}{y} f_Y(y) dy \\
 &= \int_0^{\infty} \frac{1}{y} \frac{1}{\Gamma(n)} \lambda^n y^{n-1} e^{-\lambda y} dy \\
 &= \int_0^{\infty} \frac{1}{\Gamma(n)} \lambda^n \underbrace{y^{n-2} e^{-\lambda y}}_{\substack{\text{like a} \\ \Gamma(n-1, \lambda) \\ \text{pdf}}} dy \\
 &= \frac{\Gamma(n-1)}{\Gamma(n)} \lambda \underbrace{\int_0^{\infty} \frac{1}{\Gamma(n-1)} \lambda^{n-1} y^{n-2} e^{-\lambda y} dy}_1 \\
 &= \frac{\Gamma(n-1)}{\Gamma(n)} \lambda = \frac{(n-2)!}{(n-1)!} \lambda = \frac{1}{n-1} \lambda
 \end{aligned}$$

So,

$$\mathbb{E}[\hat{\lambda}] = n \mathbb{E}\left[\frac{1}{Y}\right] = \frac{n}{n-1} \lambda.$$

Conclusion: This is not λ , so $\hat{\lambda} = 1/\bar{X}$ is not an unbiased estimator of λ .

Let us now try to find an unbiased estimator of λ .

$\hat{\lambda} = 1/\bar{X}$ came close. If we put $(n-1)/n$ in front of it, then

$$\mathbb{E}\left[\frac{n-1}{n} \frac{1}{\bar{X}}\right] = \frac{n-1}{n} \mathbb{E}\left[\frac{1}{\bar{X}}\right] = \frac{n-1}{n} \frac{n}{n-1} \lambda = \lambda.$$

Thus

$$\hat{\lambda}_2 := \frac{n-1}{n} \frac{1}{\bar{X}} = \frac{n-1}{\sum X_i}$$

is an unbiased estimator of λ !

Comparison of MSEs:

Let's find the MSE of the original $\hat{\lambda}$.

$$MSE(\hat{\lambda}) = \mathbb{E}[(\hat{\lambda} - \lambda)^2] = \mathbb{E}\left[\left(\frac{1}{\bar{X}} - \lambda\right)^2\right] = \mathbb{E}\left[\left(\frac{n}{Y} - \lambda\right)^2\right]$$

where $Y \sim \Gamma(n, \lambda)$.

So,

$$\begin{aligned} MSE(\hat{\lambda}) &= E \left[\left(\frac{n}{Y} - \lambda \right)^2 \right] \\ &= n^2 E \left[\frac{1}{Y^2} \right] - 2n\lambda E \left[\frac{1}{Y} \right] + \lambda^2 \end{aligned}$$

We already know that $E[1/Y] = \lambda/(n-1)$. We still need to find $E[1/Y^2]$. Since $Y \sim \Gamma(n, \lambda)$,

$$\begin{aligned} E \left[\frac{1}{Y^2} \right] &= \int_0^\infty \frac{1}{y^2} \cdot \frac{1}{\Gamma(n)} \lambda^n y^{n-1} e^{-\lambda y} dy \\ &= \int_0^\infty \frac{1}{\Gamma(n)} \lambda^n y^{n-3} e^{-\lambda y} dy \end{aligned}$$

That $y^{n-3}e^{-\lambda y}$ part of the integrand looks like a $\Gamma(n-2, \lambda)$ pdf. In order for that pdf to be complete, we also need $\frac{1}{\Gamma(n-2)}\lambda^{n-2}$. So, we rewrite things:

$$\begin{aligned} E \left[\frac{1}{Y^2} \right] &= \frac{\Gamma(n-2)}{\Gamma(n)} \lambda^2 \underbrace{\int_0^\infty \frac{1}{\Gamma(n-2)} \lambda^{n-2} y^{n-3} e^{-\lambda y} dy}_{\text{integral of a pdf}} \\ &= \frac{\Gamma(n-2)}{\Gamma(n)} \lambda^2 \cdot 1 = \frac{1}{(n-1)(n-2)} \lambda^2. \end{aligned}$$

Putting all of the pieces together (check this), we get

$$\boxed{MSE(\hat{\lambda}) = \frac{n+2}{(n-1)(n-2)} \lambda^2.}$$

We will now compute the MSE for the unbiased estimator $\hat{\lambda}_2$. Since $\hat{\lambda}_2$ is an unbiased estimator of λ , its MSE is simply its variance:

$$MSE(\hat{\lambda}_2) \stackrel{\text{unbiased}}{=} Var[\hat{\lambda}_2] = Var \left[\frac{n-1}{\sum X_i} \right] = (n-1)^2 Var \left[\frac{1}{Y} \right]$$

where $Y \sim \Gamma(n, \lambda)$.

Note that

$$\begin{aligned} Var \left[\frac{1}{Y} \right] &= E \left[\frac{1}{Y^2} \right] - \left(E \left[\frac{1}{Y} \right] \right)^2 \\ &= \frac{1}{(n-1)(n-2)} \lambda^2 - \left(\frac{1}{n-1} \lambda \right)^2 \\ &= \frac{1}{(n-1)^2(n-2)} \lambda^2. \end{aligned}$$

So,

$$MSE(\hat{\lambda}_2) = (n-1)^2 \text{Var} \left[\frac{1}{Y} \right] = (n-1)^2 \frac{1}{(n-1)^2(n-2)} \lambda^2 = \frac{1}{n-2} \lambda^2.$$

From two unbiased estimators of a parameter, you should choose the one with smaller variance. When comparing a biased estimator with an unbiased estimator, as in this case, it makes more sense to look at MSE. The variance of an estimator represents its spread about its mean while the MSE represents its spread about the thing you are trying to estimate!

We still don't know how to really come up with estimators of parameters. At this point we are sort of guessing and then evaluating that guess. (i.e., Is it unbiased? Does it have small variance? Small MSE?) Eventually we will learn systematic approaches for coming up with estimators. For now, we are still just going to talk about how to evaluate the quality of estimators so that we know what to aim for when talking about finding estimators.

Often we are interested in “large sample properties” of estimators. For example, when estimating the mean μ of a distribution, it seems natural to assume that the sample mean $\bar{X}$ is not only a good idea, but will become an even better estimator of the true mean for larger and larger samples. It's time to talk about limits for random variables!

In general, given a random sample $X_1, X_2, \dots, X_n$, an estimator $\hat{\theta}$ for a parameter θ will also be based on the sample size. (i.e., $\hat{\theta} = \hat{\theta}_n$) It would be nice to be able to say that $\hat{\theta}_n$ converges, in some sense, to the true value θ . Estimators are not numbers, however. They are random variables or “potential numbers”. In order to talk about convergence, we need to use probability somehow. There are several ways to do this.

2.3 Convergence in Probability

Definition: A sequence of random variables $\{X_n\}$ **converges in probability** to a random variable X if, for and, $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|X_n - X| > \varepsilon) = 0.$$

(Equivalently if $\lim_{n \rightarrow \infty} P(|X_n - X| \leq \varepsilon) = 1$.)

We write $X_n \xrightarrow{P} X$.

Note:

1. One can talk about convergence in probability of a sequence of random variables to a constant. A constant is just a very boring random variable. The constant “3” can be thought of as a random variable X where $X = 3$ with probability 1.
2. In this course, the sequence of random variables will usually be a sequence of estimators $\hat{\theta} = \hat{\theta}_n$ of a parameter θ that we hope will converge in probability to the constant θ .

In order to show convergence in probability, you might find the following inequalities useful.

2.3.1 Markov’s Inequality

Markov’s inequality is as follows.

Markov’s Inequality

For a random variable X and any numbers $r, c > 0$,

$$P(|X| \geq c) \leq \frac{E[|X|^r]}{c^r}.$$

It is sometimes stated in terms of a nonnegative random variable and thus without absolute values

In order to prove Markov's inequality, we will first prove something more general.

A “General” (Unnamed) Inequality

If X is a random variable and $g(x)$ is a non-negative real-valued function, then for any $c > 0$,

$$P(g(X) \geq c) \leq \frac{\mathbf{E}[g(X)]}{c}.$$

Once we prove this general inequality, we can establish Markov's inequality as follows.

Since $r > 0$,

$$P(|X| \geq c) = P(|X|^r \geq c^r).$$

Using $g(x) = |x|^r$ in the generalized inequality, and replacing c with c^r , we know that

$$P(|X|^r \geq c^r) \leq \frac{\mathbf{E}[|X|^r]}{c^r}.$$

So,

$$P(|X| \geq c) \leq \frac{\mathbf{E}[|X|^r]}{c^r}$$

which is Markov's inequality!

It remains to prove the generalized inequality. It will be useful to use the notation

$$\int_A f(x) dx$$

to represent the integral over all values in the set A . For example,

$$\int_2^5 f(x) dx = \int_{\{x: 2 \leq x \leq 5\}} f(x) dx.$$

The purpose of this notation is to allow us to easily and compactly denote more complicated regions of integration. For example, if we want to integrate a function $g(x)$ over all x in the region where $g(x) > 5$, we can now write

$$\int_{\{x: g(x) > 5\}} g(x) dx.$$

Note that if f is the pdf for a random variable X ,

$$\int_{\{x:2 \leq x \leq 5\}} f(x) dx = P(2 \leq X \leq 5).$$

Proof of General Inequality:

Let $f(x)$ be the pdf for X . Then,

$$\begin{aligned} E[g(X)] &= \int_{-\infty}^{\infty} g(x)f(x) dx \\ &= \int_{\{x:g(x) \geq c\}} g(x)f(x) dx + \int_{\{x:g(x) < c\}} g(x)f(x) dx \\ &\geq \int_{\{x:g(x) \geq c\}} g(x)f(x) dx \end{aligned}$$

since both f and g are non-negative.

Since $g(x) \geq c$ on this region of integration,

$$\begin{aligned} E[g(X)] &\geq \int_{\{x:g(x) \geq c\}} g(x)f(x) dx \geq \int_{\{x:g(x) \geq c\}} cf(x) dx \\ &= c \int_{\{x:g(x) \geq c\}} f(x) dx = cP(g(X) \geq c) \end{aligned}$$

as desired! □

2.3.2 Chebyshev's Inequality

Chebyshev's Inequality Let X be a random variable with mean μ and variance $\sigma^2 < \infty$. For any $k > 0$,

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}.$$

Equivalently,

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

($\sigma := \sqrt{\sigma^2} > 0$ is the “standard deviation” of X .)

Proof of Chebyshev's Inequality:

Let $g(x) = (x - \mu)^2$. Using the first inequality again with $c = k^2\sigma^2$ gives us

$$P((X - \mu)^2 \geq k^2\sigma^2) \leq \frac{E[(X - \mu)^2]}{k^2\sigma^2} = \frac{\sigma^2}{k^2\sigma^2} = \frac{1}{k^2}.$$

Note that the left hand side is equivalent to $P(|X - \mu| \geq k\sigma)$. $\square$

2.3.3 The Sample Mean and Convergence in Probability

Let $X_1, X_2, \dots, X_n$ be a random sample from any distribution with mean μ and variance $\sigma^2 < \infty$. Then

$$\overline{X} \stackrel{\text{notation}}{=} \overline{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} \mu.$$

Proof:

Recall Chebyshev's Inequality. For any $k > 0$, $P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$.

Applying Chebyshev's Inequality to the random variable $\overline{X}$, we get

$$P(|\overline{X} - \mu_{\overline{X}}| < k\sigma_{\overline{X}}) \geq 1 - \frac{1}{k^2}.$$

But, $\mu_{\overline{X}} = \mu$ and $\sigma_{\overline{X}}^2 = \sigma^2/n$, which implies that $\sigma_{\overline{X}} = \sigma/\sqrt{n}$. So, we have

$$P(|\overline{X} - \mu| < k\sigma/\sqrt{n}) \geq 1 - \frac{1}{k^2} \quad (2.1)$$

for any $k > 0$.

Let $\varepsilon > 0$. Choose k such that $k\sigma/\sqrt{n} = \varepsilon$. i.e., Choose $k = \varepsilon\sqrt{n}/\sigma$. Then (2.1) becomes

$$P(|\overline{X} - \mu| < \varepsilon) \geq 1 - \frac{\sigma^2}{\varepsilon^2 n}.$$

Therefore,

$$\lim_{n \rightarrow \infty} P(|\overline{X} - \mu| < \varepsilon) \geq \lim_{n \rightarrow \infty} \left[1 - \frac{\sigma^2}{\varepsilon^2 n} \right] = 1 - 0 = 1.$$

Since probabilities can't get greater than 1, that limit on the left-hand side must equal 1.

So, we have shown that $\bar{X} \xrightarrow{P} \mu$. □

Example:

Let $X_1, X_2, \dots, X_n$ be a random sample from the exponential distribution with rate λ , then

$$\bar{X} \xrightarrow{P} \frac{1}{\lambda}.$$

Example:

If $X_1, X_2, \dots, X_n$ is a random sample from the $\Gamma(\alpha, \beta)$ distribution, then

$$\bar{X} \xrightarrow{P} \frac{\alpha}{\beta}.$$

2.3.4 Consistent Estimators

Let $\hat{\theta}_n$ be an estimator of θ . $\hat{\theta}_n$ is a **consistent estimator** of θ if

$$\hat{\theta}_n \xrightarrow{P} \theta.$$

Here is a most useful theorem!

An unbiased estimator $\hat{\theta}_n$ of θ is a consistent estimator of θ if $\lim_{n \rightarrow \infty} \text{Var}[\hat{\theta}_n] = 0$.

Proof: By Chebyshev's Inequality, we have, for any $k > 0$

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}.$$

In this theorem, our random variable is $\hat{\theta}_n$ and, since it's unbiased, its mean is θ . Let us use σ_n^2 to denote its variance: $\sigma_n^2 = \text{Var}[\hat{\theta}_n]$. Chebyshev's Inequality becomes

$$P(|\hat{\theta}_n - \theta| < k\sigma_n) \geq 1 - \frac{1}{k^2}.$$

Take any $\varepsilon > 0$. Setting $k\sigma_n = \varepsilon$, we see that, by choosing $k = \varepsilon/\sigma_n$, we have

$$P(|\hat{\theta}_n - \theta| < \varepsilon) \geq 1 - \frac{\sigma_n^2}{\varepsilon^2}.$$

Now take the limit as $n \rightarrow \infty$ of both sides. Since are given that that $\lim_{n \rightarrow \infty} \sigma_n^2 = 0$, we get

$$\lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta| < \varepsilon) \geq 1.$$

Since that probability can't get greater than 1, we have shown that

$$\lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta| < \varepsilon) = 1.$$

Thus, $\hat{\theta}_n \xrightarrow{P} \theta$ so $\hat{\theta}_n$ is a consistent estimator of θ . □

2.3.5 Things About Convergence in Probability That Will Not Surprise You

Suppose that $\{X_n\}$ and $\{Y_n\}$ are sequences of random variables with $X_n \xrightarrow{P} a$ and $Y_n \xrightarrow{P} b$ for some constants a and b .

Then

1. $X_n + Y_n \xrightarrow{P} a + b$
2. $X_n Y_n \xrightarrow{P} ab$
3. $X_n / Y_n \xrightarrow{P} a/b$ (for $b \neq 0$)
4. For a continuous function g , $g(X_n) \xrightarrow{P} g(a)$.

Proof of 1.: We will use the Triangle Inequality: $|a + b| \leq |a| + |b|$.

$$|X_n + Y_n - (a + b)| = |(X_n - a) + (Y_n - b)| \leq |X_n - a| + |Y_n - b|$$

$\Downarrow$

$$P(|X_n + Y_n - (a + b)| > \varepsilon) \leq P(|X_n - a| + |Y_n - b| > \varepsilon) \quad (2.2)$$

Note that

$$P(|X_n - a| + |Y_n - b| > \varepsilon) \leq P(|X_n - a| > \varepsilon/2) + P(|Y_n - b| > \varepsilon/2). \quad (2.3)$$

Aside:

This is because, in order to have $|X_n - a| + |Y_n - b| > \varepsilon$, we must have at least one of the events

$$\{|X_n - a| > \varepsilon/2\}, \quad \{|Y_n - b| > \varepsilon/2\}$$

be true. i.e.,

$$\{|X_n - a| + |Y_n - b| > \varepsilon\} \quad \Rightarrow \quad \{|X_n - a| > \varepsilon/2\} \text{ or } \{|Y_n - b| > \varepsilon/2\} \text{ or both.}$$

Equivalently

$$\{|X_n - a| + |Y_n - b| > \varepsilon\} \subseteq \{|X_n - a| > \varepsilon/2\} \cup \{|Y_n - b| > \varepsilon/2\}. \quad (2.4)$$

Recall that for events A and B ,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \leq P(A) + P(B).$$

So, (2.4) implies that

$$\begin{aligned} P(|X_n - a| + |Y_n - b| > \varepsilon) &\leq P(\{|X_n - a| > \varepsilon/2\} \cup \{|Y_n - b| > \varepsilon/2\}) \\ &\leq P(|X_n - a| > \varepsilon/2) + P(|Y_n - b| > \varepsilon/2). \end{aligned}$$

Finally,

$$\begin{aligned} \lim_{n \rightarrow \infty} P(|X_n + Y_n - (a + b)| > \varepsilon) &\stackrel{(2.2)}{\leq} \lim_{n \rightarrow \infty} P(|X_n - a| + |Y_n - b| > \varepsilon) \\ &\stackrel{(2.3)}{\leq} \lim_{n \rightarrow \infty} [P(|X_n - a| > \varepsilon/2) + P(|Y_n - b| > \varepsilon/2)] \\ &= \lim_{n \rightarrow \infty} P(|X_n - a| > \varepsilon/2) + \lim_{n \rightarrow \infty} P(|Y_n - b| > \varepsilon/2) \\ &= 0 + 0 = 0 \end{aligned}$$

since $X_n \xrightarrow{P} a$ and $Y_n \xrightarrow{P} b$. Since $P(|X_n + Y_n - (a + b)| > \varepsilon)$ can not be less than zero, we have

$$\lim_{n \rightarrow \infty} P(|X_n + Y_n - (a + b)| > \varepsilon) = 0$$

and therefore $X_n + Y_n \xrightarrow{P} a + b$, as desired. $\square$

Proof of 4.: Let $\varepsilon > 0$. g continuous implies that there exists a $\delta > 0$ such that, for all x with $|x - a| < \delta$, we have $|g(x) - g(a)| < \varepsilon$.

Thus,

$$P(|g(X_n) - g(a)| < \varepsilon) \geq P(|X_n - a| < \delta)$$

Since $X_n \xrightarrow{P} a$ as $n \rightarrow \infty$, the right-hand side goes to 1. Hence

$$\lim_{n \rightarrow \infty} P(|g(X_n) - g(a)| < \varepsilon) = 1$$

so $g(X_n) \xrightarrow{P} g(a)$. □

(Note: We didn't need g to be continuous everywhere, only at a !)

Example: If $X_n \xrightarrow{P} b$ and a is a constant, $aX_n \xrightarrow{P} ab$.

This is either a result of part 2 of the Theorem, using the random variable sequence $\{Y_n\}$ with $Y_n = a$ with probability 1 or as a result of part 4 of the Theorem with $g(x) = ax$.

Note: With very minor adjustments, all statements in this documents about convergence in probability to a constant can be replaced with convergence in probability to a random variable. For example, $X_n \xrightarrow{P} X$ and $Y_n \xrightarrow{P} Y \Rightarrow X_n Y_n \xrightarrow{P} XY$.

In this course we are mainly concerned with convergence to constants. (Specifically, estimators of parameters converging to the parameters.)

2.4 Convergence In Distribution

Definition: Let $\{X_n\}$ be a sequence of random variables with cdfs $F_n(x) = P(X_n \leq x)$. Let X be a random variable with cdf $F(x)$. X_n **converges in distribution** to X if

$$\lim_{n \rightarrow \infty} F_n(x) = F(x)$$

for all x where F is continuous.

We write $X_n \xrightarrow{\mathcal{D}} X$.

Other notations are $X_n \xrightarrow{L} X$ as this is sometimes called “convergence in law”, and $X_n \Rightarrow X$. Convergence in distribution is weaker in convergence in probability in the sense that $X_n \xrightarrow{P} X \Rightarrow X_n \xrightarrow{\mathcal{D}} X$ but not necessarily the other way around. We will prove this.

Example:

Let $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} \text{Pareto}(1)$. Investigate the convergence in distribution of $Y_n := nX_{(1)}$ where $X_{(1)} = \min(X_1, X_2, \dots, X_n)$.

The $\text{Pareto}(1)$ pdf is

$$f(x) = \frac{\gamma}{(1+x)^{\gamma+1}} I_{(0,\infty)}(x) \stackrel{\gamma=1}{=} \frac{1}{(1+x)^2} I_{(0,\infty)}(x).$$

The cdf is

$$F(x) = \int_0^x (1+u)^{-2} du = \left. \frac{-1}{1+u} \right|_0^x = 1 - \frac{1}{1+x}$$

We need to find the cdf of Y_n and then let $n \rightarrow \infty$.

$$\begin{aligned} F_{Y_n}(y) &= P(Y_n \leq y) = P(nX_{(1)} \leq y) = P(X_{(1)} \leq y/n) \\ &= 1 - [P(X_1 > y/n)]^n = 1 - \left(\frac{1}{1+y/n} \right)^n \\ &= 1 - (1+y/n)^{-n} \rightarrow 1 - e^{-y} \end{aligned}$$

as $n \rightarrow \infty$.

This is the cdf of the exponential distribution with rate 1. (If you don't recognize it, take the derivative to find the pdf.)

So,

$$Y_n \xrightarrow{\mathcal{D}} Y \sim \exp(\text{rate} = 1).$$

Example:

Let $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} \text{unif}(0, 1)$. Consider $Y_n = X_{(n)} = \max(X_1, X_2, \dots, X_n)$.

The cdf for the $\text{unif}(0, 1)$ distribution is $F(x) = P(X \leq x) = \int_0^x 1 \, du = x$, for $0 \leq x < 1$. Note that the cdf is 0 for $x < 0$ and is 1 for $x \geq 1$.

The cdf of Y_n is

$$\begin{aligned} F_{Y_n}(y) &= P(Y_n \leq y) = P(X_{(n)} \leq y) \\ &= [P(X_1 \leq y)]^n = y^n \end{aligned}$$

for $0 \leq y < 1$.

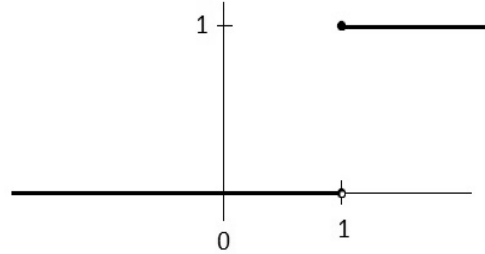
The complete cdf is

$$F_{Y_n}(y) = \begin{cases} 0 & , \quad y < 0 \\ y^n & , \quad 0 \leq y < 1 \\ 1 & , \quad y \geq 1 \end{cases}$$

Letting $n \rightarrow \infty$, we get the limiting cdf

$$F(y) = \begin{cases} 0 & , \quad y < 1 \\ 1 & , \quad y \geq 1 \end{cases}$$

which looks like this



This is the cdf of a random variable that equals 1 with probability 1. We say

$$Y_n \xrightarrow{\mathcal{D}} Y \quad \text{where } Y = 1 \text{ w.p. } 1$$

or simply

$$Y_n \xrightarrow{\mathcal{D}} 1.$$

Important Note: Cdfs are always right-continuous functions. This means that if and when you have a jump, you will have a “hole” on the lower piece and a “filled in point” on the higher piece. The proof of this is not that difficult but requires a lot more “machinery” in place than we have here. Sometimes, when trying to show convergence in distribution, you will end up with a left-continuous function. This is not a cdf but it is the same as the function you would get if you switched out the point and the hole which would be a cdf. Fortunately, to show convergence in distribution, we only need to show convergence of the sequence of cdfs to a cdf F “for all x where F is continuous.” In other words, we don’t care about these particular trouble points!

2.4.1 Convergence in Probability is Stronger

Theorem: $X_n \xrightarrow{P} X \Rightarrow X_n \xrightarrow{\mathcal{D}} X$

Proof: Let $F_n(x) = P(X_n \leq x)$. Let $F(x) = P(X \leq x)$. Let $\varepsilon > 0$. Let x be a point of continuity of F .

Note that

$$F_n(x) = P(X_n \leq x) = P(X_n \leq x, X \leq x + \varepsilon) + P(X_n \leq x, X > x + \varepsilon)$$

Also, note that the “event” that $\{X_n \leq x, X > x + \varepsilon\}$ is contained in the event that $\{|X_n - X| > \varepsilon\}$. That is, if the first event is true then the second is too. (Not convinced? Trying sketching it out on a number line.)

Thus,

$$F_n(x) \leq P(X_n \leq x, X \leq x + \varepsilon) + P(|X_n - X| > \varepsilon).$$

This is “trivially”

$$\leq P(X \leq x + \varepsilon) + P(|X_n - X| > \varepsilon)$$

since we have removed a restriction. So,

$$F_n(x) = F(x + \varepsilon) + P(|X_n - X| > \varepsilon).$$

On the other hand,

$$\begin{aligned} F(x - \varepsilon) &= P(X \leq x - \varepsilon) = P(X \leq x - \varepsilon, X_n \leq x) + P(X \leq x - \varepsilon, X_n > x) \\ &\leq P(X_n \leq x) + P(|X_n - X| > \varepsilon) \\ &= F_n(x) + P(|X_n - X| > \varepsilon). \end{aligned}$$

So, we have

$$F(x - \varepsilon) - P(|X_n - X| > \varepsilon) \leq F_n(x) + F(x + \varepsilon) + P(|X_n - X| > \varepsilon).$$

Letting $n \rightarrow \infty$ all the way across, and using the fact that $X_n \xrightarrow{P} X$, we get

$$F(x - \varepsilon) - 0 \leq \lim_{n \rightarrow \infty} F_n(x) \leq F(x + \varepsilon) + 0$$

or

$$F(x - \varepsilon) \leq \lim_{n \rightarrow \infty} F_n(x) \leq F(x + \varepsilon). \quad (2.5)$$

Notice that I did not write $F(x)$ in the middle— we are not yet sure that $F_n(x)$ has a limit! But.... letting ε become arbitrarily small in (2.5) makes both the right and left-hand sides go to $F(x)$, squeezing in of the limit in the center. So,

$$\lim_{n \rightarrow \infty} F_n(x) = F(x).$$

Therefore, $X_n \xrightarrow{D} X$, as desired! □

Counter-Example:

In general, $X_n \xrightarrow{\mathcal{D}} X \not\Rightarrow X_n \xrightarrow{P} X$. For example, consider $X_1, X_2, \dots$ all iid with distribution

$$P(X_i = 1) = 1/2$$

$$P(X_i = -1) = 1/2$$

Let X be a random variable with the exact same distribution. Clearly (since the cdfs are all the same and are not changing) $X_n \xrightarrow{\mathcal{D}} X$.

However,

$$P(|X_n - X| \geq 2) = 1/2 \not\rightarrow 0$$

so $X_n \not\xrightarrow{P} X$. □

The Reverse is Sometimes True

We have already proven that

$$X_n \xrightarrow{P} X \text{ implies } X_n \xrightarrow{\mathcal{D}} X$$

and we have seen an example showing that

$$X_n \xrightarrow{\mathcal{D}} X \text{ does **not** imply } X_n \xrightarrow{P} X.$$

However, if X is a constant, we do have the reverse implication!

Theorem: Suppose that $X \xrightarrow{\mathcal{D}} c$ where c is a constant. Then $X_n \xrightarrow{P} c$.

Proof: Let $\varepsilon > 0$.

$$\begin{aligned} P(|X_n - c| > \varepsilon) &= P(X_n > c + \varepsilon) + P(X_n < c - \varepsilon) \\ &\leq P(X_n > c + \varepsilon) + P(X_n \leq c - \varepsilon) \\ &= 1 - F_{X_n}(c + \varepsilon) + F_{X_n}(c - \varepsilon) \end{aligned}$$

Since $X_n \xrightarrow{\mathcal{D}} c$, $\lim_{n \rightarrow \infty} F_{X_n}(c + \varepsilon) = F(c + \varepsilon)$ and $\lim_{n \rightarrow \infty} F_{X_n}(c - \varepsilon) = F(c - \varepsilon)$, where $F(x)$ is the cdf of the “constant random variable” $X = c$.

This cdf is

$$F(x) = P(X \leq x) = P(c \leq x) = \begin{cases} 0 & , \text{ if } x < c \\ 1 & , \text{ if } x \geq c \end{cases}$$

So,

$$\begin{aligned} \lim_{n \rightarrow \infty} P(|X_n - c| > \varepsilon) &\leq \lim_{n \rightarrow \infty} [1 - F_{X_n}(c + \varepsilon) + F_{X_n}(c - \varepsilon)] \\ &= 1 - 1 + 0 = 0. \end{aligned}$$

Since a probability can't get less than 0,

$$\lim_{n \rightarrow \infty} P(|X_n - c| > \varepsilon) \leq 0 \quad \Rightarrow \quad \lim_{n \rightarrow \infty} P(|X_n - c| > \varepsilon) = 0$$

so we have $X_n \xrightarrow{P} c$. □

2.4.2 Slutsky's Theorem: Mixing Convergence Types

Here is a Theorem that can be used for mixing convergence types.

Slutsky's Theorem:

Suppose that $X_n \xrightarrow{P} a$ and $Y_n \xrightarrow{d} Y$. Then

1. $X_n + Y_n \xrightarrow{d} a + Y$
2. $X_n Y_n \xrightarrow{d} aY$
3. $Y_n / X_n \xrightarrow{d} Y/a$ (if $a \neq 0$)

It should not be surprising that, in all cases we are only guaranteed the weaker type of convergence!

2.4.3 Convergence in Distribution and Moment Generating Functions

We have already claimed that moment generating functions uniquely determine a distribution. A distribution, for us, is defined by a pdf which we

have then used to define a cdf. So, you might find the following theorem, which we will not prove, highly believable.

Theorem: Let $X_1, X_2, \dots$ be a sequence of random variables.

Let $F_n(x)$ be the cdf of X_n .

Let $M_n(t)$ be the mgf of X_n .

Let X be a random variable with cdf $F(x)$ and mgf $M(t)$.

Then,

$$\lim_{n \rightarrow \infty} M_n(t) = M(t) \quad \Rightarrow \quad \lim_{n \rightarrow \infty} F_n(x) = F(x).$$

Notes:

1. Just for the record, the first limit only needs to hold only for all t in an open interval containing zero. The second limit needs to hold at all points of continuity of F .
2. This result would go in “both directions” if moment generating functions always existed, but they don’t. The mgf is defined as an expectation which is defined by an integral that sometimes doesn’t diverges to ∞ !
3. This is how we prove the Central Limit Theorem!

2.5 The Central Limit Theorem

The Central Limit Theorem (CLT):

Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with mean μ and variance $\sigma^2 < \infty$.

Consider the sequence of random variables $\{Z_n\}$ where

$$Z_n := \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}.$$

Then $Z_n \xrightarrow{\mathcal{D}} Z \sim N(0, 1)$ as $n \rightarrow \infty$.

Note:

- The CLT requires a finite variance. (This implies a finite mean.)
- We will prove the CLT in the case that the mgf exists. (Recall that the mgf is an expected value. We say that an expected value “exists” if it is finite.)
- The CLT can be proved more generally by using “characteristic functions” in place of mgfs. The characteristic function, $\phi(t)$, for a rv X is defined to be: $\phi(t) = E[e^{itX}]$ where i is the imaginary unit. Characteristic functions always exist but since complex variables are not a prerequisite for this course we will only prove the Central Limit Theorem using moment generating functions.

2.5.1 Proof of the Central Limit Theorem

First note that

$$Z_n = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} = \frac{n(\frac{1}{n} \sum_{i=1}^n X_i - \mu)}{\sqrt{n}\sigma} = \frac{(\sum_{i=1}^n X_i - n\mu)}{\sqrt{n}\sigma} = \frac{\sum_{i=1}^n (X_i - \mu)}{\sqrt{n}\sigma}$$

For notational simplicity, let $m(t)$ denote the mgf of $X_i - \mu$ for any particular i in $\{1, \dots, n\}$. Note that $m(0) = 1$, $m'(0) = E[X_i - \mu] = 0$, and $m''(0) = E[(X_i - \mu)^2] = \sigma^2$.

Expanding $m(t)$ about 0 using Taylor’s formula, there exists a ξ (“ksi”), $0 < \xi < t$ such that

$$m(t) = m(0) + m'(0)t + \frac{m''(\xi)t^2}{2}$$

Plugging in $m(0) = 1$ and $m'(0) = E[X - \mu] = 0$, we have

$$m(t) = 1 + \frac{m''(\xi)t^2}{2}$$

and by adding and subtracting $\sigma^2 t^2/2$, this becomes

$$m(t) = 1 + \frac{\sigma^2 t^2}{2} + \frac{(m''(\xi) - \sigma^2)t^2}{2}.$$

Now we are ready to find the mgf of Z_n . Since the rv's $(X_i - \mu)$ are independent, it is easy to find the mgf of the sum $\sum (X_i - \mu)$.

$$\begin{aligned}
 Z_n = \frac{\sum_{i=1}^n (X_i - \mu)}{\sqrt{n}\sigma} &\Rightarrow M_{Z_n}(t) = M_{\sum (X_i - \mu)}\left(\frac{t}{\sqrt{n}\sigma}\right) \\
 &= \prod_{i=1}^n M_{X_i - \mu}\left(\frac{t}{\sqrt{n}\sigma}\right) \\
 &= \left[M_{X_1 - \mu}\left(\frac{t}{\sqrt{n}\sigma}\right)\right]^n \\
 &= \left[m\left(\frac{t}{\sqrt{n}\sigma}\right)\right]^n \\
 &= \left[1 + \frac{\sigma^2 t^2}{2n\sigma^2} + \frac{(m''(\xi) - \sigma^2)t^2}{2n\sigma^2}\right]^n \\
 &= \left[1 + \frac{t^2}{2n} + \frac{(m''(\xi) - \sigma^2)t^2}{2n\sigma^2}\right]^n
 \end{aligned}$$

for $0 < \xi < \frac{t}{\sqrt{n}\sigma}$.

Remember that our goal is to consider $\lim_{n \rightarrow \infty} M_{Z_n}(t)$.

Well, as $n \rightarrow \infty$, $t/\sqrt{n}\sigma \rightarrow 0$ so $\xi \rightarrow 0$ since $\xi < t/\sqrt{n}\sigma$.

Now, $m''(t)$ is continuous at 0, and $m''(0) = E[(X - \mu)^2] = \sigma^2$, so

$$\lim_{n \rightarrow \infty} (m''(\xi) - \sigma^2) = 0.$$

Therefore,

$$M_{Z_n}(t) = \left[1 + \frac{t^2}{2n} + \frac{d(n)}{n}\right]^n,$$

where $d(n) \rightarrow 0$ as $n \rightarrow \infty$.

So we have that

$$\lim_{n \rightarrow \infty} M_{Z_n}(t) = e^{t^2/2}$$

which means that

$$Z_n \xrightarrow{d} Z \sim N(0, 1).$$

2.5.2 Asymptotic Normality

Definition/Notation: We say that X_n has an **asymptotically normal** distribution with mean μ_n and variance σ_n^2 if

$$\frac{X_n - \mu_n}{\sigma_n} \xrightarrow{\mathcal{D}} N(0, 1).$$

(Note that, even without the limit, that thing on the left side has mean 0 and variance 1.)

If X_n has an asymptotically normal distribution with mean μ_n and variance σ_n^2 , we will write

$$X_n \overset{asympt}{\sim} N(\mu_n, \sigma_n^2).$$

Example 1: According to the Central Limit Theorem, if $X_1, X_2, \dots, X_n$ is a random sample from any distribution with mean μ and variance $\sigma^2 < \infty$,

$$\bar{X} = \bar{X}_n \overset{asympt}{\sim} N(\mu, \sigma^2/n).$$

Example 2: According to the Central Limit Theorem, if $X_1, X_2, \dots, X_n$ is a random sample from any distribution with mean μ and variance $\sigma^2 < \infty$, we also have that $\sum_{i=1}^n X_i$ is asymptotically normal since

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} Z \sim N(0, 1)$$

and since

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{\frac{1}{n} \sum X_i - \mu}{\sigma/\sqrt{n}} = \frac{n \frac{1}{n} \sum X_i - n\mu}{n \sigma/\sqrt{n}} = \frac{\sum X_i - n\mu}{\sqrt{n}\sigma/}$$

Therefore, we see that

$$\sum_{i=1}^n X_i \overset{asympt}{\sim} N(n\mu, n\sigma^2).$$

2.5.3 A Numerical Example

Let $\bar{X}$ denote the mean of a random sample of size 100 from the $\chi^2(50)$ distribution.

Give an approximate value for $P(49 < \bar{X} < 51)$.

Answer: Just for the record, we can give an exact value since

$$X_1, X_2, \dots, X_{100} \stackrel{iid}{\sim} \chi^2(50) = \Gamma(25, 1/2) \quad \Rightarrow \quad \sum_{i=1}^{100} X_i \sim \Gamma(2500, 1/2).$$

So,

$$\begin{aligned} P(49 < \bar{X} < 51) &= P(4900 < \sum X_i < 5100) \\ &= \int_{4900}^{5100} \frac{1}{\Gamma(2500)} \left(\frac{1}{2}\right)^{2500} x^{2499} e^{-\frac{1}{2}x} dx \approx 0.6827. \end{aligned}$$

(This computation requires 2,500 iterations of integration by parts and was done by Mathematica!)

With the Central Limit Theorem, we know that the distribution of $\bar{X}$ is approximately normal for large sample sizes. In statistics textbooks, a sample size of $n \geq 30$ or sometimes $n \geq 40$, is generally considered “large”.

The mean for $\bar{X}$, normal or not, is

$$\mu_{\bar{X}} = \mu = 25/(1/2) = 50$$

and the variance is

$$\sigma_{\bar{X}}^2 = \frac{\sigma^2}{n} = \frac{25/(1/4)}{100} = 1.$$

We can “standardize” $\bar{X}$ into something with mean 0 and variance 1, which, for large samples, will be approximately a standard normal random variable. So,

$$\begin{aligned} P(49 < \bar{X} < 51) &= P\left(\frac{49-50}{10/10} < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < \frac{51-50}{10/10}\right) \\ &= P\left(-1 < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < 1\right) \\ &\approx P(-1 < Z < 1) \end{aligned}$$

where $Z \sim N(0, 1)$ is a standard normal random variable.

Now,

$$P(-1 < Z < 1) \stackrel{\text{contin}}{=} P(-1 < Z \leq 1) = P(Z \leq 1) - P(Z \leq -1)$$

$$\stackrel{z\text{-table}}{=} 0.8413 - (1 - 0.8413) = 0.6826.$$

Chapter 3

A “Mostly Normal” Introduction to Confidence Intervals

Still and seemingly forever on the horizon is how to find estimators of parameters of distributions. We have found a few “common sense estimators” just by guessing and making some adjustments to our guesses. We have looked at few properties of estimators, finding their expected values, variances, and MSEs, and we have shown their convergence in probability and/or distribution to the parameter of interest. Soon, we will stop guessing at estimators! Soon.

But first, a brief introduction to confidence intervals when we have normality.

3.1 A Confidence Interval for the Mean of a Normal Distribution: σ^2 is Known

Suppose that we have $X_1, X_2, \dots, X_n$, a random sample from the $N(\mu, \sigma^2)$ where σ^2 is known but μ is unknown.

(It may be kind of silly to assume that σ^2 is known when μ is unknown. Most likely, neither are known. This example is just a building block to more realistic cases!)

The most natural (what we might call “common sense”) estimator of μ is the sample mean:

$$\hat{\mu} = \bar{X}.$$

We are now going to move from this **point estimate** to an **interval estimate** of the form “ $\bar{X}$ plus or minus something”. There will be more about interpreting “plus or minus something” soon.

Since $X_1, X_2, \dots, X_n$ are all normal random variables, $\bar{X}$ is also normal. This is because a linear combination of normal random variables, whether iid or not, is still normal.

If $X_1, X_2, \dots, X_n$, is a random sample from any distribution with mean μ and variance σ^2 , we have seen that the sample mean $\bar{X}$ has mean

$$\mu_{\bar{X}} = E[\bar{X}] = E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n E[X_i] \stackrel{iid}{=} \frac{1}{n} \sum_{i=1}^n \mu = \frac{1}{n}(\mu + \mu + \dots + \mu) = \mu$$

and variance

$$\begin{aligned} \sigma_{\bar{X}}^2 &= Var[\bar{X}] = Var\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n^2} Var[\sum_{i=1}^n X_i] \\ &\stackrel{indep}{=} \frac{1}{n^2} \sum_{i=1}^n Var[X_i] \stackrel{ident}{=} \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{1}{n^2} \cdot n\sigma^2 = \sigma^2/n. \end{aligned}$$

So, when $X_1, X_2, \dots, X_n$ are iid $N(\mu, \sigma^2)$ distributed, we have that

$$\bar{X} \sim N(\mu, \sigma^2/n).$$

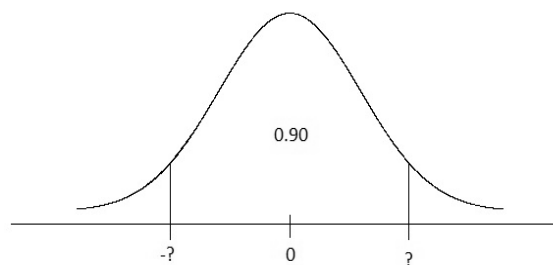
Note that this is **not** from using the Central Limit Theorem! This is using the fact that we started with normals!

We now know that we can “standardize” $\bar{X}$ into a standard normal random variable.

$$Z \stackrel{def}{=} \frac{\bar{X} - \mu_{\bar{X}}}{\sigma_{\bar{X}}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

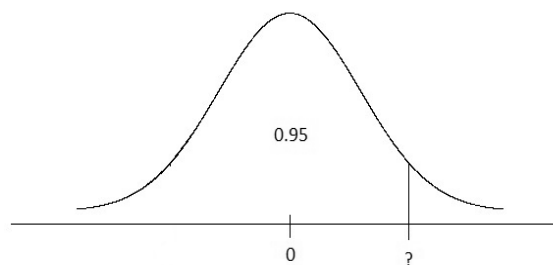
Imagine we wanted to find two values on the standard normal curve that captures, say, 90% of the area under the curve between them.

3.1. A CONFIDENCE INTERVAL FOR THE MEAN OF A NORMAL DISTRIBUTION: σ^2 IS KNOWN



I've drawn these values as symmetric around zero for a reason— more about that later.

Note that 10% of the area is outside those cutoff values, with 5% on each end. Therefore, the value “?” cuts off area 0.95 to its left.



“Reverse reading” our standard normal table, (looking for the probability in the body of the table). we see that the cutoff value “?” is about 1.645. (In this particular case, the value we were looking for fell exactly between 1.64 and 1.65 so we took the value in the middle. In general, just go for the closest value. Interpolation is not worth much in terms of accuracy.)

So, we know that $P(-1.645 < Z < 1.645) = 0.90$. Thus

$$\begin{aligned} 0.90 &= P(-1.645 < Z < 1.645) \\ &= P\left(-1.645 < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < 1.645\right) \end{aligned}$$

Solving for μ in the middle, we get

$$\begin{aligned} 0.90 &= P\left(-1.645\frac{\sigma}{\sqrt{n}} < \bar{X} - \mu < 1.645\frac{\sigma}{\sqrt{n}}\right) \\ &= P\left(-1.645\frac{\sigma}{\sqrt{n}} - \bar{X} < -\mu < 1.645\frac{\sigma}{\sqrt{n}} - \bar{X}\right) \\ &= P\left(\bar{X} - 1.645\frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 1.645\frac{\sigma}{\sqrt{n}}\right) \end{aligned}$$

We are used to seeing random variables in the middle of inequalities such as these. Note that here, μ is a constant, not a random variable and we have put it between two random endpoints with probability 0.90.

We say that a 90% confidence interval for μ is given by

$$\left(\bar{X} - 1.645\frac{\sigma}{\sqrt{n}}, \bar{X} + 1.645\frac{\sigma}{\sqrt{n}}\right),$$

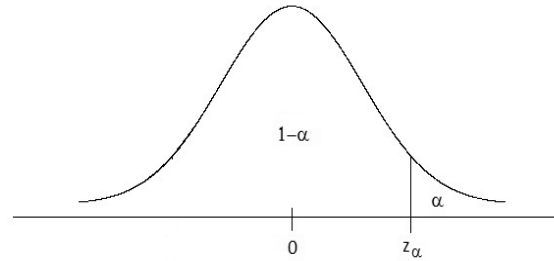
or, in shorthand notation

$$\bar{X} \pm 1.645\frac{\sigma}{\sqrt{n}}.$$

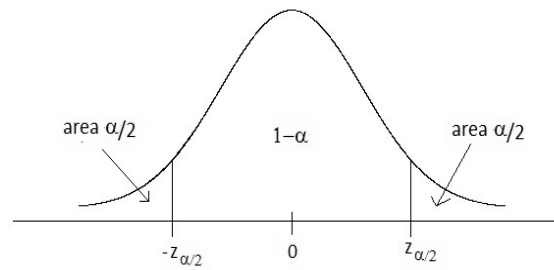
Interpretation: Once a sample of some given size n is taken, we compute $\bar{x}$ and the above confidence interval endpoints turn into numbers. Once you have done this, there is no more probability involved. The true value of μ is either in that interval or it is not. The probability comes in from the sample. If you were to take another random sample of size n from this normal “population”, you would get a different value for $\bar{x}$, which would give you different endpoints for the confidence interval. A third sample gives you yet another $\bar{x}$ which gives you a different confidence interval again. Repeating many times, you will have that the true value of μ is captured between the endpoints 90% of the time.

A More General Percentage

We will use the notation z_α to be the cutoff value for a standard normal curve that captures area α to the right. This is also called a **critical value**.



If we want to capture a certain area, say $1 - \alpha$, in the middle, the two cutoff values are $z_{\alpha/2}$ and $-z_{\alpha/2}$. (Note: In general, the values are $z_{\alpha/2}$ and $z_{1-\alpha/2}$, but, by symmetry of the $N(0, 1)$ distribution about zero, $z_{1-\alpha/2} = -z_{\alpha/2}$.)



For our 90% confidence interval, we put area 0.90 in the middle. i.e., $1 - \alpha = 0.9$. So, $\alpha/2 = 0.05$, and, in this new critical value notation, $z_{0.05} = 1.645$.

In general, if $X_1, X_2, \dots, X_n$ is a random sample from the $N(\mu, \sigma^2)$ distribution with σ^2 known, a $100(1 - \alpha)\%$ confidence interval for μ is given by

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}.$$

3.2 An Approximate Confidence Interval for the Mean of Any Distribution

Recall that the sample mean $\bar{X}$ for a random sample of size n from any distribution with mean μ and variance σ^2 has mean μ and variance σ^2/n .

Recall also that if $X_1, X_2, \dots, X_n$ is a random sample from any distribution with finite variance, the Central Limit Theorem tells us that $\bar{X}$ has an approximately normal distribution for large n . (Greater than 30 is usually considered “large”.) So, we have the following.

If $X_1, X_2, \dots, X_n$ is a random sample from any distribution with mean μ and variance $\sigma^2 < \infty$, an approximate $100(1 - \alpha)\%$ confidence interval for μ , when n is large, is given by

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}.$$

So far, we have assumed that σ^2 , the variance of the distribution from which the random sample came, is known. If it is not known, we might consider estimating it with the sample variance

$$S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}.$$

We have seen that $S^2 \xrightarrow{P} \sigma^2$. Thus, we can say that S^2 is getting “close” to σ^2 , in some sense, as the sample size n gets large. Therefore,

If $X_1, X_2, \dots, X_n$ is a random sample from any distribution with mean μ and unknown variance σ^2 , an approximate $100(1 - \alpha)\%$ confidence interval for μ , when n is large, is given by

$$\bar{X} \pm z_{\alpha/2} \frac{S}{\sqrt{n}}.$$

If the sample size is small, then we must be given a distribution since we can not appeal to the Central Limit Theorem to get an approximate distribution for $\bar{X}$. Otherwise, we can’t find a confidence interval. We will discuss the procedure for known distributions, other than the normal distribution, later in Section ???. That said, we have only one case left to consider.

3.3 A Confidence Interval for the Mean of a Normal Distribution: σ^2 is Unknown and Sample Size is Small

Our final case for this section about confidence intervals for a mean involving a normal distribution is

- normal distribution
- $n \leq 30$
- σ^2 unknown

Since the sample size is small, the sample variance S^2 may not be a very good estimator of the true variance σ^2 .

If the sample comes from a $N(\mu, \sigma^2)$ distribution, then $\bar{X} \sim N(\mu, \sigma^2/n)$. So, we can “standardize” $\bar{X}$:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

and place this between two “ z -critical values” capturing the appropriate area under the standard normal pdf.

The question now is

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim ?$$

for a small sample from a $N(\mu, \sigma^2)$ distribution.

We must digress and learn about an exciting new distribution!

3.3.1 The t Distribution

Suppose that $Z \sim N(0, 1)$ and $W \sim \chi^2(n)$ are two independent random variables.

Define

$$T \sim \frac{Z}{\sqrt{W/n}}.$$

Then we can show that the pdf of T is

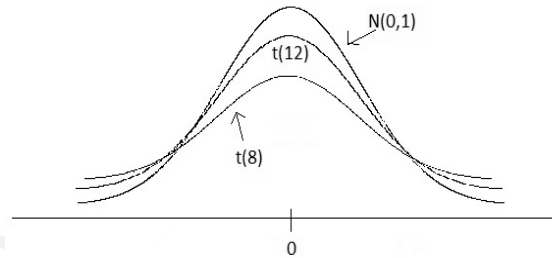
$$f_T(t) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n}{2}\right)} \frac{1}{\sqrt{\pi n}} \left(1 + \frac{t^2}{n}\right)^{-(n+1)/2}$$

for $-\infty < t < \infty$.

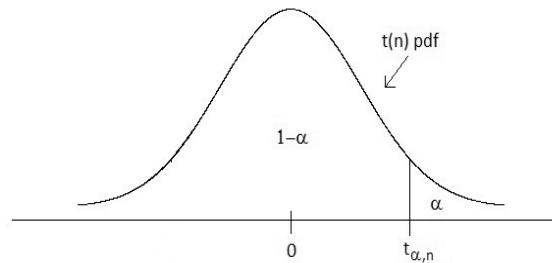
This distribution is known as the t -distribution with n “degrees of freedom”. We write

$$T \sim t(n).$$

The pdf is a symmetric bell curve centered at zero that is flatter than the $N(0, 1)$ pdf. As n increases, it becomes closer and closer to the $N(0, 1)$ pdf, as depicted in the horrible low-resolution figure below.



As with the z curve, we would like to use the notation t_α to denote a “cutoff” or “critical value” that captures area α to the right under the t curve. But, there are many t curves! We must include in our notation some reference to the degrees of freedom parameter. We will use $t_{\alpha,n}$.



3.3. A CONFIDENCE INTERVAL FOR THE MEAN OF A NORMAL DISTRIBUTION: σ^2 IS UNKNOWN

3.3.2 The Distribution of $(n-1)S^2/\sigma^2$ for a Normal Distribution

Let $X_1, X_2, \dots, X_n$ be a random sample of size n from the $N(\mu, \sigma^2)$ distribution.

Consider the quantity $\sum_{i=1}^n (X_i - \mu)^2$ expanded as

$$\begin{aligned} \sum_{i=1}^n (X_i - \mu)^2 &= \sum_{i=1}^n (X_i - \bar{X} + \bar{X} - \mu)^2 \\ &= \sum_{i=1}^n (X_i - \bar{X})^2 + 2(\bar{X} - \mu) \sum_{i=1}^n (X_i - \bar{X}) + \sum_{i=1}^n (\bar{X} - \mu)^2 \end{aligned} \quad (3.1)$$

Note that the middle term is zero since

$$\sum_{i=1}^n (X_i - \bar{X}) = \sum_{i=1}^n X_i - n\bar{X} = \sum_{i=1}^n X_i - n \frac{1}{n} \sum_{i=1}^n X_i = \sum_{i=1}^n X_i - \sum_{i=1}^n X_i = 0.$$

Note that the last term of (3.1) is $n(\bar{X} - \mu)^2$.

So, (3.1) becomes

$$\sum_{i=1}^n (X_i - \mu)^2 = \sum_{i=1}^n (X_i - \bar{X})^2 + n(\bar{X} - \mu)^2.$$

Dividing through by σ^2 , we have

$$\frac{\sum_{i=1}^n (X_i - \mu)^2}{\sigma^2} = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2} + \frac{n(\bar{X} - \mu)^2}{\sigma^2}. \quad (3.2)$$

Note that

1. The first term on the right-hand side of (3.2) is

$$\frac{(n-1)S^2}{\sigma^2}.$$

2. The term on the left-hand side has a $\chi^2(n)$ distribution since

$$\begin{aligned} X_i \sim N(\mu, \sigma^2) &\Rightarrow \frac{X_i - \mu}{\sigma} \sim N(0, 1) \Rightarrow \left(\frac{X_i - \mu}{\sigma} \right)^2 \sim \chi^2(1) \\ &\Rightarrow \frac{\sum_{i=1}^n (X_i - \mu)^2}{\sigma^2} = \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 \sim \chi^2(n). \end{aligned}$$

3. The second term on the right-hand side has a $\chi^2(1)$ distribution since

$$\begin{aligned}\bar{X} \sim N(\mu, \sigma^2/n) &\Rightarrow \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim N(0, 1) \\ &\Rightarrow \frac{n(\bar{X} - \mu)^2}{\sigma^2} = \left(\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \right)^2 \sim \chi^2(1).\end{aligned}$$

For simplicity, rewrite (3.2) as

$$W = W_1 + W_2.$$

So $W \sim \chi^2(n)$, $W_2 \sim \chi^2(n)$, and W_1 is the thing we want the distribution for. By Problem 1 on HW5, we know that W_1 and W_2 are independent.

Now

$$M_W(t) = M_{W_1+W_2}(t) \stackrel{\text{indep}}{=} M_{W_1}(t) \cdot M_{W_2}(t).$$

Using the appropriate χ^2 moment generating functions, we find that

$$M_{W_1}(t) = \frac{M_W(t)}{M_{W_2}(t)} = \frac{\left(\frac{1/2}{1/2-t}\right)^{n/2}}{\left(\frac{1/2}{1/2-t}\right)^{1/2}} = \left(\frac{1/2}{1/2-t}\right)^{(n-1)/2}$$

which is the moment generating function for the $\chi^2(n-1)$ random variable.

Therefore,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1).$$

Awesome!

3.3.3 Return to the Confidence Interval

Let $X_1, X_2, \dots, X_n$ be a random sample from the $N(\mu, \sigma^2)$ distribution with n “small”. Suppose that we do not know σ^2 . Because $X_1, X_2, \dots, X_n$ are normal, we know that $\bar{X}$ is normal, and, in particular,

$$\bar{X} \sim N(\mu, \sigma^2).$$

So,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

3.3. A CONFIDENCE INTERVAL FOR THE MEAN OF A NORMAL DISTRIBUTION: σ^2 IS UNKNOWN

Thus,

$$\begin{aligned}\frac{\bar{X}-\mu}{S/\sqrt{n}} &= \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \cdot \frac{\sigma}{S} \\ &= \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \cdot \sqrt{\frac{\sigma^2}{S^2}} \\ &= \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \bigg/ \sqrt{\frac{S^2}{\sigma^2}} \\ &= \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \bigg/ \sqrt{\frac{(n-1)S^2}{\sigma^2} / (n-1)}\end{aligned}$$

So, we have a $N(0,1)$ divided by the square root of an independent (See solutions to HW 6.) $\chi^2(n-1)$ divided by its degrees of freedom. This is how we defined a t -distribution with $n-1$ degrees of freedom. Therefore

$$X_1, X_2, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2) \quad \Rightarrow \quad \frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t(n-1).$$

<insert celebration here>

The two numbers that can capture area $1-\alpha$ in the center of a $t(n-1)$ curve are $t_{1-\alpha/2, n-1} = -t_{\alpha/2, n-1}$ and $t_{\alpha/2, n-1}$. So,

$$1-\alpha = P\left(-t_{\alpha/2, n-1} < \frac{\bar{X}-\mu}{S/\sqrt{n}} < t_{\alpha/2, n-1}\right)$$

Solving for μ “in the middle”, we get that a $100(1-\alpha)\%$ confidence interval for μ is

$$\left(\bar{X} - t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}\right),$$

or simply

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}.$$

3.3.4 Summary of Confidence Intervals for the Mean Involving Normal Distributions

Given a random sample of size n from a distribution with mean μ and variance σ^2 , a $100(1-\alpha)\%$ confidence interval for μ is given by:

		Distribution		
—		Normal	Normal but Not Known	Unknown
Sample Size	$n \leq 30$	σ^2	$\bar{X} \pm z_{\alpha/2}\sigma/\sqrt{n}$	Can't Do This!
		S^2	$\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}$	
	$n > 30$	σ^2	$\bar{X} \pm z_{\alpha/2}\sigma/\sqrt{n}$	$\bar{X} \pm z_{\alpha/2}\sigma/\sqrt{n}$
		S^2	$\bar{X} \pm z_{\alpha/2}S/\sqrt{n}$	$\bar{X} \pm z_{\alpha/2}S/\sqrt{n}$

3.4 A Difference of Means

3.4.1 Normal Samples, Variances Known

Suppose that we have two independent random samples:

- one of size n_1 from the $N(\mu_1, \sigma_1^2)$ distribution
- one of size n_2 from the $N(\mu_2, \sigma_2^2)$ distribution

Let's construct a $100(1 - \alpha)\%$ confidence interval for $\mu_1 - \mu_2$, the difference in means.

Any confidence interval starts with an estimate. Let's estimate $\mu_1 - \mu_2$ with the difference in sample means

$$\bar{X}_1 - \bar{X}_2.$$

What is the distribution of $\bar{X}_1 - \bar{X}_2$?

We know that

- $\bar{X}_1 \sim N(\mu_1, \sigma_1^2/n_1)$
- $\bar{X}_2 \sim N(\mu_2, \sigma_2^2/n_2)$

Since a linear combination of normal random variables is again normal, we know that $\bar{X}_1 - \bar{X}_2$ has a normal distribution.

The mean is

$$E[\bar{X}_1 - \bar{X}_2] = E[\bar{X}_1] - E[\bar{X}_2] = \mu_1 - \mu_2.$$

The variance is

$$\text{Var}[\bar{X}_1 - \bar{X}_2] \stackrel{\text{indep}}{=} \text{Var}[\bar{X}_1] + \text{Var}[\bar{X}_2] = \sigma_1^2/n_1 + \sigma_2^2/n_2.$$

So,

$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right).$$

We know that any normal random variable, minus its mean, and divided by its standard deviation is a standard normal ($N(0, 1)$) random variable. So, in this case, we know that

$$\frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0, 1).$$

Let $Z \sim N(0, 1)$. Suppose we want to find two numbers that capture area 0.80 in the center of the $N(0, 1)$ pdf. Using our z -table, we see that these numbers are approximately ± 1.28 . Thus,

$$\begin{aligned} 0.80 &= P(-1.28 < Z < 1.28) \\ &= P\left(-1.28 < \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} < 1.28\right) \end{aligned}$$

Solving for $\mu_1 - \mu_2$ in the middle gives

$$0.80 = P\left(\bar{X}_1 - \bar{X}_2 - 1.28\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} < \mu_1 - \mu_2 < 1.28\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}\right),$$

making an 80% confidence interval

$$\left(\bar{X}_1 - \bar{X}_2 - 1.28\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}, \bar{X}_1 - \bar{X}_2 + 1.28\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}\right)$$

or simply

$$\bar{X}_1 - \bar{X}_2 \pm 1.28\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}.$$

For the more general percentage of $100(1 - \alpha)\%$, instead of 80%, we need to replace ± 1.28 with $\pm z_{\alpha/2}$. Then we have that a $100(1 - \alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is

$$\bar{X}_1 - \bar{X}_2 \pm z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}.$$

3.4.2 Large Samples, Variances Unknown

For large n_1 and n_2 , the two population sample variances S_1^2 and S_2^2 are good approximations of σ_1^2 and σ_2^2 , respectively.

So, we have that

$$\frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \approx \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0, 1)$$

and therefore that an approximate $100(1 - \alpha)\%$ confidence interval for the difference $\mu_1 - \mu_2$ is given by

$$\bar{X}_1 - \bar{X}_2 \pm z_{\alpha/2} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}.$$

More formally, we are using the fact that

$$\frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \xrightarrow{\mathcal{D}} N(0, 1).$$

This can be shown by writing

$$\frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \bigg/ \sqrt{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right) \bigg/ \left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)} \quad (3.3)$$

It can be shown (see solution to grad problem on HW 7) that

$$\sqrt{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right) \bigg/ \left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)} \xrightarrow{P} 1.$$

For normal populations, we have that

$$\frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0, 1).$$

If we don't have normal populations, we still have

$$\frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \xrightarrow{\mathcal{D}} N(0, 1)$$

from an application of the CLT after some manipulation. This limit is not entirely obvious. Things get tricky as n_1 and n_2 may not be going to ∞ at quite the same rate.

3.4.3 At Least One Small Sample, Normality, Variances Unknown

We have two cases to consider here. The first is when $\sigma_1^2 = \sigma_2^2$ and the second is when $\sigma_1^2 \neq \sigma_2^2$. Even though we do not know the values for σ_1^2 and σ_2^2 , there are situations where we might believe them to be equal. For example, suppose that we want to find a confidence interval for $\mu_1 - \mu_2$ where μ_1 is the true mean sales volume of some breakfast cereal across all grocery stores and μ_2 is the true mean sales volume of the same cereal across all grocery stores but in an exciting new package. You might expect a new flashy package design to increase sales but maybe not necessarily to change the variance of sale prices. Then again, maybe not. You probably shouldn't bring your "expectations" into this. We will eventually learn about hypothesis testing and we will develop a statistical test to determine whether or not it is reasonable to assume that $\sigma_1^2 = \sigma_2^2$.

Case: $\sigma_1^2 = \sigma_2^2$

We begin by giving the common value of σ_1^2 and σ_2^2 a more "sample neutral name" like σ^2 . That is $\sigma^2 = \sigma_1^2 = \sigma_2^2$ and we will stop referring to σ_1^2 and σ_2^2 altogether.

We compute sample variances, S_1^2 and S_2^2 for each of the samples. (Note that we do not just combine the samples and compute a single variance

because variance is the expected square deviation from the mean and the samples can not be assumed to have the same means.)

Since S_1^2 and S_2^2 are both estimates for the common variance σ^2 , we should use information from both to come up with a single estimate for the unknown σ^2 . One possibility is the average value

$$\frac{S_1^2 + S_2^2}{2}.$$

This is maybe not the best estimate. It would be better to use some sort of weighted average to take into account the different sample sizes n_1 and n_2 . After all, if n_1 is much larger than n_2 , the sample variance from that first population (S_1^2) is probably much more accurate than the sample variance from the second population (S_2^2) and thus it should carry more weight.

Here is a weighted average:

$$\frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2}.$$

This solves the issue of giving more weight to the sample variance from the larger sample but it is not mathematically convenient. Recall that, in order to develop a confidence interval, we must be able to find the distributions of certain quantities.

To this end consider a different weighted average, denoted by S_p^2 and known as the **pooled variance**.

$$S_p^2 := \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}.$$

Note that

$$W := \frac{(n_1 + n_2 - 2)S_p^2}{\sigma^2} = \frac{(n_1 - 1)S_1^2}{\sigma^2} + \frac{(n_2 - 1)S_2^2}{\sigma^2}$$

To be continued after HW 7 is turned in... this is a problem on the HW. You want to find the distribution of W and then ultimately of

$$\frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}.$$

Then you can put this thing between appropriate critical values to come up with a confidence interval for $\mu_1 - \mu_2$!

Chapter 4

A “Mostly Normal” Introduction to Hypothesis Testing

Suppose that we have a random sample of size 10 from a normal distribution with mean μ and variance 1, and suppose that we observe a sample mean of $\bar{x} = 3.7$.

Further suppose that we wish to test the “null hypothesis”

$$H_0 : \mu = 3$$

versus the “alternative hypothesis”

$$H_1 : \mu > 3$$

These hypotheses together are supposed to represent all possibilities for μ . I purposely didn’t write the null hypothesis as $\mu \leq 3$ because it adds a complication that I want to avoid for the moment in our introductory example. So, for now, imagine that the only possibilities for μ are that $\mu = 3$ or $\mu > 3$ and that there is no such thing as $\mu < 3$.

Just because the mean for our sample is greater than 3 does not mean that we can conclude that μ , the overall mean for the entire population from which

the sample was taken, is greater than 3. It’s possible that this is a sample with a particularly high mean and that, if we took another independent sample of size 10 we could get a very different value for $\bar{X}$.

To test

$$H_0 : \mu = 3$$

versus

$$H_1 : \mu > 3$$

we will assume that H_0 is true, but we will reject H_0 in favor of H_1 if $\bar{X}$ appears to be “significantly” larger than 3.

We need to make sense out of “significantly larger than 3.”

You may have noticed that the normal curve goes on infinitely in both directions. Hence, if H_0 is true and the sample was from the $N(3, 1)$ distribution, which means that $\bar{X}$, based on a sample of size 10 has the $N(3, 1/10)$, it is possible that we might even get that $\bar{X} > 9$. However, for the $N(3, 1/10)$ distribution, this is highly unlikely. Specifically, “under H_0 ”,

$$\begin{aligned} P(\bar{X} > 9) &= P\left(\frac{\bar{X}-3}{1/\sqrt{n}} > \frac{9-3}{1/\sqrt{10}}\right) = P(Z > 18.98) \\ &= 1 - P(Z \leq 18.98) \stackrel{Table}{\approx} 1 - 0 = 0 \end{aligned}$$

The probability that $P(\bar{X} > 9)$ is not actually zero, but it is a very very small number that will not show up in the 4 digits given in the normal table. It seems safe to say that, when observing an $\bar{X}$ greater than 9, it probably means that $H_0 : \mu = 3$ is false and that we should conclude that $H_1 : \mu > 3$ is correct. Still... there’s always that really small chance that we are wrong. If we are wrong then we have made one of two types of errors in hypothesis testing.

4.1 Errors in Hypothesis Testing

In reality, the null hypothesis H_0 is either true or it is false. When you do a hypothesis test, you assume that H_0 is true but then the data might force you to “reject” it because of some highly improbable (under H_0) outcome. If the data “does not speak strongly” against H_0 , then we will “fail to reject”

it. In the language of hypothesis testing, we either reject H_0 or we fail to reject H_0 but we will never “accept” H_0 .

If H_0 is true and we fail to reject it, we did good! Likewise, if it is false and we reject it we also did good. In the other cases, we have made an error denoted as in the following table.

		Your Decision	
		Fail to Reject H_0	Reject H_0
Reality	H_0 True	✓	Type I Error
	H_0 False	Type II Error	✓

Neither type of error is inherently worse. It depends on the context of the problem and what is at stake.

4.1.1 Hypotheses: Simple or Composite?

A **simple hypothesis** is one that “completely specifies the distribution”. For example, suppose that you have a random sample from the $N(\mu, 1)$ distribution and that the null hypothesis is

$$H_0 : \mu = 3.$$

If H_0 is true, then you know the distribution that the random sample came from the $N(3, 1)$ distribution.

A **composite hypothesis** is one that does not completely specify the distribution. For example, suppose that you have a random sample from the $N(\mu, 1)$ distribution and that the null hypothesis is

$$H_0 : \mu \leq 3.$$

If H_0 is true, then you know the distribution that the random sample came from some normal distribution with mean μ and variance 1 but you do not know which one. Was it the $N(2.912, 1)$ distribution? The $N(0, 1)$ distribution? The $N(-18.237, 1)$ distribution?

It is a common misconception that simple hypotheses have “equals signs” and composite hypotheses have inequalities. This, however, is not correct. For example, suppose that $X_1, X_2, \dots, X_n$ is a random sample from the $N(\mu, \sigma^2)$ distribution for unknown μ and σ^2 . Then

$$H_0 : \mu = 3$$

is not a simple hypothesis since, if it is true, you still don’t know exactly which normal distribution the sample came from. Was it the $N(3, 1)$? The $N(3, 8)$?

4.1.2 Level of Significance: α

Definition: For a simple null hypothesis, the level of significance of the test is the probability of a Type I error. It is denoted by an α :

$$\alpha = P(\text{Type I error}).$$

For reasons which will become apparent later, it is also known as the size of the test.

To illustrate, let us revisit the example from the beginning of this Chapter. Suppose that we have a random sample, $X_1, X_2, \dots, X_{10}$, of size 10 from the $N(\mu, 1)$ distribution and that we wish to test the hypotheses

$$H_0 : \mu = 3$$

$$H_1 : \mu > 3.$$

Suppose that I tell you that I will perform this test by looking at the sample mean $\bar{X}$ and rejecting H_0 in favor of H_1 if $\bar{X} > 4$. (It’s just a “rejection rule” that I made up. Don’t worry about where it came from yet!)

Q: What is the level of significance (size) of this test?

A:

$$\begin{aligned}
 \alpha &= P(\text{Type I Error}) \\
 &= P(\text{reject } H_0 \text{ when it's true}) \\
 &= P(\bar{X} > 4 \text{ when } \mu = 3) \\
 &\stackrel{\text{standardize}}{=} P\left(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} > \frac{4-3}{1/\sqrt{10}} \text{ when } \mu = 3\right) \\
 &= P(Z > 3.16) \text{ where } Z \sim N(0, 1) \\
 &\stackrel{\text{Table}}{=} 1 - 0.9992 = 0.0008.
 \end{aligned}$$

4.2 Finding a Test: A Simple Null Hypothesis

Usually, the level of significance is not determined from a cutoff “rejection rule” but instead is specified at the beginning by the researcher who cares about the particular hypotheses. This person might say, “I am willing to let the Type I Error probability be 0.05. What should the rejection rule be?”

Specifically, suppose that you have a random sample of size 10 from the $N(\mu, 1)$ distribution and you wish to find a test of size $\alpha = 0.05$ that is based on the sample mean $\bar{X}$ for the hypotheses

$$H_0 : \mu = 3$$

$$H_1 : \mu > 3.$$

Based on $\bar{X}$, we would want to reject H_0 in favor of H_1 if $\bar{X}$ is “large”. The form of the rejection rule (for the rejection of H_0 is therefore:

$$\text{“Reject } H_0 \text{ if } \bar{X} > c.”$$

for some c that is chosen to give a size 0.05 test.

“Aside”

The form of the rejection rule is determined by the alternative hypothesis, BUT, the “greater than” in the alternative hypothesis does not automatically translate to the “greater than” in the rejection rule!

For example, suppose that you had a random sample $X_1, X_2, \dots, X_n$ from the exponential distribution with rate λ and that you want to test

$$\begin{aligned} H_0 &: \lambda = 3 \\ H_1 &: \lambda > 3 \end{aligned}$$

based on looking at the sample mean $\bar{X}$.

You would want to reject H_0 in favor of H_1 if a larger value of λ is indicated by the data. Since $\bar{X}$ is an estimator of the mean of this distribution and since the mean is $1/\lambda$, a large value of λ is suggested by the data if $\bar{X}$ is small.

Therefore, the form of the rejection rule is to

$$\text{“Reject } H_0 \text{ if } \bar{X} < c.”$$

for some c that is chosen to give a size 0.05 test.

Back to the normal example...

We have given the “form” of the test, but to give the actual test, we still need to find the value of the cutoff c . This is done using the level of significance $\alpha = 0.05$:

$$\begin{aligned} 0.05 &= P(\text{Type I Error}) \\ &= P(\text{reject } H_0 \text{ when it's true}) \\ &= P(\bar{X} > c \text{ when } \mu = 3) \\ &\stackrel{\text{standardize}}{=} P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{c-3}{1/\sqrt{10}} \text{ when } \mu = 3\right) \\ &= P\left(Z > \frac{c-3}{1/\sqrt{10}}\right) \text{ where } Z \sim N(0, 1) \end{aligned}$$

From the normal table, we know that

$$P(Z > 1.645) = 0.05.$$

So, we must have

$$\frac{c - 3}{1/\sqrt{10}} = 1.645$$

which implies that $c = 3.52$.

In summary, we will reject H_0 in favor of H_1 at the 0.05 level of significance if $\bar{X} > 3.52$. For the original “data” at the beginning of this Chapter, since we observed $\bar{x} = 3.7$, we would indeed reject H_0 in this case.

Note that our cutoff value for the rejection rule came from a transformation of the cutoff value (1.645) on the standard normal curve. The z -critical-value 1.645 cuts off area 0.05 on the right side (upper tail) of the $N(0, 1)$ curve. This was translated to the value 3.52 on the $N(3, 1)$ curve. If we had chosen a smaller α (say $\alpha = 0.01$), the critical value (which would need to cut off a smaller area to the right) would be higher. It is then possible that we would find that we would no longer reject H_0 with a sample mean of $\bar{x} = 3.7$. In fact, this is the case, as c would be 3.74. (Check this!)

4.3 Finding a Test: A Composite Null Hypothesis

Imagine trying to find α , as defined so far, for a composite null hypothesis. Suppose, for our normal sample of size 10, that we wish to test

$$H_0 : \mu \leq 3$$

$$H_1 : \mu > 3.$$

Then, if we define the level of significance or “size” of the test to be the probability of a Type I error, we would get stuck:

$$\begin{aligned}
0.05 &= P(\text{Type I Error}) \\
&= P(\text{reject } H_0 \text{ when it's true}) \\
&= P(\bar{X} > c \text{ when } \mu \leq 3) \\
&\stackrel{\text{standardize}}{=} P\left(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} > \frac{c-??}{1/\sqrt{10}}\right)
\end{aligned}$$

Exactly what are we subtracting? I don't know! We can't proceed in this way to come up with a cutoff c if we don't know what mean to subtract. There are a whole bunch of means that are less than or equal to 3!

It is for this reason that, in the composite case we define α a little differently.

Definition: For a composite null hypothesis, the level of significance of the test is the maximum probability of a Type I error where the maximum is taken over all parameter possibilities included in the null hypothesis. It is still denoted by an α :

$$\alpha = \max_{\mu \in H_0} P(\text{Type I error when the parameter is } \mu).$$

We will abbreviate this by

$$\alpha = \max_{\mu \in H_0} P(\text{Type I error}; \mu).$$

Example: Suppose that $X_1, X_2, \dots, X_{10}$ is a random sample of size 10 from the $N(\mu, 1)$ distribution. Further suppose that we wish to find a test of size $\alpha = 0.05$ of

$$H_0 : \mu \leq 3$$

$$H_1 : \mu > 3$$

based on the sample mean $\bar{X}$.

Because of the form of the alternative hypothesis, we wish to reject H_0 if the mean comes across as “large” in our sample. If we decide to estimate μ by $\bar{X}$, a large value of μ will be reflected by a large value of $\bar{X}$. Therefore, the form of the test is again:

“Reject H_0 if $\bar{X} > c$.”

for some c that is chosen to give a size 0.05 test.

In order to find the value of c , we write

$$\begin{aligned}
 0.05 &= \max_{\mu \in H_0} P(\text{Type I Error}) \\
 &= \max_{\mu \leq 3} P(\text{reject } H_0; \mu) \\
 &= \max_{\mu \leq 3} P(\bar{X} > c \text{ when the parameter is } \mu) \\
 &\stackrel{\text{standardize}}{=} \max_{\mu \leq 3} P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{c - \mu}{1/\sqrt{10}} \text{ when the parameter is } \mu\right) \\
 &= \max_{\mu \leq 3} P\left(Z > \frac{c - \mu}{1/\sqrt{10}}\right) \text{ where } Z \sim N(0, 1) \\
 &= \max_{\mu \leq 3} \left[1 - \Phi\left(\frac{c - \mu}{1/\sqrt{10}}\right)\right]
 \end{aligned}$$

where $\Phi(\cdot)$ is the cdf of the standard normal distribution.

In order to find the maximum of

$$1 - \Phi\left(\frac{c - \mu}{1/\sqrt{10}}\right),$$

we note that it is an increasing function of μ since

- $\frac{c - \mu}{1/\sqrt{10}}$ is a decreasing function of μ ,
- which implies that $\Phi\left(\frac{c - \mu}{1/\sqrt{10}}\right)$ is decreasing since $\Phi(\cdot)$, as a cdf, is an increasing function, thereby increasing as its argument increases and decreasing as its argument decreases,
- which implies that $1 - \Phi\left(\frac{c - \mu}{1/\sqrt{10}}\right)$ is increasing in μ .

So, the maximum over all $\mu \leq 3$ occurs at the right endpoint of this interval where $\mu = 3$. So,

$$0.05 = \max_{\mu \leq 3} \left[1 - \Phi\left(\frac{c - \mu}{1/\sqrt{10}}\right)\right] = 1 - \Phi\left(\frac{c - 3}{1/\sqrt{10}}\right)$$

Moving things around, we get

$$\Phi\left(\frac{c - \mu}{1/\sqrt{10}}\right) = 0.95$$

or

$$P\left(Z \leq \frac{c - \mu}{1/\sqrt{10}}\right) = 0.95$$

which again means that

$$\frac{c - \mu}{1/\sqrt{10}} = 1.645$$

or that 3.52.

In summary, we will reject H_0 in favor of H_1 at the 0.05 level of significance if $\bar{X} > 3.52$.

4.3.1 Where There’s an α , There’s a β

Definition: For a simple alternative hypothesis, we use β to denote the probability of a Type II error:

$$\beta = P(\text{Type II error}).$$

Definition: For a composite alternative hypothesis, we use β to denote the maximum probability of a Type II error:

$$\beta = \max_{\mu \in H_1} P(\text{Type II error}).$$

In everything we’ve done so far, the rejection regions for our tests have been defined by

- the form of the alternative hypothesis
- the “test statistic” used (This is the function of the data that we use in order to make a test. Throughout this document, this has been $\bar{X}$, but we could make a test based, for example, on a minimum $X_{(1)}$ or any other statistic!)

- the level of significance (also known as “size” or probability of a Type I error) of the test

After all of this is determined, the probability of a Type II error is already “locked in”.

Example: Suppose that $X_1, X_2, \dots, X_{10}$ is a random sample of size 10 from the $N(\mu, 1)$ distribution. Suppose that we wish to find a test of size $\alpha = 0.05$ of

$$H_0 : \mu = 3$$

$$H_1 : \mu > 3$$

based on the sample mean $\bar{X}$.

We have determined that this test is to

$$\text{“Reject } H_0 \text{ if } \bar{X} > 3.52.”$$

In this case,

$$\begin{aligned} \beta &= \max_{\mu \in H_1} P(\text{Type II Error}) \\ &= \max_{\mu > 3} P(\text{fail to reject } H_0; \mu) \\ &= \max_{\mu > 3} P(\bar{X} \leq 3.52 \text{ when the parameter is } \mu) \\ &\stackrel{\text{standardize}}{=} \max_{\mu > 3} P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq \frac{3.52 - \mu}{1/\sqrt{10}} \text{ when the parameter is } \mu\right) \\ &= \max_{\mu > 3} P\left(Z \leq \frac{3.52 - \mu}{1/\sqrt{10}}\right) \text{ where } Z \sim N(0, 1) \\ &= \max_{\mu > 3} \Phi\left(\frac{3.52 - \mu}{1/\sqrt{10}}\right) \end{aligned}$$

As $\Phi\left(\frac{3.52 - \mu}{1/\sqrt{10}}\right)$ is decreasing in μ , it is maximized when μ is smallest. Technically, there is no “smallest” μ since μ is strictly greater than 3 and thus there is no “largest” Type II error probability. In this case though, we define β to be the “least upper bound” on the probability of a Type II error:

$$\beta = \sup_{\mu > 3} \Phi\left(\frac{3.52 - \mu}{1/\sqrt{10}}\right) = \Phi\left(\frac{3.52 - 3}{1/\sqrt{10}}\right) = \Phi(1.645) = 0.95.$$

“Aside”

Q: Why did we bother computing that? Isn't β always going to be $1 - \alpha$?

A: No. Type I errors and Type II errors do not exist “in the same universe”. H_0 is either true or false. There is no “probability it is true” or “probability it is false”. If it is true, we will reject it with probability α and we will fail to reject it with probability $1 - \alpha$. If it is false, we will fail to reject it with probability β and we will reject it with probability $1 - \beta$. In the example that we just completed, we saw what appears to be a “1 minus relationship” between α and β but you will not see this for all distributions/rejection rules or even for the normal distribution if we look at a simple null hypothesis versus a simple alternative hypothesis.

While there is not necessarily a “1 minus relationship” between α and β , there is an inverse relationship in the sense that if one is forced to be very small the other one will end up larger. For example, if you want α to be very small then you will make a more extreme cutoff value for rejection and make it very difficult to reject H_0 . So, if H_0 is false and you should be rejecting it, this extreme cutoff will be difficult to pass so you are risking not rejecting when you should. That is, you are increasing the probability of making a Type II error!

4.4 Power

For a given distribution, $\bar{X}$ is not the only estimator of μ . (In fact, it is not even the only unbiased estimator of μ .) Just as we may use many different statistics to estimate μ , we may make tests of hypotheses based on many different estimators of μ .

For example, suppose that $X_1, X_2, \dots, X_n$ is a random sample from the $N(\mu, 1)$ distribution.

Suppose further that we want to test

$$\begin{aligned} H_0 &: \mu = 3 \\ H_1 &: \mu > 3 \end{aligned}$$

We could make a test of size α “based on $\bar{X}$ ” which makes a lot of sense

because $\bar{X}$ is a natural (and good!) estimator of μ . The “form” of the test would be to reject H_0 in favor of H_1 if $\bar{X}$ seems large. That is, we reject H_0 in favor of H_1 if

$$\bar{X} > c$$

where c is determined so that the size (level of significance) of the test is α . That is, we would need to solve for c in

$$\alpha = P(\bar{X} > c \text{ when } \mu = 3).$$

However, we could also make a test for these hypotheses of size α based on, for example, $X_{(1)} = \min(X_1, X_2, \dots, X_n)$. Maybe this doesn't seem like a good idea but let's try to quantify what is meant by “good idea”!

Evidence for $H_1 : \mu > 3$ being true based will be reflected in the sample by seeing “large” values in the sample. This includes seeing a large value for the minimum of the sample. Therefore, the form of a test of these hypotheses, based on $X_{(1)}$ will be to reject H_0 in favor of H_1 if $X_{(1)}$ is “large”. “Large” means greater than some value which I will call c_2 .

The form of the test will be to reject H_0 in favor of H_1 if

$$X_{(1)} > c_2$$

where c_2 is determined to give a size α test. That is, we would need to solve for c_2 in

$$\alpha = P(X_{(1)} > c_2 \text{ when } \mu = 3).$$

Once you find the numbers c and c_2 , you have two tests both with the exact same (by design) probability of making a Type I error. That is, they will both reject H_0 when they shouldn't because it is true, with probability α .

These tests will differ, however, when H_1 is true. For example, if the true value of μ is 5, then these tests should reject H_0 . In fact, we will see that

$$P(\bar{X} > c \text{ when } \mu = 5) \geq P(X_{(1)} > c_2 \text{ when } \mu = 5).$$

The **power** of a test is the probability it will reject H_0 when H_0 is false. (When it should reject!)

For this example, the test based on $\bar{X}$ has “higher power” than the test based on $X_{(1)}$ when $\mu = 5$. We will eventually see that it has higher power than the test based on $X_{(1)}$ for all $\mu > 3$.

So, the test based on $\bar{X}$ is better than the test based on $X_{(1)}$ in the sense that, they both have the same probability of making the correct decision when H_0 is true, but the $\bar{X}$ test does better than the $X_{(1)}$ test when H_0 is not true.

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Chapter 5

Estimation

So far, we have estimated means for distributions with sample means. This seems like a good idea but two questions arrive.

- Why is it a good idea?
- What are we supposed to do for other types of parameters?

Since we will be estimating parameters for various distributions, we will extend our notation for pdfs to emphasize these parameters. In general, instead of $f(x)$, we will write $f(x; \theta)$. Note the semicolon. Commas will be reserved for separating multiple x 's in joint pdfs and multiple parameters. (For example if X_1, X_2 is a random sample from the $N(\mu, \sigma^2)$ distribution, we will denote the joint pdf as $f(x_1, x_2; \mu, \sigma^2)$ or as $f(\vec{x}; \mu, \sigma^2)$.)

5.1 Method of Moments Estimators (MMEs)

5.1.1 The Idea

The idea behind “method of moments estimation” is to equate “population” (or “distribution”) moments with sample moments. We need some definitions.

Definition: The k^{th} “population moment” (“distribution moment”) of X , denoted by μ_k , is defined by

$$\mu_k = E[X^k].$$

Definition: The k^{th} “sample moment” of X , denoted by M_k , is defined by

$$M_k = \frac{1}{n} \sum_{i=1}^n X_i^k.$$



Method

To find a **method of moments estimator**, equate population moments to sample moments and solve for the unknown parameter(s). Use as many moments as you need in order to solve for all parameters.

5.1.2 Examples

Example:

Let $X_1, X_2, \dots, X_n$ be a random sample from the exponential distribution with rate λ . Find the MME of λ .

The first population moment is

$$\mu_1 = E[X] = 1/\lambda.$$

The first sample moment is

$$M_1 = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}.$$

Equate:

$$\frac{1}{\lambda} \stackrel{set}{=} \bar{X}$$

and solve for λ ,

$$\lambda = 1/\bar{X}.$$

The method of moments estimator for λ is

$$\hat{\lambda} = 1/\bar{X}.$$

We saw in Section 2.2 that this is a biased estimator for λ , but it is still an estimator nonetheless! □

Example:

Let $X_1, X_2, \dots, X_n$ be a random sample from the $\Gamma(\alpha, \beta)$ distribution. Find MMEs of α and β .

The first population moment is

$$\mu_1 = \mathbb{E}[X] = \alpha/\beta.$$

The first sample moment is

$$M_1 = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}.$$

Equate:

$$\frac{\alpha}{\beta} \stackrel{\text{set}}{=} \bar{X}$$

We need to solve for both α and β . We can't do this with only one equation. So, on to second moments!

The second population moment is

$$\mu_2 = \mathbb{E}[X^2] = \text{Var}[X] + (\mathbb{E}[X])^2 = \frac{\alpha}{\beta^2} + \left(\frac{\alpha}{\beta}\right)^2.$$

The second sample moment is

$$M_2 = \frac{1}{n} \sum_{i=1}^n X_i^2.$$

Equate:

$$\frac{\alpha}{\beta^2} + \left(\frac{\alpha}{\beta}\right)^2 \stackrel{\text{set}}{=} \frac{1}{n} \sum_{i=1}^n X_i^2.$$

So we have a system of equations

$$\frac{\alpha}{\beta} = \bar{X} \quad \text{and} \quad \frac{\alpha}{\beta^2} + \left(\frac{\alpha}{\beta}\right)^2 = \frac{1}{n} \sum_{i=1}^n X_i^2$$

Now solve for α and β any way you can! I'm going to plug $\alpha/\beta = \bar{X}$ from the first equation into the second equation:

$$\frac{1}{\beta}\bar{X} + \bar{X}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2.$$

So,

$$\beta = \frac{\bar{X}}{\frac{1}{n} \sum X_i^2 - \bar{X}^2}.$$

From the first equation we have

$$\alpha = \beta\bar{X} = \frac{\bar{X}^2}{\frac{1}{n} \sum X_i^2 - \bar{X}^2}.$$

So, the method of moments estimators of α and β are

$$\hat{\alpha} = \frac{\bar{X}^2}{\frac{1}{n} \sum X_i^2 - \bar{X}^2} \quad \text{and} \quad \hat{\beta} = \frac{\bar{X}}{\frac{1}{n} \sum X_i^2 - \bar{X}^2}. \quad \square$$

5.1.3 Properties of MMEs

Method of moments estimators are highly intuitive estimators. They can even be shown to be consistent. (Recall that an estimator $\hat{\theta}_n$ of θ is consistent if $\hat{\theta}_n \xrightarrow{P} \theta$.) We have seen from our examples that they can be biased, but the consistency property gives us that they are at least asymptotically unbiased. That is, if $\hat{\theta}_n$ is an MME for θ , then

$$\lim_{n \rightarrow \infty} E[\hat{\theta}_n] = \theta.$$

A major drawback of MMEs is that often give estimators that can take values outside of the parameter space! Also, they are not going to be applicable to distributions where required moments do not exist.

5.2 Maximum Likelihood Estimators (MLEs)

While method of moments estimators are highly intuitive, maximum likelihood estimators have many desirable properties, particularly large sample properties.

**Idea**

Use, as an estimate of an unknown parameter, a value in the parameter space that makes the observed data “most likely”.

Consider the following oversimplified “motivating” example.

Example:

Suppose you have an unfair coin where $p = P(\text{“Heads”})$ is known to be one of 0.2, 0.3, or 0.8.

Let’s suppose that we toss the coin twice and use the results to try to estimate p .

If the result of both coin flips are “Heads”, we might be inclined to guess that the coin is biased towards heads and that p , the probability of getting heads is the 0.8. (Of course, we would be much more convinced if we flipped it 10 times and saw 9 or 10 heads, but we have to work with what we have!)

If the result of both coin flips are “Tails”, we would be inclined to guess that p is most likely 0.2.

If we saw one heads and one tails in our two flips, we might go with estimating p as 0.3.

Formally, the model here is that we have a random sample, X_1, X_2 , of size 2 from the Bernoulli(p) distribution. That is,

$$X_i = \begin{cases} 1 & , \text{ if “H”} \\ 0 & , \text{ if “T”} \end{cases}$$

The pdf for each X_i is

$$f(x; p) = p^x (1 - p)^{1-x} I_{\{0,1\}}(x).$$

The joint pdf for X_1 and X_2 is

$$f(x_1, x_2; p) \stackrel{\text{indep}}{=} f(x_1; p) \cdot f(x_2; p) = p^{x_1+x_2} (1-p)^{2-x_1-x_2} I_{\{0,1\}}(x_1) I_{\{0,1\}}(x_2)$$

For these discrete random variables, this is $P(X_1 = x_1, X_2 = x_2)$. For fixed observations x_1 and x_2 , we would like to know the value of p from the

simplified parameter space $\{0.2, 0.3, 0.8\}$ that maximizes the probability of observing a particular (x_1, x_2) pair.

For this simple example, since everything is discrete, I'll just tabulate the probabilities.

		(x_1, x_2)			
		(0,0)	(0,1)	(1,0)	(1,1)
p	0.2	0.64	0.16	0.16	0.04
	0.3	0.49	0.21	0.21	0.09
	0.8	0.04	0.16	0.16	0.64

For each (x_1, x_2) , the maximum likelihood estimator of p is the value of p that maximizes the probability of seeing that (x_1, x_2) :

$$\hat{p}_{mle} = \begin{cases} 0.2 & , \text{ if } (x_1, x_2) = (0, 0) \\ 0.3 & , \text{ if } (x_1, x_2) = (0, 1) \text{ or } (1, 0) \\ 0.8 & , \text{ if } (x_1, x_2) = (1, 1) \end{cases}$$

□

In general, for fixed $(x_1, x_2, \dots, x_n)$, we are maximizing the joint density $f(\vec{x}; \theta) = f(x_1, x_2, \dots, x_n; \theta)$ with respect to θ .

Note that

$$f(\vec{x}; \theta) = f(x_1, x_2, \dots, x_n; \theta) \stackrel{iid}{=} f(x_1; \theta) \cdot f(x_2; \theta) \cdots f(x_n; \theta) = \prod_{i=1}^n f(x_i; \theta).$$

When thought of as a function of θ (the x 's are thought of as fixed), $f(\vec{x}; \theta)$ is called the **likelihood function** and is denoted by $L(\theta)$.

$$L(\theta) = f(\vec{x}; \theta).$$

Actually... the value of θ that maximizes $L(\theta)$ is the same value that maximizes $3L(\theta)$, $\frac{1}{\sqrt{2\pi}}L(\theta)$, or even $(\prod_{i=1}^n x_i) \cdot L(\theta)$, because $\prod_{i=1}^n x_i$ is just another constant now.

**Definition**

A **likelihood function** is denoted by $L(\theta)$ and is defined as any function proportional to the joint pdf $f(\vec{x}; \theta)$ when thought of as a function of θ .

The **maximum likelihood estimator** of θ is the value of θ that maximizes $L(\theta)$.

5.2.1 Examples**Example:**

Let us revisit the coin toss example with n flips and with p , the probability of getting heads on any one flip to be an unknown number in the parameter space $\{p : 0 \leq p \leq 1\}$.

The data is $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} \text{Bernoulli}(p)$, so, the joint pdf is

$$\begin{aligned} f(\vec{x}; p) &\stackrel{iid}{=} \prod_{i=1}^n f(x_i; p) \\ &= \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i} I_{\{0,1\}}(x_i) \\ &= p^{\sum x_i} (1-p)^{n-\sum x_i} \prod_{i=1}^n I_{\{0,1\}}(x_i). \end{aligned}$$

Since the product of indicators does not involve the unknown p , it is a constant of proportionality and hence may be dropped to form a likelihood function

$$L(p) = p^{\sum x_i} (1-p)^{n-\sum x_i}.$$

We must maximize $L(p)$ with respect to p .

In the vast majority of cases, it is more convenient to instead maximize the log of the likelihood function. Note that $y = \ln x$ is an increasing function of x , so it will preserve the order of everything (ex: $x_1 < x_2 \Rightarrow \ln x_1 < \ln x_2$), thus $L(\theta)$ and $\ln L(\theta)$ are maximized at the same value of θ .

We will use $\ell(\theta)$ to denote the log-likelihood function $\ln L(\theta)$.

Here,

$$\ell(p) = \ln L(p) = \sum x_i \ln p + (n - \sum x_i) \ln(1-p).$$

Taking the derivative with respect to p gives

$$\frac{d}{dp}\ell(p) = \frac{\sum x_i}{p} - \frac{n - \sum x_i}{1-p} \stackrel{set}{=} 0$$

Multiply through by $p(1-p)$ to kill all denominators:

$$\begin{aligned} \sum x_i(1-p) - (n - \sum x_i)p &= 0 \\ \Downarrow \\ \sum x_i - p \sum x_i - np + p \sum x_i &= 0 \\ \sum x_i - np = 0 &\quad \Rightarrow p = \sum x_i/n. \end{aligned}$$

In the end, put capital letters back in for the x 's to give an estimator for any future sample drawn from the Bernoulli distribution rather than an estimate for fixed x 's, and put a hat on the parameter because you're done!

$$\hat{p} = \frac{\sum X_i}{n} = \bar{X}.$$

□

Example:

Let $X_1, X_2, \dots, X_n$ be a random sample with pdf $f(x; \theta) = \theta x^{\theta-1} I_{(0,\infty)}(x)$ for $\theta > 0$.

Find the maximum likelihood estimator of θ .

The joint pdf is

$$f(\vec{x}; \theta) = \prod_{i=1}^n \theta x_i^{\theta-1} I_{(0,1)}(x_i) = \theta^n \left(\prod_{i=1}^n x_i \right)^{\theta-1} \prod_{i=1}^n I_{(0,1)}(x_i)$$

A likelihood function is

$$L(\theta) = \theta^n \left(\prod_{i=1}^n x_i \right)^{\theta-1}.$$

The log-likelihood is

$$\ell(\theta) = n \ln \theta - (\theta - 1) \ln \left(\prod_{i=1}^n x_i \right) = n \ln \theta - (\theta - 1) \sum_{i=1}^n \ln x_i$$

$$\frac{\partial}{\partial \theta} \ell(\theta) = \frac{n}{\theta} - \sum_{i=1}^n \ln x_i \stackrel{\text{set}}{=} 0$$

$$\Rightarrow \boxed{\hat{\theta} = \frac{-n}{\sum_{i=1}^n \ln x_i}.$$

(Note: $\partial^2 / \partial \theta^2 \ell(\theta) = -n/\theta^2 < 0$, so we did find a maximum and not a minimum.) ■

Example: (Two Parameters)

Let $X_1, X_2, \dots, X_n$ be a random sample from the $N(\mu, \sigma^2)$ distribution. Find MLEs for μ and σ^2 .

The pdf is

$$f(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}.$$

The joint pdf is

$$f(\vec{x}; \mu, \sigma) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x_i-\mu)^2} = (2\pi\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i-\mu)^2}$$

A likelihood is

$$L(\mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i-\mu)^2}.$$

(Note: You can split $(2\pi\sigma^2)^{-n/2} = (2\pi)^{-n/2}(\sigma^2)^{-n/2}$ and drop the 2π part. I'm going to leave it. Also note that we are thinking of σ^2 as a symbol and not as a “symbol squared”. It is just a parameter with the name “sigma squared”!)

The log-likelihood is

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

Two variables, two partial derivatives:

$$\begin{aligned}\frac{\partial}{\partial \mu} \ell(\mu, \sigma^2) &= \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) \stackrel{\text{set}}{=} 0 \\ \frac{\partial}{\partial \sigma^2} \ell(\mu, \sigma^2) &= -\frac{n}{2} \frac{2\pi}{2\pi\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (x_i - \mu)^2 \stackrel{\text{set}}{=} 0\end{aligned}$$

The first equation implies that

$$\sum_{i=1}^n (x_i - \mu) = 0 \quad \Rightarrow \quad \sum_{i=1}^n x_i - n\mu = 0 \quad \Rightarrow \quad \boxed{\hat{\mu} = \bar{X}}.$$

The second equation, after cancelling the 2π and multiplying both sides by 2, becomes

$$-\frac{n}{\sigma^2} + \frac{1}{(\sigma^2)^2} \sum_{i=1}^n (x_i - \mu)^2 = 0.$$

Multiplying both sides by $(\sigma^2)^2$ gives

$$-n\sigma^2 + \sum_{i=1}^n (x_i - \mu)^2 = 0.$$

Plugging in $\hat{\mu} = \bar{X}$ for μ and solving for σ^2 gives the MLE

$$\boxed{\hat{\sigma}^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n}}.$$

□

Remember, no matter how you did the algebra, with capital letters or lower case letters, in the end, make everything capital when giving your estimator. The estimator should be a random variable unless you have actually observed numerical data. At that point though, we would call it an “estimate” as opposed to an “estimator”.

Example: (Parameters “Mixed in with the Constraints”)

Let $X_1, X_2, \dots, X_n$ be a random sample from the distribution with pdf

$$f(x; \theta, \lambda) = \theta \lambda^\theta x^{-(\theta+1)} I_{(\lambda, \infty)}(x)$$

with $\theta > 0$, $\lambda > 0$.

Find MLEs for θ and λ .

The joint pdf is

$$f(\vec{x}; \theta, \lambda) = \prod_{i=1}^n \theta \lambda^\theta x_i^{-(\theta+1)} I_{(\lambda, \infty)}(x_i) = \theta^n \lambda^{n\theta} \left(\prod_{i=1}^n x_i \right)^{-(\theta+1)} I_{(\lambda, \infty)}(x_{(1)})$$

A likelihood function is

$$L(\theta, \lambda) = \theta^n \lambda^{n\theta} \left(\prod_{i=1}^n x_i \right)^{-(\theta+1)} I_{(\lambda, \infty)}(x_{(1)}) \quad (5.1)$$

but we must keep in mind that $I_{(\lambda, \infty)}(x_{(1)})$. For this problem, the indicator is not dropped when writing down a likelihood function, because it is not purely a function of the x 's and therefore is not a constant when we think of the joint pdf as a function of the parameters. However, it can be cumbersome to bring through the log-derivative process. Let's not forget about it but keep it "on the side".

The log-likelihood is

$$\ell(\theta, \lambda) = n \ln \theta + n\theta \ln \lambda - (\theta + 1) \sum_{i=1}^n \ln x_i.$$

The derivatives are

$$\begin{aligned} \frac{\partial}{\partial \theta} \ell(\theta, \lambda) &= \frac{n}{\theta} + n \ln \lambda - \sum_{i=1}^n \ln x_i \stackrel{set}{=} 0 \\ \frac{\partial}{\partial \lambda} \ell(\theta, \lambda) &= \frac{n\theta}{\lambda}. \end{aligned}$$

Note that the second equation does not give us any information. The sample size can't be zero and the parameter $\theta > 0$. This (one equation giving nothing) usually happens when the parameter is in the indicator, as it was here. The goal, however, is still the same— to maximize the likelihood.

Looking at (5.1), we see that the likelihood is an increasing function of λ . So, we want λ to be as large as possible. From our indicator, we see that the minimum value in the data set must be above λ . If you did not simplify

the indicator, the product of indicators says that, equivalently, all values in the data set must be above λ . So, looking at the data as fixed, we see that the largest λ can possibly be is the minimum value of the data set. That is, the MLE for λ is

$$\hat{\lambda} = X_{(1)}.$$

Plugging this in to the first of the two partial derivative equations, we solve for θ to get

$$\hat{\theta} = \frac{n}{\sum_{i=1}^n \ln X_i - n \ln X_{(1)}}.$$

□

5.2.2 The Invariance Property of MLEs

Let $X_1, X_2, \dots, X_n$ be a random sample from the exponential distribution with rate λ .

Let's find the MLE for $\tau(\lambda) = \lambda^2$.

The pdf is

$$f(x; \lambda) = \lambda e^{-\lambda x} I_{(0, \infty)}(x).$$

We can write it in terms of $\tau = \lambda^2$:

$$f(x; \lambda) = \sqrt{\tau} e^{-\sqrt{\tau} x} I_{(0, \infty)}(x) \stackrel{\text{call it}}{=} f(x; \tau)$$

The joint pdf is then

$$f(\vec{x}; \tau) \stackrel{iid}{=} \prod_{i=1}^n f(x_i; \tau) = \prod_{i=1}^n \sqrt{\tau} e^{-\sqrt{\tau} x_i} I_{(0, \infty)}(x_i) = \tau^{n/2} e^{-\sqrt{\tau} \sum x_i} \prod_{i=1}^n I_{(0, \infty)}(x_i).$$

So, a likelihood function is

$$L(\tau) = \tau^{n/2} e^{-\sqrt{\tau} \sum x_i}.$$

The log-likelihood is

$$\ell(\tau) = \frac{n}{2} \ln \tau - \sqrt{\tau} \sum x_i.$$

The derivative gives

$$\frac{d}{d\tau}\ell(\tau) = \frac{n}{2} \frac{1}{\tau} - \frac{1}{2\sqrt{\tau}} \sum x_i \stackrel{set}{=} 0.$$

Multiplying through by 2τ ,

$$\begin{aligned} n - \sqrt{\tau} \sum x_i &= 0 \\ \Rightarrow \tau^{1/2} &= n / \sum x_i \\ \Rightarrow \tau &= \left(n / \sum x_i \right)^2 \end{aligned}$$

So the MLE of $\tau = \tau(\lambda) = \lambda^2$ is

$$\widehat{\tau(\lambda)} = (1/\bar{X})^2.$$

Recall that the MLE of λ was

$$\hat{\lambda} = 1/\bar{X}.$$

Since $\tau(\lambda) = \lambda^2$, we see that

$$\widehat{\tau(\lambda)} = \tau(\hat{\lambda}).$$

This was no accident! This will always happen for MLEs and is known as the **invariance property of MLEs**.



Definition

If $\hat{\theta}$ is the MLE of θ , then the MLE of a function $\tau(\theta)$ is

$$\widehat{\tau(\theta)} = \tau(\hat{\theta}).$$

This is known as the **invariance property** of MLEs.

In words, the MLE of a function of a parameter is the function with the MLE of the parameter plugged in.

We will now prove that MLEs are invariant.

Let $L(\theta)$ denote the likelihood function and let L^* denote the reparameterized likelihood function. In other words, the relationship is

$$L(\theta) = L^*(\tau(\theta)).$$

Then by the chain rule we get the second equality below:

$$\frac{\partial}{\partial \theta} L(\theta) = \frac{\partial}{\partial \theta} L^*(\tau(\theta)) = \frac{\partial}{\partial \tau} L^*(\tau(\theta)) \cdot \frac{\partial \tau}{\partial \theta}.$$

We can now see that if $\frac{\partial \tau}{\partial \theta} \neq 0$, then $\frac{\partial}{\partial \tau} L^*(\tau(\theta)) = 0$ whenever $\frac{\partial}{\partial \theta} L(\theta) = 0$.

Well, $\frac{\partial}{\partial \theta} L(\theta) = 0$ at $\theta = \hat{\theta}$, because that's the definition of the MLE $\hat{\theta}$. ie:

$$\left. \frac{\partial}{\partial \theta} L(\theta) \right|_{\theta=\hat{\theta}} = 0$$

Hence

$$\left. \frac{\partial}{\partial \tau} L^*(\tau(\theta)) \right|_{\theta=\hat{\theta}} = \left. \frac{\partial}{\partial \theta} L(\theta) \right|_{\theta=\hat{\theta}} = 0$$

Therefore $\tau(\hat{\theta})$ is a solution to $\frac{\partial}{\partial \tau} L^*(\tau(\theta)) = 0$ which is the equation that comes up in finding an MLE in the reparameterized case. Well $\widehat{\tau(\theta)}$ is, by definition, our solution to $\frac{\partial}{\partial \tau} L^*(\tau(\theta)) = 0$. So

$$\widehat{\tau(\theta)} = \tau(\hat{\theta}).$$

“Aside” A Note About “One-to-One-ness”:

I have been abusing notation a bit by writing $\tau = \tau(\theta)$. Here, τ represents the value of the function $\tau(\cdot)$ when it is evaluated at θ . In the following explanation, I’m going to really need to make the distinction, so I’m going to not use τ as both a point and a function, but rather, I will assume that τ is some function evaluated at θ like

$$\tau = g(\theta).$$

If g is one-to-one, then we may invert it, so $\theta = g^{-1}(\tau)$ and hence

$$L^*(\tau) = L(g^{-1}(\tau)).$$

Therefore, τ will maximize $L^*(\cdot)$ if $g^{-1}(\tau)$ maximizes $L(\cdot)$, but $L(\cdot)$ is maximized at $\hat{\theta}$, so $g^{-1}(\hat{\tau}) = \hat{\theta}$ and $g(\hat{\theta}) = \hat{\tau} = g(\hat{\theta})$.

In the case that g is not one-to-one, there may not necessarily be a unique solution to $g(\theta) = \tau$ for every value of τ . In this case, one usually extends the definition of $L^*(\tau)$ as follows:

If for every value τ , $L(\theta)$ attains a maximum over a subset of the parameter space $\{\theta : g(\theta) = \tau\}$, we redefine $L^*(\tau)$ to be this maximum value.

(This is kind of like a “generalized inverse problem”.)

We are not finished about properties of MLEs. In order to talk about some of their asymptotic properties, we need some machinery that is substantial enough to get its own section!

5.3 The Cramér-Rao Lower Bound (CRLB)

The Cramér-Rao Lower Bound (CRLB) for an parameter θ (or for a function $\tau(\theta)$) is a lower bound on the variance of all unbiased estimators of θ (or $\tau(\theta)$). Recall that, when we have two unbiased estimators for θ , say $\hat{\theta}_1$ and $\hat{\theta}_2$, we say (all else being equal) that $\hat{\theta}_1$ is the better estimator if $Var[\hat{\theta}_1] \leq Var[\hat{\theta}_2]$. (These variances are functions of θ so really we could only say that $\hat{\theta}_1$ is the better estimator if $Var[\hat{\theta}_1] \leq Var[\hat{\theta}_2]$ for all θ .) You might be able to find a third unbiased estimator with even lower variance. The CRLB tells us how low we could possibly go with these variances. It is not always achievable. That is, the unbiased estimator with the lowest possible variance might still have a variance that is greater than the CRLB. Still, I think the UMVUE is near...

5.3.1 Definition and Derivation

Let $X_1, X_2, \dots, X_n$ be a random sample from the distribution with pdf $f(x; \theta)$.

Consider estimating some function $\tau(\theta)$.

Let $T = t(\vec{X})$ be any unbiased estimator of $\tau(\theta)$.

Then

$$\text{Var}[T] \geq \frac{[\tau'(\theta)]^2}{\mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \ln f(\vec{X}; \theta) \right)^2 \right]}. \quad (5.2)$$

That thing on the right is known as the Cramér-Rao Lower Bound (CRLB) for the variance of all unbiased estimators of $\tau(\theta)$.

Note that the X 's from the random sample appear only in the denominator of the CRLB. We can think of this denominator as containing all of the “information” from the sample. In fact, the denominator is known as the **Fisher Information** and we will denote it by $I_n(\theta)$.

So, we can write the CRLB for the variance of all unbiased estimators of $\tau(\theta)$ as:

$$\text{CRLB}_{\tau(\theta)} = \frac{[\tau'(\theta)]^2}{I_n(\theta)}.$$

In order to prove (5.2) we need to use the **Cauchy-Schwarz inequality**:

$$\left(\int f(x)g(x) dx \right)^2 \leq \left(\int f^2(x) dx \right) \cdot \left(\int g^2(x) dx \right)$$

You may also have seen the Cauchy-Schwarz inequality written in an equivalent form for sums, or, more generally, inner products.

Proof of C-S:

First note that

$$\int \int [f(x)g(y) - f(y)g(x)]^2 dy dx \geq 0.$$

Multiplying out the left-hand-side then gives us

$$\int \int f^2(x)g^2(y) dy dx - 2 \int \int f(x)f(y)g(x)g(y) dy dx + \int \int f^2(y)g^2(x) dy dx \geq 0 \quad (5.3)$$

Note that the first and third terms are exactly the same. As for the middle integral, pulling the “ x stuff” out of the inner y -integral gives us

$$\begin{aligned} \int \int f(x)f(y)g(x)g(y) dy dx &= \int f(x)g(x) \int f(y)g(y) dy dx \\ &= \int f(x)g(x) \left(\int f(y)g(y) dy \right) dx \\ &= \left(\int f(y)g(y) dy \right) \int f(x)g(x) dx \\ &= \left(\int f(x)g(x) dx \right)^2 \end{aligned}$$

So, (5.3) becomes

$$2 \int \int f^2(x)g^2(y) dy dx - 2 \left(\int f(x)g(x) dx \right)^2 \geq 0$$

$\Downarrow$

$$\int \int f^2(x)g^2(y) dy dx - \left(\int f(x)g(x) dx \right)^2 \geq 0$$

$\Downarrow$

$$\left(\int f(x)g(x) dx \right)^2 \leq \left(\int f^2(x) dx \right) \cdot \left(\int g^2(x) dx \right). \quad \checkmark$$

We are now ready to prove (5.2). We will show that

$$\tau'(\theta) = \mathbb{E} \left[(T - \tau(\theta)) \left(\frac{\partial}{\partial \theta} \ln f(\vec{X}; \theta) \right) \right]. \quad (5.4)$$

Then we will have the CRLB since expectation operator is actually an integral to which we can apply the Cauchy-Schwarz inequality:

$$\begin{aligned} [\tau'(\theta)]^2 &= \left(\mathbb{E} \left[(T - \tau(\theta)) \left(\frac{\partial}{\partial \theta} \ln f(\vec{X}; \theta) \right) \right] \right)^2 \\ &\stackrel{C-S}{\leq} \mathbb{E}[(T - \tau(\theta))^2] \cdot \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \ln f(\vec{X}; \theta) \right)^2 \right] \\ &= \text{Var}[T] \cdot \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \ln f(\vec{X}; \theta) \right)^2 \right] \end{aligned}$$

We then have the CRLB, provided that

$$0 < \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \ln f(\vec{X}; \theta) \right)^2 \right] < \infty$$

so that we can divide it to the other side.

Proof of (5.4): Since $\mathbb{E}[T] = \tau(\theta)$

$$\tau'(\theta) = \frac{\partial}{\partial \theta} \tau(\theta) = \frac{\partial}{\partial \theta} \mathbb{E}[T] = \frac{\partial}{\partial \theta} \int t(\vec{x}) f(\vec{x}; \theta) d\vec{x}$$

Since we are trying to see something like $T - \tau(\theta)$, note that this is equal to

$$\begin{aligned} &\frac{\partial}{\partial \theta} \int t(\vec{x}) f(\vec{x}; \theta) d\vec{x} - \tau(\theta) \frac{\partial}{\partial \theta} 1 \\ &= \frac{\partial}{\partial \theta} \int t(\vec{x}) f(\vec{x}; \theta) d\vec{x} - \tau(\theta) \frac{\partial}{\partial \theta} \int f(\vec{x}; \theta) d\vec{x} \\ &= \int t(\vec{x}) \frac{\partial}{\partial \theta} f(\vec{x}; \theta) d\vec{x} - \int \tau(\theta) \frac{\partial}{\partial \theta} f(\vec{x}; \theta) d\vec{x} \end{aligned}$$

if we assume that we are able to pull that derivative into the integral.

Continuing, we now have

$$\tau'(\theta) = \int (t(\vec{x}) - \tau(\theta)) \frac{\partial}{\partial \theta} f(\vec{x}; \theta) d\vec{x}$$

We want to see an expectation, which means that we want the integrand to end in $f(\vec{x}; \theta)$ (as opposed to $\frac{\partial}{\partial \theta} f(\vec{x}; \theta)$). Note that

$$\frac{\partial}{\partial \theta} \ln f(\vec{x}; \theta) = \frac{\frac{\partial}{\partial \theta} f(\vec{x}; \theta)}{f(\vec{x}; \theta)},$$

so

$$\frac{\partial}{\partial \theta} f(\vec{x}; \theta) = \left(\frac{\partial}{\partial \theta} \ln f(\vec{x}; \theta) \right) \cdot f(\vec{x}; \theta).$$

Therefore,

$$\tau'(\theta) = \int (t(\vec{x}) - \tau(\theta)) \left(\frac{\partial}{\partial \theta} \ln f(\vec{x}; \theta) \right) f(\vec{x}; \theta) d\vec{x}$$

and this integral is the definition of

$$\mathbb{E}[(T - \tau(\theta)) \left(\frac{\partial}{\partial \theta} \ln f(\vec{X}; \theta) \right)].$$

Yay!

The CRLB is said to hold under certain “regularity conditions”. These are the things we needed to hold in order to complete the above proof. They are

- $0 < \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \ln f(\vec{X}; \theta) \right)^2 \right] < \infty$
- $\frac{\partial}{\partial \theta} \int f(\vec{x}; \theta) d\vec{x} = \int \frac{\partial}{\partial \theta} f(\vec{x}; \theta) d\vec{x}$
- $\frac{\partial}{\partial \theta} \ln f(\vec{x}; \theta)$ exists

All integrals are evaluated over the domain of $f(\vec{x}; \theta)$. **Note that the second of these conditions is violated whenever the parameter θ is involved in the limits of integration.** For example, for the $unif(0, \theta)$ distribution the integrals would go from 0 to θ and therefore we would not be able to pass the derivative with respect to θ to the inside of the integral!

Example:

Let $X_1, X_2, \dots, X_n$ be a random sample from the Bernoulli distribution with parameter p .

- a) Find the CRLB for the variance of all unbiased estimators of p . ($\tau(p) = p$)

- a) Find the CRLB for the variance of all unbiased estimators of $\tau(p) = p(1-p)$.

$$\begin{aligned}
 f(x; p) &= p^x (1-p)^{1-x} I_{\{0,1\}}(x) \\
 &\Downarrow \\
 f(\vec{x}; p) &= p^{\sum x_i} (1-p)^{n-\sum x_i} \prod_{i=1}^n I_{\{0,1\}}(x_i)
 \end{aligned}$$

Ignoring the indicator (it just becomes cumbersome when dealing with the logs and derivatives), we have

$$\begin{aligned}
 \ln f(\vec{x}; p) &= \sum x_i \ln p + (n - \sum x_i) \ln(1-p) \\
 &\Downarrow \\
 \frac{\partial}{\partial p} \ln f(\vec{x}; p) &= \frac{\sum x_i}{p} - \frac{n - \sum x_i}{1-p} = \frac{\sum x_i - np}{p(1-p)}
 \end{aligned}$$

So,

$$\begin{aligned}
 I_n(p) &= \mathbb{E} \left[\left(\frac{\partial}{\partial p} \ln f(\vec{X}; p) \right)^2 \right] \\
 &= \mathbb{E} \left[\left(\frac{\sum X_i - np}{p(1-p)} \right)^2 \right] \\
 &= \frac{1}{p^2(1-p)^2} \mathbb{E} \left[(\sum X_i - np)^2 \right]
 \end{aligned}$$

Since the sum of n Bernoulli(p) random variables has the $\text{bin}(n, p)$ distribution and since the mean of the binomial is np , this expectation is simply the variance of the $\text{bin}(n, p)$ distribution. This is $np(1-p)$. So, the n -dimensional Fisher information is

$$I_n(p) = \frac{1}{p^2(1-p)^2} \cdot np(1-p) = \frac{n}{p(1-p)}.$$

So, the CRLB for the variance of all unbiased estimators of p ($\tau(p) = p$) is

$$CRLB_p = \frac{[\tau'(p)]^2}{I_n(p)} = \frac{[1]^2}{\frac{n}{p(1-p)}} = \frac{p(1-p)}{n}.$$

For part (b), $\tau(p) = p(1 - p)$ and

$$CRLB_{p(1-p)} = \frac{[\tau'(p)]^2}{I_n(p)} = \frac{[1 - 2p]^2}{\frac{n}{p(1-p)}} = \frac{(1 - 2p)^2 p(1 - p)}{n}.$$

□

5.3.2 Computational Simplifications for the CRLB

In this section, we will discuss some computational simplifications for the CRLB. I'll number them starting from 0 since the first one is only there to prove the others but not something you would actually want to use in computation.

$$\boxed{0} \quad \mathbb{E} \left[\frac{\partial}{\partial \theta} \ln f(\vec{X}; \theta) \right] = 0$$

Proof:

$$\begin{aligned} \mathbb{E} \left[\frac{\partial}{\partial \theta} \ln f(\vec{X}; \theta) \right] &= \int \frac{\partial}{\partial \theta} \ln f(\vec{x}; \theta) \cdot f(\vec{x}; \theta) d\vec{x} = \int \frac{\partial}{\partial \theta} f(\vec{x}; \theta) d\vec{x} \\ &= \frac{\partial}{\partial \theta} \int f(\vec{x}; \theta) d\vec{x} = \frac{\partial}{\partial \theta} 1 = 0 \quad \checkmark \end{aligned}$$

$$\boxed{1} \quad \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \ln f(\vec{X}; \theta) \right)^2 \right] = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \ln f(\vec{X}; \theta) \right]$$

Proof: From $\boxed{0}$, we have that

$$\mathbb{E} \left[\frac{\partial}{\partial \theta} \ln f(\vec{X}; \theta) \right] = 0$$

which is

$$\int \frac{\partial}{\partial \theta} \ln f(\vec{x}; \theta) \cdot f(\vec{x}; \theta) d\vec{x} = 0$$

Taking the derivative of both sides with respect to θ ,

$$\frac{\partial}{\partial \theta} \int \frac{\partial}{\partial \theta} \ln f(\vec{x}; \theta) \cdot f(\vec{x}; \theta) d\vec{x} = \frac{\partial}{\partial \theta} 0.$$

The right-hand side is still zero. On the left-hand side, we bring the derivative inside the integral and use the product rule to get

$$\int \left[\frac{\partial}{\partial \theta} \ln f(\vec{x}; \theta) \cdot \frac{\partial}{\partial \theta} f(\vec{x}; \theta) + \frac{\partial^2}{\partial \theta^2} \ln f(\vec{x}; \theta) \cdot f(\vec{x}; \theta) \right] d\vec{x} = 0$$

$$\Downarrow$$

$$\int \frac{\partial}{\partial \theta} \ln f(\vec{x}; \theta) \cdot \underbrace{\frac{\partial}{\partial \theta} f(\vec{x}; \theta)}_{\frac{\partial}{\partial \theta} \ln f(\vec{x}; \theta) \cdot f(\vec{x}; \theta)} d\vec{x} + \int \frac{\partial^2}{\partial \theta^2} \ln f(\vec{x}; \theta) \cdot f(\vec{x}; \theta) d\vec{x} = 0$$

$$\Downarrow$$

$$\int \left(\frac{\partial}{\partial \theta} \ln f(\vec{x}; \theta) \right)^2 \cdot f(\vec{x}; \theta) d\vec{x} + \int \frac{\partial^2}{\partial \theta^2} \ln f(\vec{x}; \theta) \cdot f(\vec{x}; \theta) d\vec{x} = 0$$

$$\Downarrow$$

$$\mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \ln f(\vec{X}; \theta) \right)^2 \right] + \mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \ln f(\vec{X}; \theta) \right] = 0$$

$$\Downarrow$$

$$\mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \ln f(\vec{X}; \theta) \right)^2 \right] = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \ln f(\vec{X}; \theta) \right] \quad \checkmark$$

2 For $X_1, X_2, \dots, X_n$ iid,

$$\mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \ln f(\vec{X}; \theta) \right)^2 \right] = n \cdot \mathbb{E} \left[(\ln f(X_1; \theta))^2 \right].$$

i.e. $I_n(\theta) \stackrel{iid}{=} nI_1(\theta)$

Proof:

$$\begin{aligned}
 \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \ln f(\vec{X}; \theta) \right)^2 \right] &= \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \ln \prod_{i=1}^n f(X_i; \theta) \right)^2 \right] \\
 &= \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \sum_{i=1}^n \ln f(X_i; \theta) \right)^2 \right] \\
 &= \mathbb{E} \left[\left(\sum_{i=1}^n \frac{\partial}{\partial \theta} \ln f(X_i; \theta) \right)^2 \right] \\
 &= \mathbb{E} \left[\left(\sum_{i=1}^n \frac{\partial}{\partial \theta} \ln f(X_i; \theta) \right) \left(\sum_{j=1}^n \frac{\partial}{\partial \theta} \ln f(X_j; \theta) \right) \right] \\
 &= \mathbb{E} \left[\sum_{i=1}^n \sum_{j=1}^n \frac{\partial}{\partial \theta} \ln f(X_i; \theta) \cdot \frac{\partial}{\partial \theta} \ln f(X_j; \theta) \right] \\
 &= \sum_{i=1}^n \sum_{j=1}^n \mathbb{E} \left[\frac{\partial}{\partial \theta} \ln f(X_i; \theta) \cdot \frac{\partial}{\partial \theta} \ln f(X_j; \theta) \right]
 \end{aligned}$$

Now when $j \neq i$, X_i and X_j imply that $\frac{\partial}{\partial \theta} \ln f(X_i; \theta)$ is independent of $\frac{\partial}{\partial \theta} \ln f(X_j; \theta)$, and, in this case

$$\mathbb{E} \left[\frac{\partial}{\partial \theta} \ln f(X_i; \theta) \frac{\partial}{\partial \theta} \ln f(X_j; \theta) \right] = \mathbb{E} \left[\frac{\partial}{\partial \theta} \ln f(X_i; \theta) \right] \cdot \mathbb{E} \left[\frac{\partial}{\partial \theta} \ln f(X_j; \theta) \right] \stackrel{\text{by } \boxed{0}}{=} 0 \cdot 0 = 0.$$

Thus,

$$\begin{aligned}
 \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \ln f(\vec{X}; \theta) \right)^2 \right] &= \sum_{i=1}^n \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \ln f(X_i; \theta) \right)^2 \right] \\
 &\stackrel{\text{ident.}}{=} n \cdot \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \ln f(X_1; \theta) \right)^2 \right], \quad \text{as desired.}
 \end{aligned}$$

Example:

Let $X_1, X_2, \dots, X_n$ be a random sample from the $\Gamma(2, \beta)$ distribution.

Find the CRLB for the variance of all unbiased estimators of β .

$$f(x; \beta) = \beta^2 x e^{-\beta x} \quad \text{for } x > 0$$

$\Downarrow$

$$Lr f(x; \beta) = 2 \ln \beta + \ln x - \beta x$$

$$\Downarrow$$

$$\frac{\partial}{\partial \beta} f(x; \beta) = \frac{2}{\beta} - x$$

So,

$$I_1(\beta) = \mathbb{E} \left[\left(\frac{\partial}{\partial \beta} \ln f(X_1; \beta) \right)^2 \right] = \mathbb{E} \left[\left(X_1 - \frac{2}{\beta} \right)^2 \right] = \text{Var}[X] = \frac{2}{\beta^2}$$

Then

$$I_n(\beta) \stackrel{iid}{=} n \cdot I_1(\beta) = \frac{2n}{\beta^2}$$

and

$$CRLB_\beta = \frac{[1]^2}{I_n(\beta)} = \frac{\beta^2}{2n}.$$

(Here, $\tau(\beta) = \beta$.)

■

5.4 Asymptotic Properties of MLEs



Definition

An unbiased estimator whose variance attains the CRLB is said to be **efficient**.



Definition

The **efficiency** of an estimator T of $\tau(\theta)$ is the ratio

$$\frac{CRLB_{\tau(\theta)}}{\text{Var}[T]}.$$

Note that an efficient estimator has efficiency 1.

Under certain “regularity conditions”, involving the existence of certain derivative, integral, logs, and “the parameter is not mixed in with the constraints”, we have, for an MLE $\hat{\theta} = \hat{\theta}_n$ of θ that

1. $\hat{\theta}_n$ **exists** and is **unique**.
2. $\hat{\theta}_n$ is a **consistent** estimator of θ . ($\hat{\theta}_n \xrightarrow{P} \theta$)
3. $\hat{\theta}_n$ is **asymptotically unbiased** for θ . ($\lim_{n \rightarrow \infty} E[\hat{\theta}_n] = \theta$)
4. $\hat{\theta}_n$ is **asymptotically efficient**. ($\lim_{n \rightarrow \infty} CRLB_\theta / Var[\hat{\theta}_n] = 1$)
- 5.

$$\hat{\theta}_n \stackrel{asympt}{\sim} N(\theta, CRLB_\theta)$$

i.e.

$$\frac{\hat{\theta}_n - \theta}{\sqrt{CRLB_\theta}} \xrightarrow{\mathcal{D}} N(0, 1)$$

Example:

Suppose that $X_1, X_2, \dots, X_n$ is a random sample from the $\Gamma(2, \beta)$ distribution.

- a) Find the MLE of β .

The pdf is

$$f(x; \beta) = \beta^2 x e^{-\beta x} I_{(0, \infty)}(x).$$

The joint pdf is

$$\begin{aligned} f(\vec{x}; \beta) &\stackrel{iid}{=} \prod_{i=1}^n f(x_i; \beta) \\ &= \prod_{i=1}^n \beta^2 x_i e^{-\beta x_i} I_{(0, \infty)}(x_i) \\ &= \beta^{2n} \left(\prod_{i=1}^n x_i \right) e^{-\beta \sum x_i} \left(\prod_{i=1}^n I_{(0, \infty)}(x_i) \right). \end{aligned}$$

So, a likelihood function is

$$L(\beta) = \beta^{2n} e^{-\beta \sum x_i}.$$

The log-likelihood is

$$\ell(\beta) = 2n \ln \beta - \beta \sum x_i.$$

The derivative with respect to β is

$$\frac{\partial}{\partial \beta} \ell(\beta) = \frac{2n}{\beta} - \sum x_i = 0.$$

This implies that the MLE for β is

$$\hat{\beta} = \frac{2n}{\sum_{i=1}^n X_i}.$$

This can also be written in terms of $\bar{X}$, but leaving it as a sum is convenient for the next part.

b) Is this MLE an unbiased estimator of β ?

$$\mathbb{E}[\hat{\beta}] = 2n \mathbb{E} \left[\frac{1}{\sum X_i} \right] = 2n \mathbb{E} \left[\frac{1}{W} \right]$$

where $W \sim \Gamma(2n, \beta)$.

Continuing,

$$\begin{aligned} \mathbb{E} \left[\frac{1}{W} \right] &= \int_{-\infty}^{\infty} \frac{1}{w} \cdot f_W(w) dw \\ &= \int_0^{\infty} \frac{1}{w} \cdot \frac{1}{\Gamma(2n)} \beta^{2n} w^{2n-1} e^{-\beta w} dw \\ &= \int_0^{\infty} \frac{1}{\Gamma(2n)} \beta^{2n} \underbrace{w^{2n-2} e^{-\beta w}}_{\substack{\text{looks like} \\ \Gamma(2n-1, \beta) \text{ pdf}}} dw \\ &= \beta \frac{\Gamma(2n-1)}{\Gamma(2n)} \underbrace{\int_0^{\infty} \frac{1}{\Gamma(2n-1)} \beta^{2n-1} w^{2n-2} e^{-\beta w} dw}_1 \\ &= \beta \frac{\Gamma(2n-1)}{\Gamma(2n)} = \frac{1}{2n-1} \beta. \end{aligned}$$

So,

$$\mathbb{E}[\hat{\beta}] = 2n \mathbb{E} \left[\frac{1}{W} \right] = \frac{2n}{2n-1} \beta.$$

The estimator is not unbiased. However, we see that it is asymptotically unbiased since

$$\lim_{n \rightarrow \infty} \mathbb{E}[\hat{\beta}] = \lim_{n \rightarrow \infty} \frac{2n}{2n-1} \beta = 1 \cdot \beta = \beta.$$

(You might prefer to stress the dependence on n by using $\hat{\beta}_n$ to denote the estimator instead of $\hat{\beta}$.)

c) Consider the unbiased estimator

$$\hat{\beta}_2 := \frac{2n-1}{2n} \hat{\beta} = \frac{2n-1}{2n} \frac{2n}{\sum X_i} = \frac{2n-1}{\sum X_i}.$$

Is this an efficient estimator?

In the previous section, we saw that the CRLB for the variance of all unbiased estimators of β (often shortened to the inaccurate “the CRLB for β ”) is

$$CRLB_{\beta} = \frac{\beta^2}{2n}.$$

We need to compute the variance for our estimator $\hat{\beta}_2$ of β and see if it is equal to the CRLB. Note that

- The CRLB is associated with the parameter and not the estimator.
- We are not going to compare the CRLB “for β ” to the variance of $\hat{\beta}$ because $\hat{\beta}$ is a biased estimator for β and the CRLB for β is a lower bound on the variance of all unbiased estimators of β .

$$\begin{aligned} \text{Var}[\hat{\beta}_2] &= \text{Var}\left[\frac{2n-1}{\sum X_i}\right] = (2n-1)^2 \text{Var}\left[\frac{1}{W}\right] \\ &= (2n-1)^2 \left\{ \mathbb{E}\left[\frac{1}{W^2}\right] - \left(\mathbb{E}\left[\frac{1}{W}\right]\right)^2 \right\} \\ \mathbb{E}\left[\frac{1}{W^2}\right] &= \int_{-\infty}^{\infty} \frac{1}{w^2} \cdot f_W(w) dw \\ &= \int_0^{\infty} \frac{1}{w^2} \cdot \frac{1}{\Gamma(2n)} \beta^{2n} w^{2n-1} e^{-\beta w} dw \\ &= \int_0^{\infty} \frac{1}{\Gamma(2n)} \beta^{2n} \underbrace{w^{2n-3} e^{-\beta w}}_{\text{looks like } \Gamma(2n-2, \beta) \text{ pdf}} dw \\ &= \beta^2 \frac{\Gamma(2n-2)}{\Gamma(2n)} \underbrace{\int_0^{\infty} \frac{1}{\Gamma(2n-2)} \beta^{2n-2} w^{2n-2} e^{-\beta w} dw}_1 \\ &= \beta^2 \frac{\Gamma(2n-2)}{\Gamma(2n)} = \frac{1}{(2n-1)(2n-2)} \beta^2. \end{aligned}$$

Putting it all together, we have

$$\begin{aligned}
 \text{Var}[\hat{\beta}_2] &= (2n-1)^2 \left\{ \mathbb{E} \left[\frac{1}{W^2} \right] - \left(\mathbb{E} \left[\frac{1}{W} \right] \right)^2 \right\} \\
 &= (2n-1)^2 \left\{ \frac{1}{(2n-1)(2n-2)} \beta^2 - \left(\frac{1}{2n-1} \beta \right)^2 \right\} \\
 &= \frac{1}{2n-2} \beta^2.
 \end{aligned}$$

Note that

$$\text{Var}[\hat{\beta}_2] = \frac{1}{2n-2} \beta^2 > \frac{1}{2n} \beta^2 = \text{CRLB}_\beta.$$

Our estimator did not achieve the CRLB, so it is not an efficient estimator of β . It is, however, asymptotically efficient since

$$\lim_{n \rightarrow \infty} \frac{\text{CRLB}_\beta}{\text{Var}[\hat{\beta}_2]} = \lim_{n \rightarrow \infty} \frac{[1/(2n)\beta^2]}{[1/(2n-1)]\beta^2} = \lim_{n \rightarrow \infty} \frac{2n-1}{2n} = 1.$$

d) According to the property list for MLEs, we should have that

$$\hat{\beta} \stackrel{\text{asympt}}{\sim} N(\beta, \text{CRLB}_\beta) = N(\beta, \beta^2/(2n)).$$

We will now give a sketch of the proof for a general MLE.

Asymptotic Normality of MLEs

< Need to finish >

5.5 Uniformly Minimum Variance Unbiased Estimators (UMVUEs)

We are close... to the estimator you've all been waiting for. Consider estimating some $\tau(\theta)$. Imagine finding an unbiased one whose variance is smaller than that of anyone else's unbiased estimator. But wait... when we compute variances of estimators for θ or a function of θ , it almost always depends on θ . When comparing the variance for two unbiased estimators, you could easily get that one is smaller for some values of θ and larger for

others. Since θ is unknown, we wouldn't know which estimator is better in terms of variance. The word “uniformly” in mathematics basically means “for all”. The uniformly minimum variance unbiased estimator (UMVUE) for $\tau(\theta)$ is an unbiased estimator that has smallest variance “uniformly in θ ”, meaning for all values of θ .

For us, “data” is a random sample $X_1, X_2, \dots, X_n$ from a distribution with pdf $f(x; \theta)$ where θ is typically unknown. Our goal is to estimate θ from the data. We often find ourselves considering some function of the data, such as the sum of “the X 's”, or the sample mean, or the minimum value in the data, just to name a few examples. Functions of the data, such as these, are known as “statistics”.

We shall often refer to the data as a vector $\vec{X} = (X_1, X_2, \dots, X_n)$ and will usually use the letter T to denote a statistic which is some function of the data:

$$T = t(\vec{X}),$$

but we will use the letter S ($S = s(\vec{X})$) in the special case of the definition in the next section.

5.5.1 Sufficient Statistics

Recall that a “statistic” is any function of the data.

Let $X_1, X_2, \dots, X_n$ be a random sample from some distribution with pdf $f(x; \theta)$, and consider, for example, the statistic $T = \sum_{i=1}^n X_i$.

This statistic condenses the n -dimensional random sample down to one dimension.

Question: If we know only this statistic instead of knowing the original sample, do we have less information about θ ?

**Definition (Informal)**

A statistic S is **sufficient for** θ if it condenses the random sample $X_1, X_2, \dots, X_n$ without sacrificing any information about θ .

“sufficient” = “enough”

If you know S , then knowing $(X_1, X_2, \dots, X_n)$ does not give you any more information about θ .

**Definition (Formal)**

Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with pdf $f(x; \theta)$. S is a **sufficient statistic** for θ if the conditional distribution of $(X_1, \dots, X_n)$ given S does not depend on θ .

Example:

An unfair coin with $p = P(\text{“Heads”})$ is tossed n times and the outcome is recorded for each toss.

i.e. We have a random sample $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Bernoulli}(p)$ where

$$X_i = \begin{cases} 1 & , \text{ if heads} \\ 0 & , \text{ if tails} \end{cases}$$

For guessing p , it would seem that knowing the total number of heads, $S = \sum_{i=1}^n X_i$, should provide as much information about p as knowing the actual 0-1 outcomes.

Let’s see...

$$X_i \sim \text{Bernoulli}(p) \Rightarrow \text{the pdf is } f(x; p) = p^x(1-p)^{1-x} I_{\{0,1\}}(x)$$

So, the joint pdf for $X_1, \dots, X_n$ is

$$f(\vec{x}; p) \stackrel{iid}{=} \prod_{i=1}^n f(x_i; p) = p^{\sum x_i} (1-p)^{n-\sum x_i} \prod_{i=1}^n I_{\{0,1\}}(x_i).$$

Note that $S = \sum_{i=1}^n X_i \sim \text{bin}(n, p)$.

The conditional pdf of $(X_1, \dots, X_n)$ is

$$\begin{aligned} f_{X_1, \dots, X_n|S}(x_1, \dots, x_n|s) &\stackrel{\text{discrete}}{=} P(X_1 = x_1, \dots, X_n = x_n|S = s) \\ &= \frac{P(X_1=x_1, \dots, X_n=x_n, S=s)}{P(S=s)} \end{aligned} \quad (5.5)$$

The Numerator: Consider the case, just as an example, where $n = 2$ and some fixed values in for x_1, x_2 , and s . Note that $S = X_1 + X_2$. The numerator might be

$$P(X_1 = 2, X_2 = 3, X_1 + X_2 = 1)$$

This probability is clearly zero since, if $X_1 = 2$ and $X_2 = 3$, we can not have $X_1 + X_2 = 1$.

On the other hand, the probability

$$P(X_1 = 2, X_2 = 3, X_1 + X_2 = 5)$$

is not necessarily zero and, furthermore, the third argument is a bit redundant. We can simply write

$$P(X_1 = 2, X_2 = 3, X_1 + X_2 = 1) = P(X_1 = 2, X_2 = 3)$$

So, in general, we have the numerator of (5.5) as

$$\begin{aligned} P(X_1 = x_1, \dots, X_n = x_n, S = s) &= \begin{cases} P(X_1 = x_1, \dots, X_n = x_n) & , \text{ if } s = \sum x_i \\ 0 & , \text{ if } s \neq \sum x_i \end{cases} \\ &= \begin{cases} p^{x_1}(1-p)^{1-x_1} I_{\{0,1\}}(x_1) \cdots p^{x_n}(1-p)^{n-x_n} I_{\{0,1\}}(x_n) & , \text{ if } s = \sum x_i \\ 0 & , \text{ if } s \neq \sum x_i \end{cases} \\ &= \begin{cases} p^{\sum x_i} (1-p)^{n-\sum x_i} \prod I_{\{0,1\}}(x_i) & , \text{ if } s = \sum x_i \\ 0 & , \text{ if } s \neq \sum x_i \end{cases} \end{aligned}$$

The Denominator: Since $S \sim \text{bin}(n, p)$,

$$P(S = s) = \binom{n}{s} p^s (1-p)^{n-s} I_{\{0,1,\dots,n\}}(s)$$

All Together Now... (5.5) becomes

$$P(X_1 = x_1, \dots, X_n = x_n | S = s) = \begin{cases} \frac{p^{\sum x_i} (1-p)^{n-\sum x_i} \prod I_{\{0,1\}}(x_i)}{\binom{n}{s} p^s (1-p)^{n-s} I_{\{0,1,\dots,n\}}(s)} & , \text{ if } s = \sum x_i \\ 0 & , \text{ if } s \neq \sum x_i \end{cases}$$

Since $s = \sum x_i$ in the first expression, we can cancel many things and are left with

$$P(X_1 = x_1, \dots, X_n = x_n | S = s) = \begin{cases} \frac{\prod I_{\{0,1\}}(x_i)}{\binom{n}{s} I_{\{0,1,\dots,n\}}(s)} & , \text{ if } s = \sum x_i \\ 0 & , \text{ if } s \neq \sum x_i \end{cases}$$

Since this does not depend on p , we have, by definition that S is a sufficient statistic for p for this Bernoulli distribution! □

Note that, for $S = s(\vec{X})$, the “ $S = s$ ” part of the numerator of the conditional pdf will always drop out like it did above when $s = s(\vec{x})$ (and that it will always be zero when $s \neq s(\vec{x})$), so we will always have

$$f_{X_1, X_2, \dots, X_n | S}(x_1, x_2, \dots, x_n | s) = \begin{cases} \frac{f(\vec{x}; \theta)}{f_S(s; \theta)} & , \text{ } s = s(\vec{x}) \\ 0 & , \text{ } s \neq s(\vec{x}) \end{cases}$$

Since this must be “ θ -free”, we require (for sufficiency) that

$$\frac{f(\vec{x}; \theta)}{f_S(s; \theta)} = h_1(\vec{x}),$$

or equivalently, and using the fact that $s = s(\vec{x})$,

$$f(\vec{x}; \theta) = h_1(\vec{x}) f_S(s(\vec{x}); \theta)$$

where, as the notation suggests, $h_1(\vec{x})$ is some function of $\vec{x}$ only.

Assuming that we don’t know the pdf for S , or even what S is ahead of time, we at least are looking for the joint pdf of the X ’s to split up into an “ x part” and a part with x ’s and θ ’s mixed together and where the x ’s are appearing through some function that we will call $s(\vec{x})$.

★ **The Factorization Criterion for Sufficiency**

If the joint pdf for $X_1, X_2, \dots, X_n$ can be written as

$$f(\vec{x}; \theta) = h(\vec{x})g(s(\vec{x}); \theta)$$

for some function h that does not depend on θ and some function g that depends on the x 's only through some function $s(\vec{x})$, then

$S = s(\vec{X})$ is sufficient for estimating θ .

Example:

For the random sample from the Bernoulli distribution,

$$\begin{aligned} f(\vec{x}; p) &= \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i} I_{\{0,1\}}(x_i) \\ &= p^{\sum x_i} (1-p)^{n-\sum x_i} \prod_{i=1}^n I_{\{0,1\}}(x_i) \\ &= \underbrace{\prod_{i=1}^n I_{\{0,1\}}(x_i)}_{h(\vec{x})} \underbrace{p^{\sum x_i} (1-p)^{n-\sum x_i}}_{g(s(\vec{x}); p)} \end{aligned}$$

In the part where the p 's and x 's are “mixed”, the x 's appear as the sum $s(\vec{x}) = \sum x_i$. By the Factorization Criterion, $S = \sum X_i$ is sufficient for estimating p . ■

Example:

Let $X_1, X_2, \dots, X_n$ be a random sample from the $unif(0, \theta)$ distribution. Find a sufficient statistic for θ .

The joint pdf is

$$f(\vec{x}; \theta) = \prod_{i=1}^n \frac{1}{\theta} I_{(0,\theta)}(x_i) = \frac{1}{\theta^n} \prod_{i=1}^n I_{(0,\theta)}(x_i).$$

That product of indicators could be written in terms of the min and max of the sample as $I_{(0,\theta)}(x_{(1)}) I_{(0,\theta)}(x_{(n)})$ or even as $I_{(0,x_{(n)})}(x_{(1)}) I_{(0,\theta)}(x_{(n)})$. In

the latter case, we would have

$$f(\vec{x}; \theta) = \underbrace{I_{(0, x_{(n)})}}_{h(\vec{x})} \cdot \underbrace{\frac{1}{\theta^n} I_{(0, \theta)}(x_{(n)})}_{g(s(\vec{x}); \theta)}$$

where $s(\vec{x}) = x_{(n)}$.

So, by the Factorization Criterion, we have that $S = s(\vec{X}) = X_{(n)}$ is sufficient for estimating θ .

Note that there were other ways to factor the joint pdf. If we used

$$f(\vec{x}; \theta) = \frac{1}{\theta^n} \underbrace{I_{(0, \theta)}(x_{(1)}) I_{(0, \theta)}(x_{(n)})}_{g(s(\vec{x}); \theta)}$$

(here $h(\vec{x}) = 1$), we would say that the vector $S = (X_{(1)}, X_{(n)})$ is sufficient for θ (or that $X_{(1)}$ and $X_{(n)}$ are “jointly sufficient”).

Indeed, for estimating the upper endpoint of possible values for the X 's, it makes sense that looking only at the maximum in the sample is “enough”. It is also “enough” to look both at the max and min though this information is “more than enough”. In fact, since a sufficient statistic can be vector-valued, we can always say that $S = (X_1, X_2, \dots, X_n)$ is sufficient for estimating θ . Without going into a formal explanation of the concept, we say for this example that $X_{(n)}$ is “minimal sufficient”. It is sufficient and of the lowest possible dimension. Typically the dimension of a minimal sufficient statistic is equal to the dimension of θ . □

★ Important Note

We have been using the phrase “sufficient for θ ”, or, “sufficient for estimating θ ”. If you find yourself wanting to estimate a function like, for example, $\tau(\theta) = e^{-\theta}$, and you will be using a procedure that requires a sufficient statistic, nothing changes. If a statistic is sufficient for estimating θ , it is sufficient for estimating $e^{-\theta}$ because if the conditional distribution of $(X_1, X_2, \dots, X_n)$ given S doesn't depend on θ it sure as heck doesn't depend on $e^{-\theta}$!

So, you want to estimate $\tau(\theta)$ and you have some unbiased estimator in the form of a statistic T . If you find a sufficient statistic S , you can make

your estimator even better (smaller variance) using the **Rao-Blackwell Theorem**. (Before reading further, you might want to review conditional expectation!)

★ The Rao-Blackwell Theorem

Suppose we want to estimate some function of θ (which may be θ itself) that we have named $\tau(\theta)$ based on some random sample $\vec{X}$. Suppose that we have some unbiased estimator $T = t(\vec{X})$ so that

$$E[T] = \tau(\theta).$$

Suppose that we have some sufficient statistic S for the distribution. Then

$$T^* \stackrel{\text{def}}{=} E[T|S]$$

is also unbiased for $\tau(\theta)$ and that $\text{Var}[T^*] \leq \text{Var}[T]$.

Furthermore, T^* is a function of S and the variance is strictly lower than the variance of T unless T^* and T are the same!

Proof:

1. T^* is unbiased for $\tau(\theta)$ since

$$E[T^*] = E[E[T|S]] = E[T] = \tau(\theta).$$

2. $\text{Var}[T^*] \leq \text{Var}[T]$ since

$$\begin{aligned} \text{Var}[T] &= \text{Var}[E(T|S)] + E[\text{Var}(T|S)] \\ &\geq \text{Var}[E[T|S]] = \text{Var}[T^*] \end{aligned}$$

since $E[\text{Var}(T|S)] \geq 0$ since that variance inside is non-negative.

3. $T^* = E[T|S]$ is a function of S by the definition of that expectation.
4. Equality of variances is achieved if and only if the thing we dropped to show the inequality is zero. That is, equality is achieved if and only if

$$E[\text{Var}(T|S)] = 0.$$

Since the variance inside is non-negative, this probability weighted average is zero if and only if $\text{Var}(T|S) = 0$. With no variability

at all, this means that T , given S is a constant. If it is constant, it's expectation (given S) is just T itself. That is, the equality of variances is achieved if and only if

$$T^* = E[T|S] = T.$$

The Rao-Blackwell Theorem tells us that, while searching for the UMVUE for $\tau(\theta)$, we can limit our attention to functions of sufficient statistics because estimators that are functions of non-sufficient statistics can be “Rao-Blackwellized” into better (lower variance) estimators!!!

(Note: All of the claims made by the Rao-Blackwell Theorem are true regardless of whether or not S is sufficient. However, we need sufficiency to ensure that $T = E[T|S]$ does not depend on θ . Otherwise, it would not technically be a statistic, by our definition, but, by any definition, it would not make a very good estimator of the unknown θ if we define it in terms of θ !)

5.5.2 Complete Statistics



Definition

Suppose that $X_1, X_2, \dots, X_n$ is a random sample from the distribution with pdf $f(x; \theta)$. We say that a statistic $S = s(\vec{X})$ is **complete** for the distribution if, for any function g such that $E[g(S)] = 0$ for all θ in the parameter space, we find that $g(S)$ must be 0 with probability 1.

Huh? Why would we care about that?

The main consequence of “completeness” is uniqueness. Suppose that we have a statistic S and two unbiased estimators of $\tau(\theta)$ that are functions of S :

$$T_1 = g_1(S) \quad \text{and} \quad T_2 = g_2(S)$$

with

$$E[T_1] = E[g_1(S)] = \tau(\theta) \quad \text{and} \quad E[T_2] = E[g_2(S)] = \tau(\theta).$$

For example, in the case of the uniform distribution on $(0, \theta)$, if we let S be the n -dimensional statistic $S = (X_1, X_2, \dots, X_n)$, we have (among others) the two unbiased estimators $T_1 = g_1(S) = \frac{n+1}{n}X_{(n)}$ and $T_2 = g_2(S) = 2\bar{X}$.

Note that

$$0 = \tau(\theta) - \tau(\theta) = E[T_1] - E[T_2] = E[g_1] - E[g_2(S)] = E[\underbrace{g_1(S) - g_2(S)}_{\substack{\text{call this} \\ g(S)}}].$$

If S is complete, since $E[g(S)] = 0$ we must have that $g(S) = 0$. Thus, we must have that $g_1(S) = g_2(S)$ or, equivalently, $T_1 = T_2$. So, **there can only be one function of a complete statistic that is unbiased for $\tau(\theta)$!**

(Note that for the above uniform example, $S = (X_1, X_2, \dots, X_n)$ is not complete for the distribution as we were able to find two distinct unbiased estimators for $\tau(\theta) = \theta$.)

Example:

We saw that $S = \sum_{i=1}^n X_i$ is sufficient for the Bernoulli distribution, let us also show that it is complete.

Take any function g such that $E[g(S)] = 0$ for all $0 \leq p \leq 1$.

Since $S \sim \text{Bernoulli}(p)$, we can write out this expectation as

$$0 = E[g(S)] = \sum_{s=0}^n g(s) \binom{n}{s} p^s (1-p)^{n-s}$$

Note that this is a polynomial in p of degree n (or $< n$ depending on whether or not some of the coefficients involving g are zero). It will have n (or less) zeros. That is, it will have n (or less) values of p that make it zero. However, we have assumed that it is zero for all p in the interval $[0, 1]$! Thus, it must be identically zero meaning that all of its coefficients must be zero! This forces $g(s) = 0$ for each of $s = 0, 1, \dots, n$. While g might be non-zero for different values of s , we can conclude that $g(S) = 0$ with probability 1 since $S = \sum_{i=1}^n X_i$ will only take on values in $\{0, 1, \dots, n\}$.

Therefore, $S = \sum_{i=1}^n X_i$ is complete for the Bernoulli distribution. □

Example:

We saw that $S = X_{(n)}$ is sufficient for the $unif(0, \theta)$ distribution, let us also show that it is complete.

Take any function g such that $E[g(S)] = 0$ for all $\theta > 0$.

Since $S = X_{(n)}$ has pdf $f_S(s) = \frac{n}{\theta^n} s^{n-1} I_{(0, \theta)}(s)$,

$$0 = E[g(S)] = \int_0^\theta g(s) \frac{n}{\theta^n} s^{n-1} ds$$

$$\frac{n}{\theta^n} \int_0^\theta g(s) s^{n-1} ds = 0 \quad \Rightarrow \quad \int_0^\theta g(s) s^{n-1} ds = 0$$

Taking the derivative of both sides with respect to θ , we get

$$g(\theta) \theta^{n-1} = 0.$$

Since this has to hold for all $\theta > 0$, we can conclude that $g(u) = 0$ for all $u > 0$. Therefore,

$$g(S) = g(X_{(n)}) = 0.$$

Thus, $S = X_{(n)}$ is complete for the $unif(0, \theta)$ distribution. □

5.5.3 The UMVUE

UMVUE stands for Uniformly Minimum Variance Unbiased Estimator. In mathematics, the word “uniformly” can be translated as “for all”. When estimating $\tau(\theta)$ with some estimator $\widehat{\tau(\theta)}$, the variance of $\widehat{\tau(\theta)}$ will depend on θ which is unknown! If you were to graph the variances of two unbiased estimators as a function of θ , you might see that one estimator is better (in the sense that it has smaller variance) for certain values of θ while the other is better for other values of θ . The UMVUE for $\tau(\theta)$ is the unbiased estimator that has smaller variance than all other unbiased estimators for all values of θ !

We find the UMVUE for $\tau(\theta)$ using the **Lehmann-Scheffé Theorem**.

★ **The Lehmann-Scheffé Theorem**

Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with pdf $f(x; \theta)$ and suppose that $S = s(\vec{X})$ is a complete and sufficient statistic for this distribution. If $g(S)$ is some function of S that is unbiased for $\tau(\theta)$ then $g(S)$ is the UMVUE for $\tau(\theta)$.

Proof: Let T be any other unbiased estimator of $\tau(\theta)$. We want to show that $\text{Var}[g(S)]$ is smaller than $\text{Var}[T]$.

Case One: T is not a function of S .

In this case, we can use the Rao-Blackwell Theorem to compute $T^* = E[T|S]$ to get another unbiased estimator with smaller variance than that of T . Note that, if the variance were the same, then we would have that $T^* = T$ (see proof of the Rao-Blackwell Theorem) but this can't happen since T^* is a function of S and T was assumed to not be a function of S .

Case Two: T is a function of S .

Since S is complete, we know there is only one function of S that is unbiased for $\tau(\theta)$. So, T (a function of S) must be equal to $g(S)$.

Example:

Let $X_1, X_2, \dots, X_n$ be a random sample from the Poisson distribution with parameter λ . Find the UMVUE for λ .

By the Lehmann-Scheffé Theorem, we must do two things:

1. Find a complete and sufficient statistic S .
2. Find a function of S that is unbiased for $\tau(\lambda) = \lambda$.

As of yet, we don't have a way to find complete statistics, so we will use the Factorization Criterion to find a sufficient statistic and hope that we can show it is complete. In the next Section we will have a way to get many complete and sufficient statistics easily!

For the Poisson distribution,

$$\begin{aligned} f(\vec{x}; \lambda) &= \frac{e^{n\lambda} (n\lambda)^{\sum x_i}}{\prod (x_i!)} \prod_{i=1}^n I_{\{0,1,2,\dots\}}(x_i) \\ &= \underbrace{\frac{\prod_{i=1}^n I_{\{0,1,2,\dots\}}(x_i)}{\prod (x_i!)}}_{h(\vec{x})} \underbrace{e^{n\lambda} (n\lambda)^{\sum x_i}}_{g(s(\vec{x}); \lambda)} \end{aligned}$$

By the Factorization Criterion, $S = s(\vec{X}) = \sum_{i=1}^n X_i$. Is sufficient for the Poisson distribution.

In order to show that $S = \sum X_i$ is also complete, we first note that $S \sim \text{Poisson}(n\lambda)$. So, if g is any function such that $E[g(S)] = 0$ for all $\lambda > 0$, then

$$0 = E[g(S)] = \sum_{s=0}^{\infty} g(s) \frac{e^{-n\lambda} (n\lambda)^s}{s!} = e^{-n\lambda} \sum_{s=0}^{\infty} g(s) \frac{(n\lambda)^s}{s!}.$$

Thus,

$$\sum_{s=0}^{\infty} \frac{g(s) n^s}{s!} \lambda^s = 0 \quad \text{for all } \lambda > 0.$$

This is a power series in λ . In order to it to be zero for all $\lambda > 0$, it must be identically zero. That, we must have the coefficients $a_s = g(s) n^s / s!$ equal to zero for $s = 0, 1, 2, \dots$. This implies that $g(s) = 0$ for $s = 0, 1, 2, \dots$. So, we have that $g(S) = 0$ with probability 1 since S is Poisson and can only take on values in $\{0, 1, 2, \dots\}$. Therefore, S is not only sufficient but complete as well!

To get the UMVUE for λ , we must find a function of $S = \sum_{i=1}^n X_i$ that is unbiased for λ . Trying S itself gives

$$E[S] = E\left[\sum_{i=1}^n X_i\right] = nE[X_1] = n\lambda.$$

So, $\frac{1}{n}S = \frac{1}{n} \sum X_i = \bar{X}$ is a function of the complete and sufficient statistic S that is unbiased for λ . By the Lehman-Scheffé Theorem, we have that

$$\hat{\lambda} = \bar{X}$$

is the UMVUE for λ . □

Example:

Let $X_1, X_2, \dots, X_n$ be a random sample from the Poisson distribution with parameter λ . Find the UMVUE for λ^2 .

We already know that $S = \sum_{i=1}^n X_i$ is complete and sufficient for the Poisson distribution. (It is complete and sufficient “for λ ” and is not actually dependent of the specific function of λ we are estimating. See the “Important Note” right before the Rao-Blackwell Theorem.) By the Lehmann-Scheffé Theorem, we need to find a function of S that is unbiased for λ^2 .

Let’s try $E[S^2]$. Recall that $S \sim \text{Poisson}(n\lambda)$ so

$$E[S^2] = \text{Var}[S] + (E[S])^2 = n\lambda + (n\lambda)^2.$$

We know that $\bar{X} = S/n$ is unbiased for λ so we now have that

$$E[S^2] = nE[S/n] + n^2\lambda^2 = E[S] + n^2\lambda^2$$

so we can get λ^2 as the expectation of $(S^2 - S)/n^2$. Since this is a function of the complete and sufficient statistic S that is unbiased for λ^2 , it is the UMVUE for λ^2 . Bringing this back to the X ’s, we have that

$$\hat{\lambda}_{\text{UMVUE}} = \frac{(\sum X_i)^2 - \sum X_i}{n^2}.$$

□

5.5.4 Exponential Families: Finding Complete and Sufficient Statistics

We have been fortunate so far that our sufficient statistics found using the Factorization Criterion have also been complete. However, there is a different kind of factorization of the joint pdf from which we can find statistics that are sufficient and complete all in one step!

**Definition**

$X_1, X_2, \dots, X_n$ is a random sample from a **one-parameter exponential family** if θ is one-dimensional and if $X_1, X_2, \dots, X_n$ has a joint pdf that can be written as

$$f(\vec{x}; \theta) = a(\theta)b(\vec{x}) \exp[c(\theta)d(\vec{x})]$$

for some functions $a(\theta)$, $b(\vec{x})$, $c(\theta)$, and $d(\vec{x})$. (As the notation suggests, a and c depend only on θ and not on $\vec{x}$ while b and d depend only on $\vec{x}$ and not on θ .)

**Theorem**

If $X_1, X_2, \dots, X_n$ is a random sample from a one-parameter exponential family distribution, then $S = d(\vec{X})$ is complete and sufficient for the distribution.

The proof of the completeness part of this claim is omitted. It is a bit “beyond the scope of this course”. I would include a sketch of a proof but I need to get this handout out to you now so you can finish your homework and, given that we are in the end of the semester “death spiral” as I like to call it, the chances are slim that I will get back to finish this. The sufficiency part of this claim follows directly from the Factorization Criterion for sufficiency since

$$f(\vec{x}; \theta) = \underbrace{b(\vec{x})}_{h(\vec{x})} \cdot \underbrace{a(\theta) \exp[c(\theta)d(\vec{x})]}_{g(s(\vec{x}); \theta)}$$

we can see that $S = s(\vec{X}) = d(\vec{X})$ is sufficient.

This “one-parameter exponential family factorization” will work for just about every distribution we have talk about in class, except... you guessed it... those pesky distributions where the “parameter is mixed in with the indicator”.

Example:

Let $X_1, X_2, \dots, X_n$ be a random sample from the Bernoulli distribution with parameter p . Find a complete and sufficient statistic for this distribution.

The joint pdf is

$$f(\vec{x}; p) = p^{\sum x_i} (1-p)^{n-\sum x_i} \prod_{i=1}^n I_{\{0,1\}}(x_i)$$

which may be rewritten as

$$f(\vec{x}; p) = \underbrace{(1-p)^n}_{a(p)} \underbrace{\prod_{i=1}^n I_{\{0,1\}}(x_i)}_{b(\vec{x})} \underbrace{\left(\frac{p}{1-p}\right)^{\sum x_i}}_{\text{"mixed part"}}$$

which may again be rewritten as

$$f(\vec{x}; p) = \underbrace{(1-p)^n}_{a(p)} \underbrace{\prod_{i=1}^n I_{\{0,1\}}(x_i)}_{b(\vec{x})} \exp \left[\underbrace{\sum x_i}_{d(\vec{x})} \underbrace{\ln \left(\frac{p}{1-p} \right)}_{c(\theta)} \right].$$

So, by the “one-parameter exponential family factorization”,

$$S = d(\vec{X}) = \sum_{i=1}^n X_i$$

is complete and sufficient for the Bernoulli distribution. ■



Definition

$X_1, X_2, \dots, X_n$ is a random sample from a **k -parameter exponential family** if θ is k -dimensional and if $X_1, X_2, \dots, X_n$ has a joint pdf that can be written as

$$f(\vec{x}; \theta) = a(\theta) b(\vec{x}) \exp \left[\sum_{i=1}^k c_i(\theta) d_i(\vec{x}) \right]$$

for some functions $a(\theta)$, $b(\vec{x})$, $c_1(\theta), \dots, c_k(\theta)$, and $d_1(\vec{x}), \dots, d_k(\theta)$.

Note that this definition does not require c_i to be a function of θ_i (the i th component of the vector θ). $c_i(\theta)$ is a function of the vector θ and could involve one, several, or all components of θ !



Theorem

If $X_1, X_2, \dots, X_n$ is a random sample from a k -parameter exponential family distribution, then $S = (d_1(\vec{X}), d_2(\vec{X}), \dots, d_k(\vec{X}))$ is a vector of complete and sufficient statistics for the distribution.

Example:

Let $X_1, X_2, \dots, X_n$ be a random sample from the $N(\mu, \sigma^2)$ distribution where both parameters are unknown. Find the UMVUEs for μ and σ^2

By the Lehmann-Scheffé Theorem, we must find a (set of) complete and sufficient statistic(s) for this distribution and then find functions of this (these) statistic(s) that are unbiased for μ and σ^2 .

The joint pdf for the random sample is

$$f(\vec{x}; \mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right]$$

which may be rewritten as

$$\begin{aligned} f(\vec{x}; \mu, \sigma^2) &= (2\pi\sigma^2)^{-n/2} \exp \left[-\frac{1}{2\sigma^2} \sum x_i^2 + \frac{\mu}{\sigma^2} \sum x_i - n \frac{\mu^2}{2\sigma^2} \right] \\ &= \underbrace{(2\pi\sigma^2)^{-n/2} \exp \left[-n \frac{\mu^2}{2\sigma^2} \right]}_{a(\theta)} \exp \left[\underbrace{-\frac{1}{2\sigma^2} \sum x_i^2}_{c_1(\theta)} + \underbrace{\frac{\mu}{\sigma^2} \sum x_i}_{d_1(\vec{x})} + \underbrace{\frac{mu}{\sigma^2} \sum x_i}_{c_2(\theta)} + \underbrace{\sum x_i}_{d_2(\vec{x})} \right]. \end{aligned}$$

Here $\theta = (\mu, \sigma^2)$. By the “two-parameter exponential family factorization”, $S = (\sum X_i^2, \sum X_i)$ is complete and sufficient for the $N(\mu, \sigma^2)$ distribution.

5.5. UNIFORMLY MINIMUM VARIANCE UNBIASED ESTIMATORS (UMVUES) 201

To get the UMVUEs, we must find functions of S that are unbiased for μ and σ^2 . We already know these! They are

$$\hat{\mu}_{\text{UMVUE}} = \frac{\sum X_i}{n} = \bar{X} \quad \text{and} \quad \widehat{\sigma^2}_{\text{UMVUE}} = \frac{\sum X_i^2 - (\sum X_i)^2/n}{n-1} = S^2$$

where S^2 is the sample variance, not to be confused with the S for the sufficient statistic. 

Note that, in the previous example, we used both of the complete and sufficient statistics to estimate σ^2 . In general, you should NOT associate one statistic with one parameter and the other statistic with the other parameter...

HAPPY UMVUEing!!!

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Chapter 6

General Hypothesis Testing

In Chapter 4, we had a taste of hypothesis testing that was quite restrictive. We were limited in what distributions we could handle, we had to be given the statistic we should base our tests on without any real explanation other than “this seems like a good idea”, and we could only handle hypotheses involving the comparisons in $\{=, \neq, <, \leq, >, \geq\}$. In this Chapter, we will generalize our techniques and, to that end, we must first generalize our notation.

As usual, use θ to denote a generic parameter of interest. We will use Θ to denote the parameter space (all the possible values for θ).

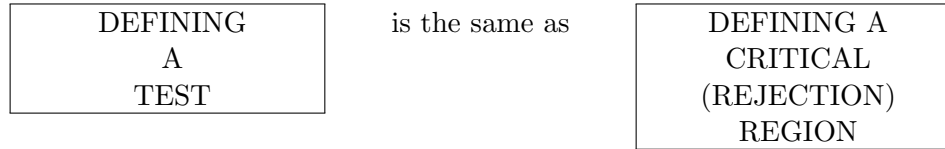
A general set of hypotheses looks like this:

$$H_0 : \theta \in \Theta_0$$

$$H_1 : \theta \in \Theta \setminus \Theta_0.$$

Here, $\Theta \setminus \Theta_0$ denotes the set of everything in Θ that is not in Θ_0 .

Note that



↓

Example: Given the sample X_1, X_2, X_3 , the test is: “Reject H_0 if $X_1^2 + X_2^2 + X_3^2 \leq 5$.”

↓

Example: Given the sample X_1, X_2, X_3 , the test is: “Reject H_0 if $\vec{X} = (X_1, X_2, X_3) \in C$ where $C = \{(x_1, x_2, x_3) : x_1^2 + x_2^2 + x_3^2 \leq 5\}$.”

Language and Notation We have used the symbol α to denote the “level of significance” of a test. It is also known as the “size” of the test. (This language comes from looking at probability as a measure.)

If we are now defining a test by whether or not a sample falls into a critical region C , we may write α , for a simple null hypothesis ($H_0 : \theta = \theta_0$), as

$$\begin{aligned}
 \alpha &= \text{the level of significance} = \text{the size of the test} \\
 &= P(\text{Type I error}) \\
 &= P(\text{reject } H_0 \text{ when it's true}) \\
 &= P((X_1, X_2, \dots, X_n) \in C \text{ when } H_0 \text{ is true}) \\
 &\stackrel{\text{notation}}{=} P((X_1, X_2, \dots, X_n) \in C; H_0) \\
 &\stackrel{\text{notation}}{=} P((X_1, X_2, \dots, X_n) \in C; \theta_0)
 \end{aligned}$$

For a non-simple null hypothesis $H_0 : \theta \in \Theta_0$, the level of significance or size of the test is defined by

$$\alpha = \max_{\theta \in \Theta_0} P((X_1, X_2, \dots, X_n) \in C; \theta)$$

6.1 A Best Test

In this Section, we will deal only with a “simple versus simple” test:

$$\begin{aligned} H_0 &: \theta = \theta_0 \\ H_1 &: \theta = \theta_1 \end{aligned} \tag{6.1}$$

As with any hypothesis test, these represent all possibilities for θ . ie: $\Theta = \{\theta_0, \theta_1\}$.

We want a test of size α . Rather than give a “rejection rule”, we will, equivalently, give a critical region C of size α . That is, we will find a set C such that

$$P((X_1, X_2, \dots, X_n) \in C; \theta_0) = \alpha.$$

Just as there may be several different rejection rules (for example, one based on $\bar{X}$, one based on $X_{(1)}$, etc...), there may be several such sets C .

Let’s suppose that Kermit and Uli both have tests of size α for (6.1). This means, by definition, that they each have a set C_K and C_U (K=Kermit, U=Uli) such that

$$P((X_1, X_2, \dots, X_n) \in C_K; \theta_0) = \alpha$$

$$P((X_1, X_2, \dots, X_n) \in C_U; \theta_0) = \alpha.$$

How can we compare their tests?

Looking at, for example, the probabilities that the sample $X_1, X_2, \dots, X_n$ is not in the respective sets won’t help. It will be $1 - \alpha$ in either case. I guess that the only way we can differentiate between the tests will be in the case that the alternative hypothesis is true. So, we will consider

$$P((X_1, X_2, \dots, X_n) \in C_K; \theta_1)$$

and

$$P((X_1, X_2, \dots, X_n) \in C_U; \theta_1).$$

Suppose that

$$P((X_1, X_2, \dots, X_n) \in C_K; \theta_1) > P((X_1, X_2, \dots, X_n) \in C_U; \theta_1). \tag{6.2}$$

Question: Whose test is better?

Answer: Kermit's. We are looking at probabilities that the sample falls in a critical (rejection) region when H_1 is true. When H_1 , we should be rejecting H_0 . Kermit has a higher probability than Uli of rejecting H_0 in the case when H_1 is true, so Kermit's test is better.

What if the inequality in (6.2) holds using Kermit's C_K in the left hand side and **any** other critical region of size α in the entire world in the right hand side? Then Kermit would have the best test in the world!



Definition

For testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$, with a random sample $X_1, X_2, \dots, X_n$, C is called a best critical region of size α (or the corresponding test is a best test of size α) if:

1. $P((X_1, X_2, \dots, X_n) \in C; \theta_0) = \alpha$
(This makes it "of size α ".)
2. If A is any other set such that $P((X_1, X_2, \dots, X_n) \in A; \theta_0) = \alpha$, then

$$P((X_1, X_2, \dots, X_n) \in C; \theta_1) \geq P((X_1, X_2, \dots, X_n) \in A; \theta_1).$$

(This makes it "best".)

Language and Notation Recall the definition of the power function:

$$\gamma(\theta) = P(\text{reject } H_0 \text{ when the parameter is } \theta).$$

Since rejection depends on the critical (rejection) region C , we will now emphasize this by writing the power function with a subscript: $\gamma_C(\theta)$.

So, putting all of our notation together, we have:

$$\gamma_C(\theta) = P((X_1, X_2, \dots, X_n) \in C; \theta).$$

Note that part 2 of the definition of a best test can now be written as

$$\gamma_C(\theta_1) \geq \gamma_1(\theta_1).$$

So, the test based on C has “higher power”, when $\theta = \theta_1$, than the test based on A .



A best test of size α is often referred to as a most powerful test of size α .

How can one derive a best or most powerful test?



The Neyman-Pearson Lemma

Suppose that $X_1, X_2, \dots, X_n$ have joint pdf $f(\vec{x}; \theta)$.

Consider the ratio

$$\lambda(\vec{x}; \theta_0, \theta_1) = \frac{f(\vec{x}; \theta_0)}{f(\vec{x}; \theta_1)}$$

and let C be the set $C = \{(x_1, \dots, x_n) : \lambda(\vec{x}; \theta_0, \theta_1) \leq k\}$ where k is a constant such that $P((X_1, \dots, X_n) \in C; H_0) = \alpha$.

Then C is a best critical region of size α for testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$.

Comments about the Neyman-Pearson Lemma:

1. The idea here is to put points into the critical region until it reaches size α .
2. In order to maximize the power of the test, points that are more likely under H_1 than under H_0 should be put into the critical region.
3. If a point $(x_1, \dots, x_n)$ is more likely under H_1 than under H_0 , then the denominator of $\lambda(\vec{x}; \theta_0, \theta_1)$ is larger and so $\lambda(\vec{x}; \theta_0, \theta_1)$ is smaller. This is why the requirement for putting $(x_1, \dots, x_n)$ into C is of the form $\lambda(\vec{x}; \theta_0, \theta_1) \leq k$.

4. Let $L(\theta)$ be a likelihood function for the model under consideration. Recall that $L(\theta)$ is proportional to the joint pdf $f(\vec{x}; \theta)$ when it is thought of as a function of θ . This means that $L(\theta) = c \cdot f(\vec{x}; \theta)$ where $c = c(\vec{x})$ may depend on $\vec{x}$. The ratio from the Neymann-Pearson Lemma is called a **likelihood ratio** since

$$\frac{f(\vec{x}; \theta_0)}{f(\vec{x}; \theta_1)} = \frac{(1/c)L(\theta_0)}{(1/c)L(\theta_1)} = \frac{L(\theta_0)}{L(\theta_1)}.$$

For this reason, the test given by the Neyman-Pearson Lemma is one example of a **likelihood ratio test**, discussed more generally in Section 6.3.

Proof of the Neyman-Pearson Lemma:

We will prove the N-P Lemma for a random sample from a continuous distribution. The proof for a discrete distribution is similar.

1. Let C be the so-called “best” critical region described by the N-P Lemma.
2. Let $\vec{X} = (X_1, X_2, \dots, X_n)$. Note that for any set A , $P(\vec{X} \in A; \theta)$ may be written as

$$P(\vec{X} \in A; \theta) = \int_A f(\vec{x}; \theta) d\vec{x}$$

which is shorthand notation for an n -dimensional integral with respect to $dx_1 dx_2 \dots dx_n$ where all the limits of integration run around describing points $(x_1, x_2, \dots, x_n) \in A$.

3. Note that if $A \subseteq C$,

$$P(\vec{X} \in A; \theta_0) \leq kP(\vec{X} \in A; \theta_1)$$

since, in C , and hence in $A \subseteq C$, $f(\vec{x}; \theta_0) \leq k \cdot f(\vec{x}; \theta_1)$, so

$$P(\vec{X} \in A; \theta_0) = \int_A f(\vec{x}; \theta_0) d\vec{x} \leq k \cdot \int_A f(\vec{x}; \theta_1) d\vec{x} = k \cdot P(\vec{X} \in A; \theta_1).$$

4. On the other hand, by the same reasoning, if $A \subseteq C^c$ (the complement of C),

$$P(\vec{X} \in A; \theta_0) \geq kP(\vec{X} \in A; \theta_1)$$

5. For the remainder of this proof, let A denote any critical region of size α :

$$P(\vec{X} \in A; \theta_0) = \alpha.$$

6. Note that we may write

$$C = (C \cap A) \cup (C \cap A^c)$$

and that $C \cap A$ and $C \cap A^c$ are disjoint sets.

So,

$$\gamma_C(\theta) = P(\vec{X} \in C; \theta) = P(\vec{X} \in C \cap A; \theta) + P(\vec{X} \in C \cap A^c; \theta).$$

7. On the flip side, we may write

$$A = (A \cap C) \cup (A \cap C^c)$$

and that $A \cap C$ and $A \cap C^c$ are disjoint sets.

So,

$$\gamma_A(\theta) = P(\vec{X} \in A; \theta) = P(\vec{X} \in A \cap C; \theta) + P(\vec{X} \in A \cap C^c; \theta).$$

8. In order to prove the N-P Lemma, we want to show that $\gamma_C(\theta_1) \geq \gamma_A(\theta_1)$. So, we consider the difference:

$$\gamma_C(\theta) - \gamma_A(\theta) = P(\vec{X} \in C \cap A^c; \theta) - P(\vec{X} \in A \cap C^c; \theta).$$

9. Specifically, we consider the difference at θ_1 and use the fact that $C \cap A^c \subseteq C$ and $A \cap C^c \subseteq C^c$:

$$C \cap A^c \subseteq C \quad \Rightarrow \quad P(\vec{X} \in C \cap A^c; \theta_1) \geq (1/k)P(\vec{X} \in C \cap A^c; \theta_0)$$

$$A \cap C^c \subseteq C^c \quad \Rightarrow \quad P(\vec{X} \in A \cap C^c; \theta_1) \leq (1/k)P(\vec{X} \in A \cap C^c; \theta_0)$$

So, by step 8,

$$\gamma_C(\theta_1) - \gamma_A(\theta_1) \geq (1/k) \left[P(\vec{X} \in C \cap A^c; \theta_0) - P(\vec{X} \in A \cap C^c; \theta_0) \right]$$

$$= (1/k) [\gamma_C(\theta_0) - \gamma_A(\theta_0)] \quad \text{by step 8 again}$$

$$= (1/k)(\alpha - \alpha) = 0$$

Therefore, $\gamma_C(\theta_1) \geq \gamma_A(\theta_1)$. Since this hold for any size α region A , we have shown that C is the best critical region of size A ! $\square$

Example 6.1.1

Let $X_1, X_2, \dots, X_n$ be a random sample from the $\exp(\text{rate} = \theta)$ distribution.
Test

$$H_0 : \theta = \theta_0$$

$$H_1 : \theta = \theta_1$$

where $\theta_0 < \theta_1$.

The N-P Theorem says to reject H_0 if

$$\lambda(\vec{x}; \theta_0, \theta_1) \leq k$$

where k is such that $P(\lambda(\vec{X}; \theta_0, \theta_1) \leq k; \theta_0) = \alpha$.

Now

$$\lambda(\vec{x}; \theta_0, \theta_1) = \frac{\theta_0^n e^{-\theta_0 \sum x_i} \prod I_{(0, \infty)}(x_i)}{\theta_1^n e^{-\theta_1 \sum x_i} \prod I_{(0, \infty)}(x_i)} = \left(\frac{\theta_0}{\theta_1}\right)^n e^{-(\theta_0 - \theta_1) \sum x_i}.$$

So, we want:

$$\begin{aligned} \alpha &= P\left(\left(\frac{\theta_0}{\theta_1}\right)^n e^{-(\theta_0 - \theta_1) \sum X_i} \leq k; \theta_0\right) \\ &= P\left(-(\theta_0 - \theta_1) \sum X_i \leq \ln\left(\left(\frac{\theta_1}{\theta_0}\right)^n k\right)\right) \\ &\stackrel{\theta_0 < \theta_1}{=} P(\sum X_i \leq k_1) \end{aligned}$$

where $k_1 = \frac{\ln\left(\left(\frac{\theta_1}{\theta_0}\right)^n k\right)}{-(\theta_0 - \theta_1)}.$

So, for this example, a best critical region of size α has the form:

$$C = \left\{(x_1, \dots, x_n) : \sum x_i \leq k_1\right\}$$

for some k_1 which must be chosen so that

$$P((X_1, \dots, X_n) \in C; \theta_0) = \alpha.$$

(Note: If you are asked for the form of a test, you may stop here!)

Suppose we want to get the actual test. That is, suppose we want the value of k_1 .

$$\begin{aligned}\alpha &= P((X_1, \dots, X_n) \in C; \theta_0) \\ &= P(\sum X_i \leq k_1; \theta_0)\end{aligned}$$

Well, if $\theta = \theta_0$, $\sum X_i \sim \Gamma(n, 1/\theta_0)$.

So, for a given α , n , and θ_0 , we could find k_1 by solving

$$\alpha = P(G \leq k_1) \quad \text{where} \quad G \sim \Gamma(n, 1/\theta_0).$$

In practice, the gamma pdf is not easy to integrate (except if you're going all the way to ∞), and it is not easy to invert the resulting integral in order to solve for k_1 . I'm sure you can coax Mathematica, or some such software, to do it for you.

However, as chi-squared tables have been around for a long time, we will try to state this test in terms of a χ^2 -critical value.

Recall that $\chi^2(n) = \Gamma(n/2, 2)$ and that $X \sim \Gamma(\alpha, \beta) \Rightarrow cX \sim \Gamma(\alpha, \beta/c)$, provided $c > 0$

For this problem, we want to find k_1 such that $P(G \leq k_1) = \alpha$ where $G \sim \Gamma(n, 1/\theta_0)$.

$$\begin{aligned}\alpha &= P(G \leq k_1) \\ &= P(2\theta_0 G \leq 2\theta_0 k_1) \\ &= P(W \leq 2\theta_0 k_1)\end{aligned}$$

where $W \sim \chi^2(2n)$.

So, $2\theta_0 k_1 = \chi_{1-\alpha}^2(2n)$ which means that

$$k_1 = \frac{\chi_{1-\alpha}^2(2n)}{2\theta_0}.$$

In summary, if $X_1, X_2, \dots, X_n$ is a random sample from the $\exp(\text{rate} = \theta)$ distribution, the best test of size α of $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$ is to reject H_0 , in favor of H_1 if $\sum X_i \leq \frac{\chi_{1-\alpha}^2(2n)}{2\theta_0}$. ■

Example 6.1.2

Consider the distribution with pdf $f(x; \theta) = \theta x^{\theta-1} I_{(0,1)}(x)$. Based on a random sample of size $n = 1$, find the best test of $H_0 : \theta = 1$ versus $H_1 : \theta = 2$ with $\alpha = 0.05$.

$$\lambda(x_1; 1, 2) = \frac{f(x_1; 1)}{f(x_1; 2)} = \frac{1 \cdot I_{(0,1)}(x_1)}{2x_1 \cdot I_{(0,1)}(x_1)} = \frac{1}{2x_1}$$

So,

$$C = \left\{ x : \frac{1}{2x} \leq k \right\}$$

where k is such that

$$P(X_1 \in C; H_0) = 0.05$$

$$P\left(\frac{1}{2X_1} \leq k; H_0\right) = 0.05$$

When H_0 is true, $X_1 \sim \text{unif}(0, 1)$! So, when H_0 is true,

$$0.05 = P\left(\frac{1}{2X_1} \leq k\right) = P\left(X_1 \geq \frac{1}{2k}\right) = P(X_1 \geq k_1) \quad \Rightarrow \quad k_1 = 0.95$$

So, the test of size 0.05 of $H_0 : \theta = 1$ versus $H_1 : \theta = 2$ is “Reject H_0 if $X_1 > 0.95$ ”.

6.2 Uniformly Most Powerful Tests (UMPs)

In this section, we will consider a simple versus composite hypothesis of the form:

$$H_0 : \theta = \theta_0$$

$$H_1 : \theta \in \Theta \setminus \{\theta_0\}.$$

(H_1 will usually be one of: $\theta \neq \theta_0$, $\theta < \theta_0$, or $\theta > \theta_0$.)

In mathematics, the word “uniformly” can be loosely translated to “for all”. (In this case, we’re going to be doing something “for all” θ .)

For example, you could have a convergence of functions $f_n(x) \rightarrow f(x)$ as $n \rightarrow \infty$ for one particular x , or you could say that $f_n \rightarrow f$ uniformly in x which means that the convergence holds for all arguments x .



Definition

For simple H_0 and composite H_1 , the critical region C is a **uniformly most powerful critical region of size α** if C is the most powerful (best) critical region for testing H_0 against every simple hypotheses in H_1 .

The corresponding test is a **uniformly most powerful (UMP) test** with level of significance α for testing the simple H_0 versus the composite (or simple) H_1 .

Comments:

1. A UMP test may not exist. (There is usually trouble with the two-sided alternative hypothesis $H_1 : \theta \neq \theta_0$.)
2. H_1 could be simple. Then the most powerful or best test described in Section 6.1 is, by default, uniformly most powerful. It is the most powerful (best) test for all θ in the singleton set $\{\theta_1\}$!
3. A UMP test may be easily defined for the composite versus composite case:

$$H_0 : \theta \in \Theta_0 \quad \text{versus} \quad H_1 : \theta \in \Theta \setminus \Theta_0$$

C^* is a UMP critical region of size α if

$$\alpha = \max_{\theta \in \Theta_0} \gamma_{C^*}(\theta)$$

and

$$\gamma_{C^*}(\theta) \geq \gamma_C(\theta) \quad \forall \theta \in \Theta \setminus \Theta_0$$

and for all critical regions C of size α .

6.2.1 Finding a UMP Test

One possible way to find a UMP test for $H_0 : \theta = \theta_0$ versus $H_1 : \theta \in \Theta \setminus \{\theta_0\}$ is to derive the Neyman-Pearson test for a particular alternative value in the set $\Theta \setminus \theta_0$ and then to show that the test does not depend on the specific alternative value.

Example 6.2.1

Let $X_1, X_2, \dots, X_n$ be a random sample from the exponential distribution with rate θ .

Suppose that we want to test,

$$H_0 : \theta = \theta_0$$

$$H_1 : \theta > \theta_0.$$

We could consider the simple versus simple hypothesis:

$$H_0 : \theta = \theta_0$$

$$H_1 : \theta = \theta_1,$$

for some $\theta_1 > \theta_0$.

This is exactly what we tested in the previous section on best (most powerful) tests.

A best or most powerful or best test for this simple versus simple hypothesis is to reject H_0 if

$$\sum X_i \leq \chi_\alpha^2(2n)/(2\theta_0).$$

The intermediate steps may have depended on θ_1 , but the ultimate decision rule does not. Recall though that it was necessary that $\theta_1 > \theta_0$ or else the inequality in the decision rule would have flipped.

So, basically, we get this best test for any $\theta_1 > \theta_0$. So, by definition of a UMP test, this decision rule gives us the UMP test of $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$. □

Example 6.2.2

Let $X_1, X_2, \dots, X_n$ be a random sample from the distribution with pdf

$$f(x; \theta) = \frac{3x^2}{\theta} e^{-x^3/\theta} I_{(0, \infty)}(x).$$

Let's find a UMP test for

$$H_0 : \theta = \theta_0$$

$$H_1 : \theta > \theta_0$$

We begin by reducing the problem to the simple versus simple hypotheses:

$$H_0 : \theta = \theta_0$$

$$H_1 : \theta = \theta_1$$

for some fixed $\theta_1 > \theta_0$.

We then use the Neyman-Pearson Lemma for the simple versus simple hypotheses:

$$\lambda(\vec{x}; \theta_0, \theta_1) = \frac{\prod_{i=1}^n \left[\frac{3x_i^2}{\theta_0} e^{-x_i^3/\theta_0} \right]}{\prod_{i=1}^n \left[\frac{3x_i^2}{\theta_1} e^{-x_i^3/\theta_1} \right]} = \left(\frac{\theta_1}{\theta_0} \right)^n e^{-\left(\frac{\theta_1 - \theta_0}{\theta_0 \theta_1} \right) \sum x_i^3}$$

We will reject H_0 if $\lambda(\vec{x}; \theta_0, \theta_1) \leq k$ where k is such that $P(\lambda(\vec{X}; \theta_0, \theta_1) \leq k; H_0) = \alpha$.

$$\begin{aligned} P(\lambda(\vec{X}; \theta_0, \theta_1) \leq k; H_0) &= P\left(\left(\frac{\theta_1}{\theta_0} \right)^n e^{-\left(\frac{\theta_1 - \theta_0}{\theta_0 \theta_1} \right) \sum X_i^3} \leq k; H_0 \right) \\ &= P\left(\sum X_i^3 \geq -\frac{\theta_0 \theta_1}{\theta_1 - \theta_0} \ln \left[\left(\frac{\theta_1}{\theta_0} \right)^n k \right]; H_0 \right) \end{aligned}$$

(The inequality flipped along the way since $\theta_1 > \theta_0$ had us dividing by a negative when we moved the $-\frac{\theta_1 - \theta_0}{\theta_0 \theta_1}$ to the right side of the inequality.)

So, the critical region has the form

$$C = \left\{ (x_1, x_2, \dots, x_n) : \sum x_i^3 \geq k_1 \right\}.$$

To find the actual critical region, we need to find the k_1 such that

$$P(\sum X_i^3 \geq k_1; H_0) = \alpha.$$

What is the distribution of $\sum X_i^3$ if H_0 is true?

Let's first consider just one of the X 's:

Let $Y = X^3$. Then $y = g(x) = x^3 \Rightarrow x = g^{-1}(y) = y^{1/3}$.

Then

$$\begin{aligned} f_Y(y) &= f_X(g^{-1}(y)) \cdot \left| \frac{\partial}{\partial y} g^{-1}(y) \right| \\ &= \frac{3(y^{1/3})^2}{\theta_0} e^{-(y^{1/3})^3/\theta_0} \cdot I_{(0,\infty)}(y^{1/3}) \cdot \left| \frac{1}{3} y^{-2/3} \right| \\ &= \frac{1}{\theta_0} e^{-y/\theta_0} \cdot I_{(0,\infty)}(y). \end{aligned}$$

(Note: The absolute value went away because $y^{-2/3} = (y^{-1/3})^2 > 0$. Also, the indicator kept us in a region where $y > 0$.)

So, when H_0 is true (which means that $\theta = \theta_0$) $Y = X^3 \sim \exp(\text{rate} = 1/\theta_0)$.

Therefore,

$$\sum_{i=1}^n X_i^3 \sim \Gamma(n, \theta_0)$$

So,

$$\begin{aligned} \alpha &= P(\sum X_i^3 \geq k_1; H_0) \\ &= P(G \geq k_1) \end{aligned}$$

where $G \sim \Gamma(n, \theta_0)$.

As usual, this is not analytically tractable, so let's go for a chi-squared critical value.

Recall that

$$G \sim \Gamma(n, \theta_0) \quad \Rightarrow \quad 2G/\theta_0 \sim \chi^2(2n).$$

So,

$$\begin{aligned} \alpha &= P(G \geq k_1) \\ &= P(2G/\theta_0 \geq 2k_1/\theta_0) \\ &= P(W \geq 2k_1/\theta_0) \end{aligned}$$

where $W \sim \chi^2(2n)$.

Therefore

$$2k_1/\theta_0 = \chi_\alpha^2(2n) \quad \Rightarrow \quad k_1 = \theta_0 \chi_\alpha^2(2n)/2$$

Let us review where we are so far in this example:

We have determined that the best or most powerful test of $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$ where $\theta_1 > \theta_0$ is to

$$\text{reject } H_0 \text{ if } \sum X_i^3 \geq \theta_0 \chi_\alpha^2(2n)/2.$$

This test does not depend on the specific value of θ_1 , although it did depend on the fact that $\theta_1 > \theta_0$ (otherwise an inequality would have flipped). So, it will work for any $\theta_1 > \theta_0$.

Hence, the test:

$$\text{reject } H_0 \text{ if } \sum X_i^3 \geq \theta_0 \chi_\alpha^2(2n)/2,$$

is a UMP test of size α for $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$. □

Example 6.2.3

Let us consider the last example again with the hypotheses: $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$.

- First we consider $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$ for any $\theta_1 > \theta_0$. Since we just did this in the previous example, we know that this leads us to a UMP critical region for $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$ of the form

$$C = \left\{ (x_1, x_2, \dots, x_n) : \sum x_i^3 \geq c \right\}$$

where c is the solution of $\alpha = P(G \geq c)$ for $G \sim \Gamma(n, \theta_0)$.

- If we now consider $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$ for any $\theta_1 < \theta_0$, and go through the steps of the previous example, we will end up showing that a UMP critical region for $H_0 : \theta = \theta_0$ versus $H_1 : \theta < \theta_0$ of the form

$$C = \left\{ (x_1, x_2, \dots, x_n) : \sum x_i^3 \leq c_1 \right\}$$

where c_1 is the solution of $\alpha = P(G \leq c_1)$ for $G \sim \Gamma(n, \theta_0)$.

- Since a best test for $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$ for $\theta_0 < \theta_1$ is different than a best test for $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$ for $\theta_0 > \theta_1$, there can't be a uniformly best test (most powerful) for $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$.

Therefore, there is no UMP test for this example. That is often, but not always, the case for such a two-sided alternate hypothesis. □



Note

All of the tests that we discussed in this section can be easily expanded to composite null hypotheses. This affects only the definition of α when finding the cutoff value k . (Or k_1 , or k_2 , or whatever you ended up calling it!)

6.2.2 Exponential Families and UMP Tests

★ **Theorem**

Suppose that $X_1, X_2, \dots, X_n$ have a joint pdf in the one-parameter exponential family:

$$f(\vec{x}; \theta) = a(\theta)b(\vec{x}) \exp [c(\theta)d(\vec{x})]$$

where $c(\theta)$ is an increasing function of θ . Then

1. A UMP test of size α for $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$ is to reject H_0 if $d(\vec{x}) \geq k$ where k is chosen so that $P(d(\vec{X}) \geq k; \theta_0) = \alpha$.
2. A UMP test of size α for $H_0 : \theta = \theta_0$ versus $H_1 : \theta < \theta_0$ is to reject H_0 if $d(\vec{x}) \leq k$ where k is chosen so that $P(d(\vec{X}) \leq k; \theta_0) = \alpha$.

Proof: (of #1)

This is not too difficult to show. For example, to show #1, we consider the simple versus simple hypotheses

$$H_0 : \theta = \theta_0 \quad \text{versus} \quad H_1 : \theta = \theta_1$$

for $\theta_1 > \theta_0$. We apply the N-P Lemma and get that

$$\lambda(\vec{x}; \theta_0, \theta_1) = \frac{f(\vec{x}; \theta_0)}{f(\vec{x}; \theta_1)} = \frac{a(\theta_0)}{a(\theta_1)} \cdot \exp [(c(\theta_0) - c(\theta_1)) d(\vec{x})].$$

Setting $\lambda(\vec{x}; \theta_0, \theta_1) \leq k$ and “solving down to the x-stuff on the left hand side” will give

$$d(\vec{x}) \geq k_1,$$

and we solve for k_1 by setting $P(d(\vec{X}) \geq k_1) = \alpha$.

Note that the inequality flips at the point when we divide through by $c(\theta_0) - c(\theta_1)$ which is negative since $\theta_1 > \theta_0$ and c is, by assumption, increasing in θ .

Finding k_1 will involve the distribution of $d(\vec{X})$ under the assumption that the null hypothesis ($H_0 : \theta = \theta_0$) and will not involve θ_1 . Hence, this most

powerful (best) test provided by the N-P Lemma for $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$ will be UMP for $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$. $\square$

Here are a couple of things to note.

- If you read and understood the proof, it is very easy to see that if $c(\theta)$ is decreasing in θ , all the inequalities directly following $d(\vec{x})$ and $d(\vec{X})$ in the statement of the theorem would flip.
- If we have $H_0 : \theta \leq \theta_0$ in #1 or $H_0 : \theta \geq \theta_0$ in #2, nothing would change because we could show, under the assumptions of this theorem, that for #1

$$\alpha = \max_{\theta \leq \theta_0} P(d(\vec{X}) \geq k; \theta) = P(d(\vec{X}) \geq k; \theta_0)$$

and, for #2

$$\alpha = \max_{\theta \geq \theta_0} P(d(\vec{X}) \leq k; \theta) = P(d(\vec{X}) \leq k; \theta_0).$$

Example 6.2.4

Let $X_1, X_2, \dots, X_n$ be a random sample from the $N(0, \sigma^2)$ distribution.

Find a UMP test of size α for $H_0 : \sigma = \sigma_0$ versus $H_1 : \sigma > \sigma_0$.

$$\begin{aligned} f(x; \sigma) &= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}x^2} \\ \Rightarrow f(\vec{x}; \sigma) &= (2\pi)^{-n/2} (\sigma^2)^{-n/2} \cdot \exp \left[-\frac{1}{2\sigma^2} \sum x_i^2 \right] \\ \Rightarrow c(\sigma) &= -\frac{1}{2\sigma^2} \quad \text{and} \quad d(\vec{x}) = \sum x_i^2. \end{aligned}$$

Note that $c(\sigma)$ is an increasing function of σ .

So, a UMP test of size α is to reject H_0 if $\sum X_i^2 \geq k$ where k is chosen such that $P(\sum X_i^2 \geq k; H_0) = \alpha$.

Now let's find k .

When H_0 is true, $\sigma = \sigma_0$. Under this assumption, we need the distribution of $\sum X_i^2$.

A standard normal squared is a chi-squared, so let's standardize it:

$$\begin{aligned}\alpha &= P(\sum X_i^2 \geq k; H_0) \\ &= P(\sum X_i^2 \geq k; \sigma_0) \\ &= P\left(\frac{\sum X_i^2}{\sigma_0^2} \geq \frac{k}{\sigma_0^2}; \sigma_0\right) \\ &= P\left(\sum \left(\frac{X_i}{\sigma_0}\right)^2 \geq \frac{k}{\sigma_0^2}; \sigma_0\right) \\ &= P(W \geq k_1)\end{aligned}$$

where $W \sim \chi^2(n)$ and $k_1 = k/\sigma_0^2$.

We know that $k_1 = \chi_\alpha^2(n)$.

So, the UMP test of size α is to

$$\text{reject } H_0 \text{ if } \frac{\sum X_i^2}{\sigma_0^2} \geq \chi_\alpha^2(n)$$

OR

$$\text{reject } H_0 \text{ if } \sum X_i^2 \geq \sigma_0^2 \chi_\alpha^2(n).$$

□

Example 6.2.5 (Numerical)

Just as a quick check to make sure that we can read the χ^2 -table, let's put some actual numbers in here!

Suppose we observe a random sample of size 10 from the $N(0, \sigma^2)$ distribution. Further suppose that we wish to test $H_0 : \sigma = 1$ versus $H_1 : \sigma > 1$ at 0.05 level of significance.

We will reject H_0 if

$$\frac{\sum_{i=1}^{10} X_i^2}{\sigma_0^2} = \sum_{i=1}^{10} X_i^2 \geq \chi_{0.05}^2(10) = 18.3.$$

□

6.3 Generalized Likelihood Ratio Tests (GLRTs)

As has been mentioned, the hypothesis tests already described in this Chapter are examples of “likelihood ratio tests” since they involve ratios of joint pdfs which are equivalent to ratios of likelihood functions. As we saw in Example 6.2.3, it is not always possible to find a uniformly most powerful test. In this Section we will consider a **generalized likelihood ratio test** (GLRT) that will be applicable in a much wider range of situations. Consider it to the UMP test as the MLE estimator is to the UMVUE. While the UMVUE is awesome in theory, in real life you are probably going to reach for an MLE!

6.3.1 The Generalized Likelihood Ratio

Recall that our general hypotheses statement is

$$H_0 : \theta \in \Theta_0$$

$$H_1 : \theta \in \Theta \setminus \Theta_0$$



Definition

Let $L(\theta)$ be a likelihood for a random sample having joint pdf $f(\vec{x}; \theta)$ for $\theta \in \Theta$.

The **(generalized) likelihood ratio (GLR)** is defined to be

$$\lambda = \lambda(\vec{x}) = \frac{\max_{\theta \in \Theta_0} L(\theta)}{\max_{\theta \in \Theta} L(\theta)} = \frac{L(\hat{\theta}_0)}{L(\hat{\theta})}$$

where $\hat{\theta}$ denotes the usual “unrestricted MLE” and $\hat{\theta}_0$ denotes the “restricted MLE” which is the MLE when H_0 is true.

(Note: For the more “mathy folk”, yes— those maximums should really be supremums!)

Are you wondering why the maximum in the denominator of the GLR is taken over the entire parameter space and not just the alternate hypothesis

part $\Theta \setminus \Theta_0$? The fact of the matter is that one absolutely could define a likelihood ratio test in this way as well as many other ways. The test of this section is not claiming to be optimal in any sense. However, by using the given definition for the GLR, we will get a nice asymptotic result that will allow us to develop an approximate large sample test that will be useful for otherwise intractable problems.

Example 6.3.1 (Restricted versus unrestricted MLEs)

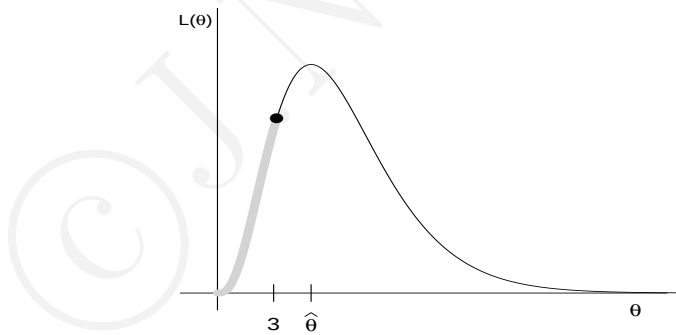
Suppose we wish to test $H_0 : \theta \leq 3$ versus $H_1 : \theta > 3$ and we consider the likelihood function $L(\theta)$.

We maximize $L(\theta)$ by setting $\frac{d}{d\theta}L(\theta) = 0$.

It is usually the case that there is only one solution to $\frac{d}{d\theta}L(\theta) = 0$ and that it is indeed a max and not a min. Call this solution $\hat{\theta}$.

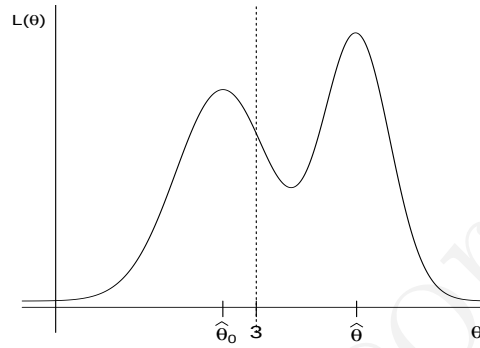
If $\hat{\theta}$ is the unique solution, there will be no other “turning points” on the graph of $L(\theta)$. Suppose the graph of $L(\theta)$ is as shown in Figure 6.1.

Figure 6.1: A Unimodal Likelihood



From Figure 6.1, we see that the location of the standard unrestricted MLE and then we see that the restricted maximum in the numerator of the GLR, $\max_{\theta \leq 3} L(\theta)$, is simply $\hat{\theta}_0 = 3$.

Figure 6.2: A Bimodal Likelihood



Of course it's possible (though not probable) that the likelihood function looks like that shown in Figure 6.2. (This behavior would be reflected in multiple solutions to $\frac{d}{d\theta}L(\theta) = 0$.) In this case, the standard unrestricted ($\hat{\theta}$) and restricted ($\hat{\theta}_0$) are shown. ■

6.3.2 The Generalized Likelihood Ratio Test

We are now ready to define a generalized likelihood ratio test.

★ **The Generalized Likelihood Ratio Test**
 Reject H_0 if $\lambda \leq k$ where k is chosen to give a size α test.
 (ie: $P(\lambda(\vec{X}) \leq k; H_0) = \max_{\theta \in \Theta_0} P(\lambda(\vec{X}) \leq k; \theta) = \alpha$)

Remarks:

1. This is just like the Neyman-Pearson test in the simple versus simple case, the only change here is the addition of the “maxes” when defining the GLR.

2. The original Neyman-Pearson test can be thought of as a simple likelihood ratio test (or SLRT, pronounced “slirt”) since

$$\max_{\theta \in \Theta_0} L(\theta) = L(\theta_0) \quad \text{when} \quad \Theta_0 = \{\theta_0\}.$$

Example 6.3.2

Let $X_1, X_2, \dots, X_n$ be a random sample from the $N(\mu, \sigma^2)$ distribution where σ^2 is known.

Test $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$.

(Note: There is no UMP for this problem as is often the case for a two-sided alternative hypothesis.)

Derive a GLRT of size α .

The joint pdf is

$$f(\vec{x}, \mu) = (2\pi\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2} \sum (x_i - \mu)^2}.$$

Since a likelihood is any function proportional to the joint pdf, let's take

$$L(\mu) = e^{-\frac{1}{2\sigma^2} \sum (x_i - \mu)^2}.$$

We already know the usual (unrestricted) MLE for μ : $\hat{\mu} = \bar{X}$.

Question: Now what maximizes $L(\mu)$ when H_0 is true?

Answer: That's easy since H_0 contains only one point! (μ_0)

So,

$$\max_{\mu = \mu_0} L(\mu) = L(\mu_0)!$$

(exciting not factorial...)

So,

$$\lambda = \frac{e^{-\frac{1}{2\sigma^2} \sum (x_i - \mu_0)^2}}{e^{-\frac{1}{2\sigma^2} \sum (x_i - \bar{x})^2}} = e^{-\frac{1}{2\sigma^2} [\sum (x_i - \mu_0)^2 - \sum (x_i - \bar{x})^2]}$$

Since we're going to have to compute a probability $P(\lambda(\vec{X}) \leq k; H_0)$, let's simplify λ :

$$\begin{aligned} \sum (x_i - \mu_0)^2 - \sum (x_i - \bar{x})^2 &= \sum x_i^2 - 2\mu_0 \sum x_i + n\mu_0^2 - \sum x_i^2 + 2\bar{x} \sum x_i - n\bar{x}^2 \\ &= -2\mu_0 \sum x_i + n\mu_0^2 + 2\bar{x} \sum x_i - n\bar{x}^2 \end{aligned}$$

Hey! This sort of looks like something squared...

In fact, if we pull the n out:

$$\begin{aligned} n(-2\mu_0 \frac{1}{n} \sum x_i + \mu_0^2 + 2\bar{x} \frac{1}{n} \sum x_i - \bar{x}^2) &= n(-2\mu_0 \bar{x} + \mu_0^2 + 2\bar{x}^2 - \bar{x}^2) \\ &= n(-2\mu_0 \bar{x} + \mu_0^2 + \bar{x}^2) \\ &= n(\bar{x} - \mu_0)^2 \end{aligned}$$

(Cool!)

So,

$$\lambda = \exp \left[\frac{-n(\bar{x} - \mu_0)^2}{2\sigma^2} \right]$$

Recall that we will reject H_0 if $\lambda \leq k$ where k is such that $P(\lambda(\vec{X}) \leq k; H_0) = \alpha$:

$$\begin{aligned} P \left(\exp \left[\frac{-n(\bar{X} - \mu_0)^2}{2\sigma^2} \right] \leq k; H_0 \right) &= P \left(\frac{-n(\bar{X} - \mu_0)^2}{2\sigma^2} \leq \ln k; H_0 \right) \\ &= P \left(\frac{n(\bar{X} - \mu_0)^2}{\sigma^2} \geq -2 \ln k; H_0 \right) \\ &= P \left(\left(\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \right)^2 \geq k_1; H_0 \right) \end{aligned}$$

where k_1 is such that this probability is α .

Now if H_0 is true and μ is indeed μ_0 , then $\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$ and so

$$\left(\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \right)^2 \sim \chi^2(1)$$

So

$$P\left(\left(\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}\right)^2 \geq k_1; H_0\right) = P(W \geq k_1)$$

where $W \sim \chi^2(1)$.

Hence,

$$k_1 = \chi_\alpha^2(1).$$

So, we will reject H_0 if

$$\left(\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}\right)^2 \geq \chi_\alpha^2(1).$$

This is the GLRT of size α ! □

Example 6.3.3

(Note: I could only think of one example of a composite versus composite that isn't a computational nightmare, so I'll save this easy problem for you!)

Let $X_1, X_2, \dots, X_n$ be a random sample from the $unif(0, \theta]$ distribution.

(Note that I closed the right side of the interval. I only did this so that we won't have a problem with a "max", but this wouldn't matter at all if we were using the more general definition of the GLR that uses supremums.)

Find the GLRT of size α for $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$.

$$\begin{aligned} f(x; \theta) &= \frac{1}{\theta} I_{(0, \theta]}(x) \\ \Rightarrow L(\theta) &= \theta^{-n} \prod_{i=1}^n I_{(0, \theta]}(x_i) = \theta^{-n} I_{(0, \theta]}(x_{(n)}) \end{aligned}$$

We already know that the plain old unrestricted MLE is $\hat{\theta} = X_{(n)}$. (This is because the derivative of θ^{-n} with respect to θ set equal to zero gives no

information, so since $0 < x_i \leq \theta$ for $i = 1, 2, \dots, n$, the smallest θ can be is $x_{(n)}$ which then maximizes the decreasing $L(\theta) = \theta^{-n}$.)

Since H_0 consists of only one point: $\theta = \theta_0$, the maximum of $L(\theta)$ restricted to this one point set is simply $L(\theta_0)$.

So,

$$\lambda = \frac{\theta_0^{-n} I_{(0, \theta_0]}(x_{(n)})}{x_{(n)}^{-n} I_{(0, x_{(n)}]}(x_{(n)})} = \left(\frac{x_{(n)}}{\theta_0} \right)^n I_{(0, \theta_0]}(x_{(n)})$$

As usual, we will reject H_0 if

$$\left(\frac{x_{(n)}}{\theta_0} \right)^n I_{(0, \theta_0]}(x_{(n)}) \leq k$$

where k is such that

$$P \left(\left(\frac{X_{(n)}}{\theta_0} \right)^n I_{(0, \theta_0]}(X_{(n)}) \leq k; H_0 \right) = \alpha$$

Well, under H_0 , that indicator is always 1, so we can drop it:

$$\alpha = P \left(\left(\frac{X_{(n)}}{\theta_0} \right)^n \leq k; H_0 \right) = P \left(X_{(n)} \leq \theta_0 k^{1/n}; H_0 \right) = P \left(X_{(n)} \leq k_1; H_0 \right)$$

Finally, we solve for k_1 :

$$\begin{aligned} \alpha &= P \left(X_{(n)} \leq k_1; H_0 \right) \\ &= [P \left(X_1 \leq k_1; H_0 \right)]^n \\ &= \frac{k_1}{\theta_0} \\ \Rightarrow \quad k_1 &= \theta_0 \alpha^{1/n}. \end{aligned}$$

So, the GLRT of size α , for $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$ for a random sample of size n from the $unif(0, \theta]$ distribution, is to reject H_0 in favor of H_1 if

$$X_{(n)} \leq \theta_0 \alpha^{1/n}.$$

Wait a minute... something doesn't seem quite right. Does that rejection rule make sense? One would certainly think that it is not true that θ , the upper limit of the support set for the sample, is equal to θ_0 if we happened to observe $x_{(n)} > \theta_0$. Shouldn't we be rejecting for some values of $X_{(n)}$ that are too large? We don't just want to say: "Well of course we would automatically reject if $X_{(n)} > \theta_0$ " because making up rules to suit our needs will mess with the size of the test that we worked so hard to obtain.

The answer is "yes" but I didn't mention it as we were going through the example because I didn't want to muck up the steps of a standard GLRT procedure with this special case of having the indicators mixed with the parameters. Normally,

- We set something less than or equal to k .
- We simplify this to something else less (greater) than or equal to some k_1 .
- We say that the original k doesn't matter anymore.

With indicators mixed with parameters, it does matter... this is a weird sticky little point that I will not hold you accountable for in this course, but, for the record, I will expound upon it here.

Going back to the original rejection rule, we reject H_0 if

$$\left(\frac{X_{(n)}}{\theta_0}\right)^n I_{(0, \theta_0]} \leq k.$$

Since $k = (k_1/\theta_0)^n$, this becomes, "reject H_0 if":

$$\left(\frac{X_{(n)}}{\theta_0}\right)^n I_{(0, \theta_0]}(X_{(n)}) \leq \left(\frac{k_1}{\theta_0}\right)^n = \left(\frac{\theta_0 \alpha^{1/n}}{\theta_0}\right)^n = \alpha.$$

For a non-trivial α (ie: $\alpha > 0$), if $X_{(n)}$ is greater than θ_0 , the left hand side of this inequality will be zero, hence less than α , hence we will reject, as desired. ■

6.3.3 Wilks' Theorem

For most of this course, we have been pretty fortunate in that we have been able to find the distribution of various statistics we have used in estimation and hypothesis testing. The Central Limit Theorem has helped us out of a few tough spots though. Here is another asymptotic result that can be quite useful!

Suppose that $X_1, X_2, \dots, X_n$ is a random sample from the distribution with pdf $f(x; \theta)$. Let Θ be the parameter space. Denote, as usual, the joint pdf as $f(\vec{x}; \theta)$. Consider the following “regularity conditions”.

- The true parameter that generated the data lies in the interior of the parameter space Θ .
- The derivative $\frac{\partial}{\partial \theta} f(\vec{x}; \theta)$ exists and $\frac{\partial}{\partial \theta} \int f(\vec{x}; \theta) d\vec{x} = \int \frac{\partial}{\partial \theta} f(\vec{x}; \theta) d\vec{x}$.
- $\frac{\partial}{\partial \theta} \ln f(\vec{x}; \theta)$ exists.
- The third derivative of the log-likelihood function exists and is bounded.
- The Fisher information $I_n(\theta)$ satisfies $0 < I_n(\theta) < \infty$.

Under these conditions, we have the following result.



Wilks' Theorem

Under the assumption that $H_0 : \theta \in \Theta_0$ is true,

$$-2 \ln \lambda(\vec{X}) \xrightarrow{d} \chi^2(r)$$

where $r = \dim(\Theta) - \dim(\Theta_0)$.

Here $\dim(\Theta)$ and $\dim(\Theta_0)$ refer to the dimensions of the parameter space and restricted parameter space, respectively.

Example 6.3.4

(Finding Dimensions)

Suppose that $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$.

- (a) Suppose that σ^2 is known and that we are testing

$$H_0 : \mu = \mu_0 \text{ versus } H_1 : \mu \neq \mu_0.$$

The parameter space is $\{\mu : -\infty < \mu < \infty\}$ which is the real line and is one dimensional. The null hypothesis restricts μ to a single point, which is considered zero-dimensional. Thus, by Wilks' Theorem

$$-2 \ln \lambda(\vec{X}) \xrightarrow{d} \chi^2(1).$$

- (b) Suppose that σ^2 is unknown and that we are testing

$$H_0 : \mu = \mu_0, \sigma^2 = \sigma_0^2 \text{ versus } H_1 : \text{Not } H_0.$$

Here, the parameter space is $\{(\mu, \sigma^2) : -\infty < \mu < \infty, \sigma^2 \geq 0\}$. It is two-dimensional.

The null hypothesis says that $(\mu, \sigma^2) = (\mu_0^2, \sigma_0^2)$, which is a single point, again considered zero-dimensional. So, for Wilks' Theorem, $r = 2 - 0 = 2$ and we have

$$-2 \ln \lambda(\vec{X}) \xrightarrow{d} \chi^2(2).$$

- (c) Suppose that σ^2 is unknown but that we are only testing

$$H_0 : \mu = \mu_0 \text{ versus } H_1 : \mu \neq \mu_0.$$

The parameter space is still two dimensional but now the null hypothesis space is

$$\{(\mu_0, \sigma^2) : \sigma^2 \geq 0\}$$

which is a line in $\mathbb{R}^2$ and hence is one-dimensional. So, for Wilks' Theorem, $r = 2 - 1 = 1$ and we have

$$-2 \ln \lambda(\vec{X}) \xrightarrow{d} \chi^2(1).$$

Now that we are adept at finding the dimensions for Wilks' Theorem, let's try to apply it.

Example 6.3.5

Suppose that $X_1, X_2, \dots, X_n$ is a random sample from the $\Gamma(3, \beta)$ distribution. Find an approximate large sample GLRT of size α for

$$H_0 : \beta = \beta_0 \text{ versus } H_1 : \beta \neq \beta_0$$

for some fixed $\beta_0 > 0$.

The pdf is

$$f(x; \beta) = \frac{1}{\Gamma(3)} \beta^3 x^2 e^{-\beta x} I_{(0, \infty)}(x) = \frac{1}{2} \beta^3 x^2 e^{-\beta x} I_{(0, \infty)}(x).$$

The joint pdf is

$$f(\vec{x}; \beta) \stackrel{iid}{=} \prod_{i=1}^n f(x_i; \beta) = \frac{1}{2^n} \beta^{3n} \left[\prod_{i=1}^n x_i^2 \right] e^{-\beta \sum_{i=1}^n x_i} \prod_{i=1}^n I_{(0, \infty)}(x_i).$$

Let's form the GLR. The usual unrestricted MLE for β is $\hat{\beta} = (3n)/\sum X_i$. The restricted MLE is $\hat{\beta}_0 = \beta_0$.

The GLR is

$$\lambda(\vec{x}) = \frac{L(\hat{\beta}_0)}{L(\hat{\beta})} = \dots = \left(\frac{\sum x_i}{3n} \right)^{3n} \beta_0^{3n} e^{-\beta_0 \sum x_i}.$$

Plugging the random variables in, the GLRT is to reject H_0 , in favor of H_1 if $\lambda(\vec{X}) \leq k$ where k is chosen to give a size α test.

Now, for this simple null hypothesis,

$$\begin{aligned} \alpha &= P(\text{Reject } H_0; \beta_0) \\ &= P(\lambda(\vec{X}) \leq k; \beta_0) \\ &= P(\ln \lambda(\vec{X}) \leq \ln k; \beta_0) \\ &= P(-2 \ln \lambda(\vec{X}) \geq -2 \ln k; \beta_0) \\ &= P(-2 \ln \lambda(\vec{X}) \geq k_1; \beta_0) \end{aligned}$$

where $k_1 = -2 \ln k$.

By Wilks' Theorem, we have that $-2 \ln \lambda(\vec{X}) \stackrel{approx}{\sim} \chi^2(1)$ for a large sample. Thus, we can take $k_1 = \chi_{\alpha, 1}^2$.

The approximate large sample test is to reject H_0 , in favor of H_1 if

$$-2 \ln \lambda(\vec{X}) \geq \chi_{\alpha, 1}^2.$$

(Equivalently, we could solve for k and say that the test is to reject H_0 if $\lambda(\vec{X}) \leq \exp[-\frac{1}{2}\chi_{\alpha,1}^2]$. That's cool.)

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Appendix A

Tables of Distributions

Table A.1: Common Discrete Distributions

Name	pdf	Parameter Space	Mean	Variance	Moment Generating Function = $E[e^{tX}]$
Uniform	$\frac{1}{n+1}I_{\{0,1,\dots,n\}}(x)$	$n = 1, 2, \dots$	$n/2$	$n(n+2)/12$	$\sum_{j=0}^n \frac{1}{n+1} e^{jt} = \frac{1-e^{(n+1)t}}{(n+1)(1-e^t)}$
Bernoulli	$p^x(1-p)^{1-x}I_{\{0,1\}}(x)$	$0 \leq p \leq 1$	p	$p(1-p)$	$(1-p) + pe^t$
Binomial	$\binom{n}{x} p^x(1-p)^{n-x}I_{\{0,1,\dots,n\}}(x)$	$0 \leq p \leq 1$ $n = 1, 2, \dots$	np	$np(1-p)$	$[(1-p) + pe^t]^n$
Hyper-geometric	$\frac{\binom{K}{x} \binom{M-K}{n-x}}{\binom{M}{n}} I_{\{0,1,\dots,n\}}(x)$	$M = 1, 2, \dots$ $K = 0, \dots, M$ $n = 1, \dots, M$	$n \frac{K}{M}$	$n \frac{K}{M} (1 - \frac{K}{M}) \frac{M-n}{M-1}$	not useful
Poisson	$\frac{e^{-\lambda} \lambda^x}{x!} I_{\{0,1,\dots\}}(x)$	$\lambda > 0$	λ	λ	$\exp[\lambda(e^t - 1)]$
Geometric	$p(1-p)^x I_{\{0,1,\dots\}}(x)$	$0 < p < 1$	$(1-p)/p$	$(1-p)/p^2$	$\frac{1-(1-p)e^t}{1-(1-p)e^t}$ for $t < -\ln(1-p)$
Geometric	$p(1-p)^{x-1} I_{\{1,2,\dots\}}(x)$	$0 < p < 1$	$1/p$	$(1-p)/p^2$	$\frac{pe^t}{1-(1-p)e^t}$ for $t < -\ln(1-p)$
Negative Binomial	$\binom{r+x-1}{x} p^r(1-p)^x I_{\{0,1,\dots\}}(x)$	$0 < p < 1$ and $r > 0$	$r(1-p)/p$	$r(1-p)/p^2$	$\left[\frac{p}{1-(1-p)e^t} \right]^r$ for $t < -\ln(1-p)$
Negative Binomial	$\binom{x-1}{r-1} p^r(1-p)^{x-r} I_{\{r,r+1,\dots\}}(x)$	$0 < p < 1$ and $r > 0$	r/p	$r(1-p)/p^2$	$\left[\frac{pe^t}{1-(1-p)e^t} \right]^r$ for $t < -\ln(1-p)$
Beta-binomial	$\binom{n}{x} \frac{\mathcal{B}(x+a, n-x+b)}{\mathcal{B}(a, b)} I_{\{0,\dots,n\}}(x)$	$a > 0, b > 0$ $n = 1, 2, \dots$	$\frac{na}{a+b}$	$\frac{nb(a+b)}{(a+b)^2(a+b+1)}$	not useful
Logarithmic	$\frac{(1-p)^x}{(-x \ln p)} I_{\{1,2,\dots\}}(x)$	$0 < p < 1$	$\frac{(1-p)}{(-p \ln p)}$	$\frac{(1-p)(1-p+\ln p)}{(-p \ln p)^2}$	$\frac{\ln[1-(1-p)e^t]}{\ln p}$ for $t < -\ln(1-p)$
Discrete Pareto	$\sum_{j=1}^{\infty} \frac{(1/x)^{j+1}}{(1/j)^{\gamma+1}} I_{\{1,2,\dots\}}(x)$	$\gamma > 0$	$\frac{\sum_{j=1}^{\infty} (1/j)^{\gamma}}{\sum_{j=1}^{\infty} (1/j)^{\gamma+1}}$ for $\gamma > 1$		does not exist

Table A.2: Common Continuous Distributions

Name	pdf = $f(x)$ cdf = $F(x)$	Parameter Space	Mean	Variance	Moment Generating Function = $E[e^{tX}]$
Uniform	$f(x) = \frac{1}{\beta} I_{(\alpha, \alpha+\beta)}(x)$	$-\infty < \alpha < \infty$ $\beta > 0$	$\alpha + \frac{\beta}{2}$	$\frac{\beta^2}{12}$	$\frac{e^{(\alpha+\beta)t} - e^{\alpha t}}{\beta t}$
Normal	$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$	$-\infty < \mu < \infty$ $\sigma > 0$	μ	σ^2	$\exp[\mu t + \frac{1}{2}\sigma^2 t^2]$
Exponential (rate λ)	$f(x) = \lambda e^{-\lambda x} I_{(0, \infty)}(x)$	$0 < \lambda < \infty$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{\lambda}{\lambda - t}$ for $t < \lambda$
Bilateral exponential	$f(x) = \frac{1}{2}\beta e^{-\beta x-\alpha }$	$-\infty < \alpha < \infty$ $0 < \beta < \infty$	α	$\frac{2}{\beta^2}$	$e^{t\alpha}/(1 - t^2/\beta^2)$ for $ t < \beta$
Gamma	$f(x) = \frac{1}{\Gamma(\alpha)} \beta^\alpha x^{\alpha-1} e^{-(\beta x)} I_{(0, \infty)}(x)$	$\alpha > 0$ $\beta > 0$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$	$\left(\frac{\beta}{\beta - t}\right)^\alpha$ for $t < \beta$
Weibull	$f(x) = \frac{\gamma}{\beta} \left(\frac{x-\alpha}{\beta}\right)^{\gamma-1} e^{-\left(\frac{x-\alpha}{\beta}\right)^\gamma} I_{(\alpha, \infty)}(x)$	$-\infty < \alpha < \infty$ $\beta > 0, \gamma > 0$	$\alpha + \beta \Gamma\left(1 + \frac{1}{\gamma}\right)$	$\beta^2 \left[\Gamma\left(1 + \frac{2}{\gamma}\right) - \Gamma^2\left(1 + \frac{1}{\gamma}\right) \right]$	not useful $E[(X - \alpha)^k] = \beta^k \Gamma\left(1 + \frac{k}{\gamma}\right)$
Beta	$f(x) = \frac{1}{B(a,b)} x^{a-1} (1-x)^{b-1} I_{(0,1)}(x)$	$a > 0$ $b > 0$	$\frac{a}{a+b}$	$\frac{ab}{(a+b)^2(a+b+1)}$	not useful $E[X^k] = \frac{B(a+k,b)}{B(a,b)}$
Pareto	$f(x) = \frac{\gamma}{(1+x)^{\gamma+1}} I_{(0, \infty)}(x)$	$\gamma > 0$	$1/(\gamma - 1)$ for $\gamma > 1$	$\gamma/[(\gamma - 2)(\gamma - 1)^2]$ for $\gamma > 2$	does not exist
Cauchy	$f(x) = \frac{1}{\beta\pi} \frac{1}{1 + \left(\frac{x-\alpha}{\beta}\right)^2}$	$-\infty < \alpha < \infty$ $\beta > 0$	does not exist	does not exist	does not exist
Logistic	$F(x) = \left[1 + e^{-\left(\frac{x-\alpha}{\beta}\right)} \right]^{-1}$	$-\infty < \alpha < \infty$ $\beta > 0$	α	$\beta^2 \pi^2 / 3$	$e^{\alpha t} \beta \pi t \csc(\beta \pi t)$
Gumbel	$F(x) = \exp \left[-e^{-\left(\frac{x-\alpha}{\beta}\right)} \right]$	$-\infty < \alpha < \infty$ $\beta > 0$	$\alpha + \beta \gamma$ where $\gamma \approx 0.577216$	$\beta^2 \pi^2 / 6$	$e^{\alpha t} \Gamma(1 - \beta t)$ for $t < 1/\beta$
Log normal	$F(x) = \Phi \left(\frac{\ln x - \mu}{\sigma} \right) I_{(0, \infty)}(x)$	$-\infty < \mu < \infty$ $\sigma > 0$	$\exp[\mu + \frac{1}{2}\sigma^2]$	$\exp[2\mu + 2\sigma^2] - \exp[2\mu + \sigma^2]$	does not exist $E[X^k] = \exp[k\mu + \frac{1}{2}k^2\sigma^2]$

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Appendix B

The Jacobian and a Change of Variables

B.1 The Jacobian

For a single integral, the substitution $x = x(u)$ gives us

$$\int_{x_1}^{x_2} f(x) dx = \int_{u_1}^{u_2} f(x(u)) \frac{dx}{du} du$$

For double integrals, we consider a change of variables from the x_1x_2 -plane to the y_1y_2 -plane, where

$$y_1 = g_1(x_1, x_2) \quad \text{and} \quad y_2 = g_2(x_1, x_2).$$

We assume we can solve these equations for x_1 and x_2 :

$$x_1 = g_1^{-1}(y_1, y_2) \quad \text{and} \quad x_2 = g_2^{-1}(y_1, y_2).$$

Now, consider with a small rectangle R in the x_1x_2 -plane whose lower left corner is the point $(x_1^{(0)}, x_2^{(0)})$ and whose dimensions are Δx_1 and Δx_2 .

The image of R in the y_1y_2 -plane is a region S , one of whose boundary points is $(y_1^{(0)}, y_2^{(0)}) = (g_1(x_1^{(0)}, x_2^{(0)}), g_2(x_1^{(0)}, x_2^{(0)}))$.

The vector

$$\vec{v}(x_1, x_2) = g_1(x_1, x_2)\vec{i} + g_2(x_1, x_2)\vec{j}$$

is the position vector of the image of the point (x_1, x_2) .

The equation of the lower side of R is $x_2 = x_2^{(0)}$, and the image curve of this side is given by the vector function $\vec{v}(x_1, x_2^{(0)})$.

The tangent vector at $(y_1^{(0)}, y_2^{(0)})$ to this image curve is

$$\vec{v}_{x_1} = \left. \frac{\partial g_1(x_1, x_2^{(0)})}{\partial x_1} \right|_{x_1=x_1^{(0)}} \vec{i} + \left. \frac{\partial g_2(x_1, x_2^{(0)})}{\partial x_1} \right|_{x_1=x_1^{(0)}} \vec{j}.$$

Similarly, the tangent vector at $(y_1^{(0)}, y_2^{(0)})$ to the image curve of the left side of R (given by $x_1 = x_1^{(0)}$) is given by

$$\vec{v}_{x_2} = \left. \frac{\partial g_1(x_1^{(0)}, x_2)}{\partial x_2} \right|_{x_2=x_2^{(0)}} \vec{i} + \left. \frac{\partial g_2(x_1^{(0)}, x_2)}{\partial x_2} \right|_{x_2=x_2^{(0)}} \vec{j}.$$

We can approximate the image region of R by a parallelogram determined by the secant vectors

$$\vec{a} = \vec{v}(x_1^{(0)} + \Delta x_1, x_2^{(0)}) - \vec{v}(x_1^{(0)}, x_2^{(0)}) \quad \text{and} \quad \vec{b} = \vec{v}(x_1^{(0)}, x_2^{(0)} + \Delta x_2) - \vec{v}(x_1^{(0)}, x_2^{(0)}).$$

However,

$$\vec{v}_{x_1} = \lim_{\Delta x_1 \rightarrow 0} \frac{\vec{v}(x_1^{(0)} + \Delta x_1, x_2^{(0)}) - \vec{v}(x_1^{(0)}, x_2^{(0)})}{\Delta x_1}$$

and so, for Δx_1 “small”, we have

$$\vec{v}(x_1^{(0)} + \Delta x_1, x_2^{(0)}) - \vec{v}(x_1^{(0)}, x_2^{(0)}) \approx \vec{v}_{x_1} \Delta x_1.$$

Similarly

$$\vec{v}(x_1^{(0)}, x_2^{(0)} + \Delta x_2) - \vec{v}(x_1^{(0)}, x_2^{(0)}) \approx \vec{v}_{x_2} \Delta x_2.$$

This means that we can approximate the image of R by a parallelogram determined by the vectors $\vec{v}_{x_1} \Delta x_1$ and $\vec{v}_{x_2} \Delta x_2$. Therefore, we can approximate the area of the image of R by the area of this parallelogram which is

$$|(\vec{v}_{x_1} \Delta x_1) \times (\vec{v}_{x_2} \Delta x_2)| = |\vec{v}_{x_1} \times \vec{v}_{x_2}| \Delta x_1 \Delta x_2.$$

Computing this cross product, we get

$$\vec{v}_{x_1} \times \vec{v}_{x_2} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial y_1}{\partial x_1} & \frac{\partial y_2}{\partial x_1} & 0 \\ \frac{\partial y_1}{\partial x_2} & \frac{\partial y_2}{\partial x_2} & 0 \end{vmatrix} = \begin{vmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_2}{\partial x_1} \\ \frac{\partial y_1}{\partial x_2} & \frac{\partial y_2}{\partial x_2} \end{vmatrix} \vec{k} = \begin{vmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} \end{vmatrix} \vec{k}$$

The determinant that arises in this calculation is called the Jacobian of the transformation. Call it $J(x_1, x_2)$. We can then approximate the area ΔA of the image of R by

$$\Delta A = |J(x_1, x_2)| \Delta x_1 \Delta x_2$$

So,

$$\int \int_S f(y_1, y_2) dA \approx \sum_{i=1}^n \sum_{j=1}^m f(y_1^{(i)}, y_2^{(j)}) \Delta A \approx \sum_{i=1}^n \sum_{j=1}^m f(g_1(x_1^{(i)}, x_2^{(j)}), g_2(x_1^{(i)}, x_2^{(j)})) |J(x_1, x_2)| \Delta x_1 \Delta x_2.$$

B.2 A Bivariate Transformation

We now proceed with the problem of finding the joint pdf of two functions of two continuous random variables.

Let the joint pdf for X_1 and X_2 be denoted by $f_{X_1, X_2}(x_1, x_2)$ and let $Y_1 = g_1(X_1, X_2)$ and $Y_2 = g_2(X_1, X_2)$ denote a one-to-one transformation from the $x_1 x_2$ -plane to the $y_1 y_2$ -plane.

Let $\mathcal{A}$ denote the two-dimensional set in the $x_1 x_2$ -plane for which $f_{X_1, X_2}(x_1, x_2)$ is non-zero. Let $\mathcal{B}$ denote the transformed region in the $y_1 y_2$ -plane.

Now, let $A \subset \mathcal{A}$ and let B denote the mapping of A under the one-to-one transformation.

$$P((Y_1, Y_2) \in B) = P((X_1, X_2) \in A) = \int \int_A f_{X_1, X_2}(x_1, x_2) dx_1 dx_2.$$

By “The Jacobian” section above, this is the same as

$$\int \int_B f_{X_1, X_2}(g_1^{-1}(y_1, y_2), g_2^{-1}(y_1, y_2)) |J(y_1, y_2)| dy_1 dy_2$$

B.3 More Variables

All of the above can be extended to the case of many variables. That is, if $X_1, X_2, \dots, X_n$ has joint pdf $f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n)$ and we consider the transformations

$$\begin{aligned} y_1 &= g_1(x_1, x_2, \dots, x_n) \\ y_2 &= g_2(x_1, x_2, \dots, x_n) \\ &\vdots \\ y_n &= g_n(x_1, x_2, \dots, x_n), \end{aligned}$$

then the joint pdf of $Y_1, Y_2, \dots, Y_n$ is given by

$$f_{Y_1, Y_2, \dots, Y_n}(y_1, y_2, \dots, y_n) = f_{X_1, X_2, \dots, X_n}(g_1^{-1}(y_1, y_2, \dots, y_n), \dots, g_n^{-1}(y_1, y_2, \dots, y_n)) \cdot |J(y_1, y_2, \dots, y_n)|$$

where

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} & \dots & \frac{\partial x_1}{\partial y_n} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} & \dots & \frac{\partial x_2}{\partial y_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial y_1} & \frac{\partial x_n}{\partial y_2} & \dots & \frac{\partial x_n}{\partial y_n} \end{vmatrix}.$$

Appendix C

Standard Normal, t , χ^2 and F Tables

Table C.1: Standard Normal Probabilities

 $\Phi(z) = P(Z \leq z)$ for $Z \sim N(0, 1)$

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

Table C.2: t Critical ValuesCritical Values for $T \sim t(\nu)$ (t_α captures area α to the right)

ν	$t_{0.40}$	$t_{0.333}$	$t_{0.25}$	$t_{0.20}$	$t_{0.125}$	$t_{0.10}$	$t_{0.05}$	$t_{0.025}$	$t_{0.01}$	$t_{0.005}$	$t_{0.001}$
1	0.325	0.577	1.000	1.376	2.414	3.078	6.314	12.706	31.821	63.657	318.31
2	0.289	0.500	0.816	1.061	1.604	1.886	2.920	4.303	6.965	9.925	22.327
3	0.277	0.476	0.765	0.978	1.423	1.638	2.353	3.182	4.541	5.841	10.215
4	0.271	0.464	0.741	0.941	1.344	1.533	2.132	2.776	3.747	4.604	7.173
5	0.267	0.457	0.727	0.920	1.301	1.476	2.015	2.571	3.365	4.032	5.893
6	0.265	0.453	0.718	0.906	1.273	1.440	1.943	2.447	3.143	3.707	5.208
7	0.263	0.449	0.711	0.896	1.254	1.415	1.895	2.365	2.998	3.499	4.785
8	0.262	0.447	0.706	0.889	1.240	1.397	1.860	2.306	2.896	3.355	4.501
9	0.261	0.445	0.703	0.883	1.230	1.383	1.833	2.262	2.821	3.250	4.297
10	0.260	0.444	0.700	0.879	1.221	1.372	1.812	2.228	2.764	3.169	4.144
11	0.260	0.443	0.697	0.876	1.214	1.363	1.796	2.201	2.718	3.106	4.025
12	0.259	0.442	0.695	0.873	1.209	1.356	1.782	2.179	2.681	3.055	3.930
13	0.259	0.441	0.694	0.870	1.204	1.350	1.771	2.160	2.650	3.012	3.852
14	0.258	0.440	0.692	0.868	1.200	1.345	1.761	2.145	2.624	2.977	3.787
15	0.258	0.439	0.691	0.866	1.197	1.341	1.753	2.131	2.602	2.947	3.733
16	0.258	0.439	0.690	0.865	1.194	1.337	1.746	2.120	2.583	2.921	3.686
17	0.257	0.438	0.689	0.863	1.191	1.333	1.740	2.110	2.567	2.898	3.646
18	0.257	0.438	0.688	0.862	1.189	1.330	1.734	2.101	2.552	2.878	3.610
19	0.257	0.438	0.688	0.861	1.187	1.328	1.729	2.093	2.539	2.861	3.579
20	0.257	0.437	0.687	0.860	1.185	1.325	1.725	2.086	2.528	2.845	3.552
21	0.257	0.437	0.686	0.859	1.183	1.323	1.721	2.080	2.518	2.831	3.527
22	0.256	0.437	0.686	0.858	1.182	1.321	1.717	2.074	2.508	2.819	3.505
23	0.256	0.436	0.685	0.858	1.180	1.319	1.714	2.069	2.500	2.807	3.485
24	0.256	0.436	0.685	0.857	1.179	1.318	1.711	2.064	2.492	2.797	3.467
25	0.256	0.436	0.684	0.856	1.178	1.316	1.708	2.060	2.485	2.787	3.450
26	0.256	0.436	0.684	0.856	1.177	1.315	1.706	2.056	2.479	2.779	3.435
27	0.256	0.435	0.684	0.855	1.176	1.314	1.703	2.052	2.473	2.771	3.421
28	0.256	0.435	0.683	0.855	1.175	1.313	1.701	2.048	2.467	2.763	3.408
29	0.256	0.435	0.683	0.854	1.174	1.311	1.699	2.045	2.462	2.756	3.396
30	0.256	0.435	0.683	0.854	1.173	1.310	1.697	2.042	2.457	2.750	3.385
35	0.255	0.434	0.682	0.852	1.170	1.306	1.690	2.030	2.438	2.724	3.340
40	0.255	0.434	0.681	0.851	1.167	1.303	1.684	2.021	2.423	2.704	3.307
45	0.255	0.434	0.680	0.850	1.165	1.301	1.679	2.014	2.412	2.690	3.281
50	0.255	0.433	0.679	0.849	1.164	1.299	1.676	2.009	2.403	2.678	3.261
55	0.255	0.433	0.679	0.848	1.163	1.297	1.673	2.004	2.396	2.668	3.245
60	0.254	0.433	0.679	0.848	1.162	1.296	1.671	2.000	2.390	2.660	3.232
∞	0.253	0.431	0.674	0.842	1.150	1.282	1.645	1.960	2.326	2.576	3.090

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Symbols/Notation

$\Gamma(\alpha)$	gamma function, page 68
$\Gamma(\alpha, \beta)$	gamma distribution, page 66
μ	mean, page 31
Ω	sample space, page 9
σ	standard deviation, page 33
σ^2	variance, page 32
$\sim$	specifies a distribution, page 15
$geom_0$	geometric distribution starting from 0, page 19
$geom_1$	geometric distribution starting from 1, page 19
I	indicator (Usually, see also information.), page 34
$n!$	n factorial, page 5
$P(A)$	probability A occurs, page 10

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